EMV®
Terminal Type Approval

Book C-8

Test Cases

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Revision Log – 1.1c

The following changes were done since the publication of version 1.1b:

- In the following Test Cases reference to ‘C8’ proprietary tag was replaced by ‘CF’:
  - 4M10-9081 case 5a
  - 4M25-9015 case 5a
  - 4M35-9260 case 6
  - 4M55-9140 case 5a
- Editorial updates

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1 Using this Manual

1.1 Purpose

EMV Contactless Specifications for Payment Systems, Book C-8 – Kernel 8 Specification defines the behaviour of the Kernel used in combination with cards having a Kernel Identifier indicating Kernel 8.
The present document describes a set of test cases which when applied to the Contactless Reader, are designed to determine whether the Contactless Reader meets the requirements stated in the specification.

1.2 Audience

The intended audiences for this document are EMVCo recognised laboratories, Entry Point and Contactless application providers, acquiring members or merchants. Although acquiring members and merchants may reference this document, type approval testing oriented toward Entry Point and Contactless application providers.

1.3 Reference Documents

Table 1-1—Reference Documents

Reference	Document	Version Issue Date
[C8]	EMV Contactless Specifications for Payment Systems – Book C-8 - Kernel 8 Specification	Version 1.1, June 2023
[SB302]	EMV Specification Bulletin No. 302	1st Ed, March 2024
[SB310]	EMV Specification Bulletin No. 310	1st Ed, March 2025
[Admin]	EMV Terminal Type Approval Contactless Product – Administrative Process	Latest applicable version
[Book A]	EMV Contactless Specifications for Payment Systems – Book A – Architecture and General Requirements	Latest applicable version
[Book B]	EMV Contactless Specifications for Payment Systems – Book B – Entry Point Specification	Latest applicable version

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Reference	Document	Version Issue Date
[Book E]	EMV Contactless Specifications for Payment Systems – Book E – Security and Key Management	Latest applicable version
[ICS]	EMV Kernel 8 Implementation Conformance Statement	Latest applicable version
[TestEnv]	EMV® Terminal Type Approval Contactless Product – Kernel 8 Testing Environment Requirements	Latest applicable version

1.4 Notations

Table 1-2—Notations

Term	Definition
‘0’ to ‘9’ and ‘A’ to ‘F’	16 hexadecimal digits. Values expressed in hexadecimal form are enclosed in single quotes (i.e. ‘_’).

1.5 Abbreviations

Table 1-3—Abbreviations

Abbreviation	Description
an	Alphanumeric
API	Application Priority Indicator
AUC	Application Usage Control
AVN	Application Version Number
b	Binary
CAkey	Certification Authority Public Key
CID	Cryptogram Information Data
GenAC	Generate AC
GetPO	Get Processing Options
hex.	Hexadecimal
n	Numeric

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Abbreviation	Description
NCA	Length of the Certificate Authority Public Key Modulus
OPS	Outcome Parameter Set
PK	Public Key
RFU	Reserved for Future Use
RID	Registered Application Provider Identifier
RRP	Relay Resistance Protocol. In this document, RRP also refers to the 'exchange relay resistance data' APDU
RRPcEntropy	Device RRP Entropy
RRPcEstimated	Device Estimated Transmission Time For RRP R-APDU
RRPcMaxTime	Max Time for processing RRP APDU
RRPcMinTime	Minimum Time for processing RRP APDU
RRPrAccuracyThreshold	RRP Accuracy Threshold
RRPrCounter	RRP Counter
RRPrEntropy	Terminal Relay Resistance Entropy
RRPrExpectedCAPDU	Terminal Expected Transmission Time for RRP C-APDU
RRPrExpectedRAPDU	Terminal Expected Transmission Time for RRP R-APDU
RRPrMeasuredTime	Measured RRP Time
RRPrMaxGrace	Maximum RRP Grace Period
RRPrMinGrace	Minimum RRP Grace Period
RRPrMismatchThreshold	RRP Transmission Time Mismatch Threshold
TAC	Terminal Action Code
TVR	Terminal Verification Results
UIRD	User Interface Request Data
UN	Unpredictable Number

1.6 Terminology

This section explains the terms used in this document.

Successfully complete:

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When a Pass Criteria expects the product to successfully complete the transaction, it means the transaction should either be approved offline or sent online for authorization.

Amount:

This document uses several wordings as listed below:

- "Purchase Amount": this refers to the amount entered for a purchase transaction. A cashback amount may then be entered.
- "Cashback amount": this refers to the cashback amount
- "Transaction amount" / "Amount": this refers to the amount entered for a generic transaction (purchase, withdrawal,…). It is assumed that no cashback amount is entered.

The Test Tool Suppliers must pay attention at the following:

- The '9F02' tag value is the sum of the purchase and cashback (if any) amounts
- Some Product Providers will provide a Level 3 test application that permits to enter the purchase and cashback amounts. This Level 3 application will then sum both amounts before passing the '9F02' tag to the Kernel
- Some Product Providers will not provide any Level 3 test application, in which case the operator shall enter the '9F02' value directly

The test wording must be reviewed accordingly. Please find an example below:

Please enter PURCHASE AMOUNT = 7.50 or AMOUNT AUTHORIZED = 10.00

Please enter CASHBACK AMOUNT (or AMOUNT OTHER) = 2.50

1.7 Remarks

1.7.1 DEK checking

The DEK pass criteria are like "After issuance of the first ReadRecord command, the reader shall build a DEK signal including DataToSend template containing the PAN".

The pass criteria expect the Kernel to build the DEKs at the right time, as defined in the specifications. In other words, the tests will ensure the DEK value is as expected.

The tests do not verify at what time the Kernel is sending the DEKs.

As a consequence, the test tools shall:

- Not verify when the DEK is sent (eg: after the 1st ReadRecord command).
- Ensure the DEK value is the correct one.
- Ensure the number of DEK(s) is as expected.

Note 1: Some readers may be unable to send a DEK until a R-APDU has been received.

Note 2: When automation is used, the test operators must take into account that there maybe a significant communication speed difference between the APDU interface and the TEI interface. The TEI interface will use TCP-IP so the exchange will typically last 10-20ms. The APDU (NFC) interface will use the test probes and the exchanges will typically last 100-200ms. Consequently the events received by the test tool maybe in an order different from the ones sent by the Kernel. For instance, the Kernel may send CAPDU-DEK but the test tool may

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receive DEK-CAPDU. This is why EMVCo recommends the Product Providers to use the PCD-bypass interface (TCP-IP) defined in [TestEnv].

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2 Introduction

2.1 Terminal Vendor Testing Process

The [Admin] document describes the process Kernel 8 providers must follow for their products to be approved by EMVCo.

One of the steps of this process is the Terminal Contactless Products Type Approval. The objective of this process is to ensure the Kernel 8 embedded in the Contactless Reader is compliant with requirements defined in [C8].

The compliance of a Contactless Reader to the [C8] requirements is verified by executing tests specified by Tests Cases. A Test Case:

- Covers a specific objective (for example: ensure that if, in the course of normal processing, the reader recognises that data returned by the card is incorrectly formatted, then the reader terminates processing.).
- Refers to a specific requirement from the [C8] specifications.
- Defines the conditions (card & reader state) that must be met before the tests can be started.
- Defines the procedure to execute the tests.
- Defines the main criteria that allow a test to be considered as successful.

2.2 Test Case Template Description

2.2.1 Overview

This paragraph describes the Test Case template used in this document.

Table 2-1—Test Case Template

Test Case number and name	
Objective	*Objective of the Test Case*
Applicability	*Implementation option(s) needed to run this Test Case (absent if empty)*
References	*Reference to specification section*
Conditions	*Card and reader configuration required for this Test Case*
Procedure	*Test procedure (absent if empty)*
Pass Criteria	*Main criteria for Test Case validation*

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Test Case number and name	
Note	*Note/comment (absent if empty)*

2.2.2 Naming convention

Test Cases names are structured as follow: '**4Xbb-cccc**' (e.g.: 4M60-1016).

Where:

- "4X" defines the type of Test Case:
  - "4M": Unitary Test Cases
  - "4S": System Test Cases
  - "4U": Undisclosed Test Cases (Not present in this document, refer to section 2.3.3)
- "bb" indicates a transaction stage. Value is set to '00' for all System and Undisclosed Test Cases

Table 2-2 lists the transaction stage identifiers used in this document.

Table 2-2—Transaction stage ID

Transaction stage ID	Description
00	System tests and Undisclosed tests
10	Initialization
20	DOLs
25	GetPO
30	RRP
35	ReadRecord
36	ReadData
38	Waiting for PTFWF
40	Pre-GenAC checks
50	TAA
55	GenAC
59	No local authentication
60	Certificates processing
61	IAD MAC
62	EDA MAC
65	CVM processing
70	WriteData

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Transaction stage ID	Description
80	Completion
X8	Stress tests

- "cccc" is an arbitrary number permitting to order the Test Cases. Note: '9ccc' corresponds to negative/failure tests.

2.2.3 Applicability Conditions

The Applicability Conditions are detailed in the below table.

Table 2-3—Applicability Conditions

Transaction stage ID	Description
[DE/DS]	The Test Case applies when the product implements DE/DS
NOT [DE/DS]	The Test Case applies when the product does not implement DE/DS

2.2.4 Pass Criteria

The Test Case description provides the main criteria the product must meet to pass the corresponding Test Case.

However, as the tests are run in black-box mode, they may fail for reasons not explicitly described in the test. For example, a test on Generate AC may fail before reaching the GenerateAC if the product does not implement the GetPO command correctly.

This is particularly true as all tests verify all signals and data outputted by the kernel. Typically, the Test Tool verifies that:

- C-APDUs are issued in the correct order
- DOL values are correctly populated
- DEK values are correctly populated
- the MSG signal is correctly populated
- the OUT signal is correctly populated (e.g all bits of the TVR are verified)

A consequence of this is that the debugging of the very first test will take time as the Test Tool will fail until the product implements the main sections of [C8] correctly. EMVCo recommends starting with the test 4M80-4444 which is considered as a 'nominal' test.

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2.3 Test Case execution.

2.3.1 Debug session

As mentioned above, the test 4M80-4444 may be a good start when initiating the debugging of a product. This test is using ECC certificates.

The test 4M60-0227 is another nominal test. It implements RSA certificates.

2.3.2 Type Approval session

During a Type Approval session, the laboratory will order the tests by test configuration to reduce the number of reconfiguration of the Product.

2.3.3 Undisclosed Test Cases

Undisclosed Test Cases are not public and shall not be performed during Debug session.

During the Type Approval session, Testing Laboratories shall not provide details of these Undisclosed Test Cases (e.g. logs) to Product Providers. However the Testing Laboratories are allowed to communicate the 'derivated tests' that are mentioned in the Test Case note field.

2.3.4 System Test Cases

System Test Cases are performed without TEI Interface and with Test Tool NFC Probe (PCD-bypass Interface must not be used). As a consequence:

- The Configuration and transaction related data shall be loaded manually. These data shall be provided by the Test Tool before each test.
  Default transaction related data are:
  - Transaction Currency Code= '0978'
  - Transaction Currency Exponent = '02'
  - Transaction Type = '00'
  - Amount, Authorised = '000000001500'
  - Transaction Date = Testing date
  - Transaction Time = Testing Time

- Some Acceptance Criteria must be validated by the Operator, the Operator shall follow the instructions provided by the Test Tool.

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3 Unitary Test Cases

4M10-0010: Initialization - Instantiate tags	
Objective	To ensure the reader correctly instantiates the known tags.
References	KS.1
Conditions	Test the tags that have not been tested already: - AcquirerIdentifier '9F01' (K) - MerchantCategoryCode '9F15' (K) - MerchantID '9F16' (K) - Merchant Name and Location '9F4E' (K) - TerminalID '9F1C' (K) - Track1DD '9F1F' (K/RA) - Track2DD '9F20' (K/RA) - Transaction Currency Expo '5F36' (K/ACT) - Terminal Risk Management Data '9F1D' (K)
Pass Criteria	For all cases, the reader shall return the correct values in the PDOL and CDOL1.

4M10-0011: Initialization - KernelKeyData and UN	
Objective	To ensure the reader correctly initializes the KernelKeyData and UN tags.
References	s1.13,s2.7
Conditions	PDOL and CDOL request UN - transaction#1 is performed and leads to StartB (after GenAC) - transaction#2a is performed and leads to StartC (GPO SW) - transaction#2b is performed and leads to termination (GPO: AIP missing) - transaction#3 is performed and leads to completion - no RRP Reader is powered off and on - transaction#4 is performed and leads to StartB (after GenAC) - transaction#5 is performed and leads to completion - 2 RRP exchanges Reader is powered off and on - transaction#6a is performed and leads to StartC (GPO SW) - transaction#6b is performed and leads to termination (GPO: AIP missing) - transaction#7 is performed and leads to completion - 3 RRP exchanges

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Pass Criteria	For all cases, the reader shall provide different value for the KernelKeyData tag. UN(s) shall be different at each transaction UN shall have the same value in GenAC and DataRecord. Transaction#3 UN must be the same in the GetPO and GenAC (since RRP not performed) When applicable: - UN must be different in the GetPO and the first RRP command - UN must be different in each RRP command - UN must be the same in the last (or single) RRP command and GenAC

4M10-0012: Initialization - extend TLV DB with proprietary tags	
Objective	To ensure the reader correctly extends the TLV database with proprietary tags, if any. To ensure the reader include empty proprietary tags in DiscretionaryData.
References	KS.1
Conditions	TLV DB includes proprietary tags
Pass Criteria	For all cases, the reader shall return the correct values in the PDOL and CDOL1.

4M10-0070: Initialization - DF Name partial match with AID	
Objective	To ensure terminal continues the transaction as expected when the DF Name return on final Select is longer than AID(Terminal) but DF Name begins with the AID.
References	s202122232425.16-TRUE, s202122232425.17
Conditions	The Length of the DF Name returned in Final Select is higher than the AID length. DF Name begins with the AID (partial match)
Pass Criteria	transaction must successfully complete.

4M10-0071: Initialization - Parsing Select Response - TLV length	
Objective	To ensure the reader correctly manages the TLV length when parsing the Select AID response.
References	s1.5, s1.6-TRUE
Conditions	Case 1: FCI Template ('6F') length and FCI Proprietary Template ('A5') length are coded on 1 byte. Primitive data object lengths are coded on 1 byte. Case 2: FCI Template ('6F') length is coded on 1 byte. Primitive data object lengths are coded on 2 bytes. Case 3: FCI Template ('6F') length is coded on 2 bytes while the data length is less than 128. Primitive data object lengths are coded on 1 byte.

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	Case 4: FCI Template ('6F') length is coded on 2 bytes while the data length is more than 128. Primitive data object lengths are coded on 2 bytes. Case 5: FCI Proprietary ('A5') Template length is coded on 1 byte. Primitive data object lengths are coded on 2 bytes. Case 6: FCI Proprietary ('A5') Template length is coded on 2 bytes while the data length is less than 128. Primitive data object lengths are coded on 1 byte. Case 7: FCI Proprietary ('A5') Template length is coded on 2 bytes while the data length is more than 128. Primitive data object lengths are coded on 2 bytes.
Pass Criteria	In all cases the transaction must successfully complete.

4M10-0072: Initialization - Parsing Select Response - Padding	
Objective	To ensure the reader ignores any '00' padding before, between or after TLV-coded data objects.
References	s1.5, s1.6-TRUE
Conditions	Case 1: the template contains DFName and FCI Proprietary Template with a '0000' padding in between. Case 2: the template contains API and LanguagePreference with a '00' padding of 50 bytes after the two objects. Case 3: the template contains '0000' padding between tags inside the 'BF0C' template
Pass Criteria	In all cases the EMV transaction must successfully complete.

4M10-0073: Initialization - Parsing Select Response - storing data	
Objective	To ensure the reader stores recognized data objects (Known and update conditions include RA Signal) and ignores the unrecognized ones. To ensure the reader disregards known card returned tags not having RA update conditions and the unknown tags.
References	s1.5, s1.6-TRUE
Conditions	FCI of ADF contains all mandatory tags: DFName ('6F\84'), PDOL ('6F\A5\9F38') and CardQualifier ('6F\A5\BF0C\9F2C'). The PDOL includes all the data objects (except '9F38' itself) listed in the related cases below. Case 0: (no optional tags): FCI of ADF contains no optional data objects (tested in 4M80-4444). Case1: AFL is returned in Select command instead of GPO command. Case 2a: (all optional tags, min length): FCI of ADF contains all optional data objects: ApplicationLabel ('50'), API ('87'), LanguagePreference ('5F 2D'), PDOL ('9F38'), ICTI ('9F

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11'), ApplicationPreferredName ('9F 12') and FCI issuer DiscretionaryData ('BF0C') containing: ASRPD ('9F0A' with value '0095 01 AF'), IIN and IINextended. Variable length data items have the minimum length (except '9F38').

Case 2b: (all optional tags, max length): Same as 2a, optional data items have the maximum length (except ASRPD that may keep the same value as in case 2a).

Case 3: same as case 2b plus the following EMV proprietary data objects not present in TLV database(i.e.: IsKnown=False) are returned :

- IBAN ('5F53') is returned above the 'A5' template
- Bank Identifier Code ('5F54') is returned in 'A5' before the 'BF0C' template
- IssuerCountryCode - alpha 2 ('5F55') is returned in 'BF0C'
- IssuerCountryCode - alpha 3 ('5F56') is returned in 'BF0C'

Case 4 :

- the proprietary tag '9F05' is present in the TLV empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K/RA). This tag is returned in 'BF0C'.
- The tag '9F38' is returned in the template 'A5' but after the 'BF0C'.

Case 5:Proprietary tag 'DFED08', Private Class, not present in TLV database(i.e.: IsKnown=False). This tag is returned in 'BF0C' with 3 bytes length '82 00 0x' and any value.

Case 6:

- Card returns CardQualifier in template A5 instead of BF0C
- Card returns DFname (extended) in A5 instead of 6F
- Card returns AppLabel in 6F instead of A5
- BF0C not returned

Case 7: Proprietary tag '81' is present in the TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K (not RA)). This tag is returned in 'BF0C' with different value.

Case 8: Proprietary tag 'C8', Private Class, upd=K/RA, empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True: upd=K/RA). This tag is returned in 'BF0C'with any value.

Case 9: Known tag '9F8221',not proprietary tag, upd=K (not RA), not present in TLV database

Case 10: Known tag '9F4E', not proprietary tag, upd=K (not RA), empty in TLV database

Case 11: Known tag '9F40',not proprietary tag, upd=K (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in 'BF0C' with different value and incorrect Length.

Case 12: Known tag'CE', proprietary tag ,privateClass, not RA, not empty in TLV database

Case 13: Proprietary tag '81' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in 'BF0C' with different value and incorrect Length.

Case 14: Proprietary tag '9F04' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in 'BF0C' with different value and incorrect Length.

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	Case 15: Proprietary tag '9F04' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in 'BF0C' with same value. Case 16: Proprietary tag '9F04' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in 'BF0C' with different value and correct Length. Case 17: Known tag '9F02',not proprietary tag, upd=K/ACT/DET (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in 'BF0C' with different value and incorrect Length. Case 18: Known tag '9F02',not proprietary tag, upd=K/ACT/DET (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in 'BF0C' with different value and correct Length. Case 19: Known tag '9F40',not proprietary tag, upd=K (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in 'BF0C' with different value . Case 20: UnKnown tag 'D8',Private Class (i.e.: IsKnown=False, IsPresent=False). This tag is returned in 'BF0C' . Case 21: Proprietary tag'9F08', upd=K/RA, not empty in TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/RA). This tag is returned in 'BF0C' with different value. Case 22: Proprietary tag'C4', Private Class, upd=K/ACT/DET, empty in TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K/ACT/DET (not RA)). This tag is returned in 'A5' with maximum correct length. Case 23: Proprietary tag 'C6', Private Class, upd=K/DET, empty in TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K/DET (not RA)). This tag is returned in 'A5' . Case 24: Proprietary tag 'C5', Private Class, upd=K/ACT, not empty (updated in ACT) (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT (not RA)). This tag is returned in '6F' with incorrect length . Case 25: Proprietary tag '5F19', upd=K/ACT/RA, not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K/ACT/RA). This tag is returned in '6F' with correct length . Case 26: Proprietary tag 'C9', Private Class, , with update condition K/ACT/DET/RA, not empty (updated in ACT) (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False and update conditions include RA Signal). This tag is returned in '6F' with correct length and same that value than in ACT .
Pass Criteria	The transaction must successfully complete. Cases 2a, 2b, 4, 8, 25 and 26: The PDOL populated data includes value returned in Select response for the recognized tags . Others Cases: The reader must ignore the values of unrecognized tags returned by the card in Select response, these values are not stored in TLV Database. The PDOL shall be correctly populated.

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4M10-0108: Initialization - Select Response - CardQualifier.Version	
Objective	To ensure the reader accepts CardQualifier. Version greater than 0.
References	s1.8-TRUE
Conditions	Case 0: CardQualifier.Version = '02' (already tested by default) Case 1: CardQualifier.Version = '01' Case 2: CardQualifier.Version = '03' Case 3: CardQualifier.Version = '10' Case 4: CardQualifier.Version = 'FE'
Pass Criteria	The transaction must successfully complete.

4M10-0112: Initialization - Transaction Data	
Objective	To ensure the reader correctly adds the transaction data to the TLV database.
Applicability	Cases 01b: [DE/DS] Cases 01a: NOT([DE/DS])
References	s1.12
Conditions	Case 1a (known + upd=ACT, no DEDS): The ACT signal contains the tags with update condition "/ACT/": - AccountType - Amount* - AmountOther* - TransactionCurrencyCode* - TransactionCurrencyExponent - TransactionDate* - TransactionTime - TransactionType* Case 1b (known + upd=ACT, DEDS): samse as 1a, plus the ACT signal contains the DE/DS tags with update condition "/ACT/": - DataEnvelopeToWrite** - PTFWF - TagsToRead**: non empty in the TLV database, provided in the ACT with different value (empty) Case 2 (proprietary, present,not Empty upd=K/ACT): '9F04' is present in TLV DB not empty, '9F04' is provided in ACT with different Value. Case 3 (present, upd=ACT): '9F02' present in TLV DB, '9F02' is provided in ACT Case 5 (Proprietary Tag, PrivateClass, not present): The ACT signal contains the following unknown tag: C4

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	Case 6 (Proprietary, present, upd=K/RA): '9F05' is present in TLV DB empty, it is provided in ACT Case 7 (known, not present, upd!=ACT): PAN is provided in ACT Case 8 (known, present, upd!=ACT): The ACT signal contains the following tag: '9F06 05 1122334455'. Case 9 (Proprietary Tag, present, Empty, K/ACT): '9F28' is provided in ACT
Pass Criteria	Transaction must complete successfully. Case 1a: The reader does not send any DEK with 9F0A since the Tags2Read value is modified (to '9F2C') in the ACT signal. A DEK shall be sent with '9F2C' value. Cases 1a and 1b: Reader must provide the correct values in PDOL Cases 2, 3 and 9: the reader must store the value provided in ACT Cases 5 to 8: reader must provide zero value in PDOL
Note	***: already tested by default (value verified in DataRecord)** ****: already tested**

4M10-0113: Initialization - Initialize data objects	
Objective	To ensure the reader correctly initializes data objects when FCI parsing is correct.
Applicability	Cases 02: [DE/DS] Cases 01: NOT([DE/DS])
References	s1.13
Conditions	PDOL requests: CVMResults, TVR, TerminalCapabilities, RelayResistanceTimeExcess, KernelQualifier, ReadDataStatus, WriteDataStatus, DataNeeded.
Procedure	- transaction#1 is performed and leads to completion - transaction#2 is performed and leads to StartB (after GenAC) - transaction#3a is performed and leads to StartC (GPO SW) - transaction#3b is performed and leads to termination (GPO: AIP missing) - transaction#4 is performed and leads to completion - no RRP Reader is powered off and on - transaction#5 is performed and leads to decline - transaction#6 is performed and leads to StartB (after GenAC) - transaction#7 is performed and leads to completion - 3 RRP exchanges - transaction#8 is performed and leads to completion - 1 RRP exchange
Pass Criteria	The following values must be returned by the reader during the GetPO command: - CVMResults indicates '000000' - TVR indicates '0000000080' - TerminalCapabilities byte 1 = CardDataInputCapability - TerminalCapabilities byte 2 indicates '00'

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	- TerminalCapabilities byte 3 = SecurityCapability - RelayResistanceTimeExcess = '0000' - KernelQualifier bytes 1-2= '0180' or '0100' (case 2)

4M10-0115: Initialization - setting Security Capability	
Objective	To ensure the reader correctly initializes Security Capability in Terminal Capabilities.
References	s1.13
Conditions	2 transactions are performed, the security capability ('9F820A') is part of the PDOL. - transaction#1 is performed with Security Capability set to '10' - transaction#2 is performed with Security Capability set to '20'
Pass Criteria	The transaction must successfully complete.The reader shall return the correct values in the PDOL Related Data.

4M10-0130: Initialization - setting Language Preference	
Objective	To ensure the reader correctly initializes OPS, UIRD1, UIRD2 and ErrorIndication as per KS.2. To ensure the reader correctly sets (and pads if necessary) the Language Preference in UIRD1 and UIRD2. To ensure the reader correctly behaves when SupportForFieldOffDetection is set. To ensure the reader correctly behaves when restart is requested.
References	s1.10, s1.11, s2930.3, s2930.4, s2930.5, s2930.6, s2930.7, s2930.8
Conditions	SupportForFieldOffDetection is set. RestartIndicator.restartRequested = RESTART. Several values are tested for RestartIndicator.alternateInterface. Case 1: Language Preference length is 2 - 'Message Identifier' in RestartIndicator = '00000000' Case 2: Language Preference length is 2 - 'Message Identifier' in RestartIndicator = '00100001' (TRY AGAIN) Case 3: Language Preference length is 2 - 'Message Identifier' in RestartIndicator = '00011000' (TRY ANOTHER INTERFACE) Case 4: Language Preference length is 2 - 'Message Identifier' in RestartIndicator = '00100000' (SEE PHONE) Case 5: Language Preference length is 2 - 'Message Identifier' in RestartIndicator = '00011011' (RFU) Case 6: Language Preference length is 2 - 'Message Identifier' in RestartIndicator = '11111111' (RFU)

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	Case 7: Language Preference length is 2 - 'Message Identifier' in RestartIndicator = '10000000' (RFU) Case 8: Language Preference length is 8 - 'Message Identifier' in RestartIndicator = '00000000' Case 9: Language Preference is absent
Pass Criteria	Language Preference is correctly set in UIRD1 and UIRD2 In UIRD1 and UIRD2, 'MessageID' is set to: - 'Try Again' (cases 1, 5, 6, 7, 8, 9) - 'Message ID' in Restart Indicator' (cases 2, 3, 4)

4M10-2019: Initialization - ASI identification	
Objective	To ensure the reader correctly identifies the SC ASI. To ensure the reader correctly sets the 'SC ASI' and 'C ASI List' in Kernel Qualifier.
References	C.101-FALSE, C.10, s2.6
Conditions	SC ASI returned by card in CardQualifier: Case 0: '00FFFF' (already tested) Case 1: '018800' - ECC Case 2: '880001' - ECC, RSA not supported Case 3: 'FFFF00' - RSA
Pass Criteria	Transaction must complete successfully. The reader must correctly sets the 'SC ASI' (to '00') and 'C ASI List' in Kernel Qualifier.

4M10-9017: Initialization - Mandatory Configuration Data	
Objective	To ensure the reader stops the transaction when Mandatory Configuration data object is empty or not present or with incorrect length.
References	KS3, KS.4-FALSE, KS.5, KS.6
Conditions	Case 1: Mandatory Data Objects are not present Case 2: Mandatory Data Objects are empty Case 3: Mandatory Data Objects lengths are incorrect
Pass Criteria	The reader must stop the transaction and select the next AID in the candidate list.

4M10-9019: Initialization - invalid ASIs	
Objective	To ensure the reader terminates when the card does not support ASI '00'.
References	C.101-FALSE, C.10, s2.6

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Conditions	SC ASI returned by card in CardQualifier: Case 1: '0188FF' Case 2: '01FFFF' RSA not supported Case 3: '880102' Case 4: '200102'
Pass Criteria	Transaction must terminate.

4M10-9081: Initialization - Select - parsing error	
Objective	To ensure the reader stops the transaction when the response message of the Final Select command is not correctly formatted (i.e.: data is absent from the response, parsing is not as per the encoding rules, tag already present in the TLV database with different value, tag not present in the TLV database but update conditions do not include RA, length is not in the specified range).
References	s1.7, s1.14
Conditions	The card returns several valid combinations in the PPSE response. SW12 in Final Select response is '9000'. Case 1: Final Select response does not include any data, only SW12='9000' Final Select response includes the following formatting errors: Case 2: the length value indicated in the template '6F' is greater than the actual data length Case 3: the following additional data is returned: 'DF 00' (tag indicates subsequent byte and subsequent byte is not greater than zero, plus length is absent.) Case 4: the following additional data is returned: 'DF01 02 03' (length is erroneous) Case 5a: the proprietary tag 'CF' is present in the TLV database. This tag is returned in the Final Select response with length lower than expected length range(with any value). Case 5b: the proprietary tag 'C8' is present in the TLV database. This tag is returned in the Final Select response with length higher than expected length range(with any value). Case 6: '87' is returned in 'A5' with a value of 2 bytes (instead of 1) Case 8: '84' is returned twice, with different values Case 9: AIP is returned empty Case 10: DFname is returned empty Case 11: CardQualifier is returned empty Case 16: Application Label ('50') length is 17 bytes Case 17: Application Preferred Name ('9F12') length is 17 bytes Case 18: DFName ('84') length is 17 bytes Case 19: Language Preference ('5F2D') length is 1 bytes Case 20: primitive tag is returned without length Case 21: Card Qualifier ('9F2C')length is 8 bytes Case 22: Issuer Identification Number Extended ('9F0C') length is 2 bytes case 23: Language Preference ('5F2D') length is empty

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	Case 24: Tag with Tag field length greater than 3 Bytes is returned (9FFF8F01) in 'A5'
Pass Criteria	The reader must stop the transaction upon reception of the erroneous Final Select response. The reader must build an OUT signal. In OUT\OPS, - 'Start' must indicate "C" - 'Status' must indicate "SELECT NEXT" In OUT\DiscretionaryData\ErrorIndication, 'L2' must indicate 'PARSING ERROR' (or also 'CARD DATA MISSING' for case 1). The reader shall select the next application in the candidate list.

4M10-9110: Initialization - Select - DFname or CardQualifier missing	
Objective	To ensure the reader stops the transaction when DFname or CardQualifier are empty.
References	s1.8-FALSE, s1.9, s1.14
Conditions	Case 1: DFname is absent Case 2a: CardQualifier is absent Case 2b: BF0C is empty Case 2c: BF0C is absent Case 3: CardQualifier.Version = '00'
Pass Criteria	The reader must stop the transaction upon reception of the erroneous SelectADF response, and select the next AID in the candidate list.

4M20-0310: DOL - Concatenation	
Objective	To ensure the reader concatenates a list of data elements in the sequence in which the corresponding data objects appear in the DOL. To ensure the reader correctly provides the Terminal Risk Management Data object ('9F1D') when requested in a DOL. To ensure the reader correctly builds the DOL when a tag is requested on more than 128 bytes (length>=80). To ensure the reader correctly builds the GPO command when template '83' is more than 128 bytes long.
References	s2.15-TRUE,s2.16,s2.17,"DOL Handling"
Conditions	The DOL contains the tags of several data objects which are known to the reader. Include 1-byte and 2-bytes and 3-bytes tags. Terminal Risk Management Data object value is different for Test1 and Test2 AIDs. Perform a test for each of the following DOLs: - PDOL

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	- CDOL1 Case 1: Test1 AID Case 2: Test2 AID Case 3: Test1 AID ,PDOL data > 128 Case 4: DOL requests a tag on 129 bytes Case5: long PDOL (PDOL='9F66049F02069F03069F1A0295055F2A029A039C019F37049F4E149F2B089E40') Case6: CashBack transaction with long PDOL (PDOL='9F66049F02069F03069F1A0295055F2A029A039C019F37049F4E149F2B089E40')
Pass Criteria	Within the data field of a command message containing the DOL-related data, the reader orders the data elements in the sequence in which the corresponding data objects appear in the DOL. Case 3: the template tag '83' has a length coded on 2 bytes

4M20-0340: DOL - Unknown tag	
Objective	To ensure the reader provides a data element with the length specified and a value of all hexadecimal zeroes if the tag of any data object identified in the DOL is unknown to the reader.
References	"DOL Handling"
Conditions	The DOL requests tags that are unknown to the reader: - EMV tag - C2 tag returned in Select (eg: 9F5E) - C2 tag returned in Record (eg: 8E) - 3-byte private class tag Perform a test for each of the following DOLs: - PDOL - CDOL1
Pass Criteria	The reader shall provide the DOL with portion of the DOL field representing the Data Object filled with hexadecimal zeroes (portion has the same length as the Data Object in DOL).

4M20-0355: DOL - Tag with length=0	
Objective	To ensure that, if the tag of any data object identified in the DOL has a length field coded '00' then the reader considers the tag as not present in the DOL.
References	"DOL Handling"
Conditions	The DOL includes a tag with length field coded '00'. The tag is known to the reader. Perform a test for each of the following DOLs:

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	- PDOL - CDOL1
Pass Criteria	The reader shall not send any value in the DOL for the data object having the length field coded '00' in the DOL request.

4M20-0405: DOL - Known tag, missing	
Objective	To ensure that, if the tag of any data object identified in the DOL is known to the reader and the optional static data object requested in the DOL is absent, then the reader provides a data element with the length specified and a value of all hexadecimal zeroes.
References	"DOL Handling"
Conditions	The DOL contains a tag which is known to the reader but is absent from the card The DOL contains a tag which is known to the reader but is absent from the reader. The tag update conditions include DET Perform a test for each of the following DOLs: - PDOL - CDOL1*
Pass Criteria	The reader shall provide a data element with the length specified and a value of all hexadecimal zeroes.
Note	***: when the kernel implements DE/DS, the missing DOL items are added to DataNeeded. However the configuration data (PTFWF is absent) are such that the DEK is not sent and the GenAC command is built as defined in the pass criteria.**

4M20-0410: DOL - Shorter data object length, numeric format	
Objective	To ensure that, if the length specified in the DOL entry is less than the length of the actual data object, the reader truncates the leftmost bytes of the data element if the data object has numeric (n) format.
References	"DOL Handling"
Conditions	Case1: The DOL contains a data objects which haves numeric format and a length shorter than actual Data Object Length. Case2: The DOL contains proprietary tag present in TLV database which has numeric format and a length shorter than actual Data Object Length. For each case Perform a test is performed for each of the following DOLs: - PDOL - CDOL1
Pass Criteria	All Cases: The reader shall provide the DOL with portion of the DOL field representing the Data Object correctly truncated (portion has the same length as the Data Object in DOL).

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4M20-0412: DOL - Shorter data object length, other format	
Objective	To ensure that, if the length specified in the DOL entry is less than the length of the actual data object, the reader truncates the rightmost bytes of the data for any format other than numeric.
References	"DOL Handling"
Conditions	Case1: The DOL contains a data objects which has have a format other than numeric (b, an, ans, cn) and a length shorter than actual Data Object Length. Case2: The DOL contains proprietary tag present in TLV database which has a format other than numeric (b, an, ans, cn) and a length shorter than actual Data Object Length. Perform a test for each of the following DOLs: - PDOL - CDOL1
Pass Criteria	The reader shall provide the DOL with portion of the DOL field representing the Data Object correctly truncated (portion has the same length as the Data Object in DOL).

4M20-0415: DOL - Longer data object length, other format	
Objective	To ensure that, if the length specified is greater than the actual data, the reader pads the actual data with trailing hexadecimal zeroes for any other format than numeric or compressed numeric.
References	"DOL Handling"
Conditions	Case 1: The DOL contains a data objects which has a format other than numeric or compressed numeric and a length longer than actual Data Object Length. Case 2: The DOL contains proprietary tags present in TLV database which has a format other than numeric or compressed numeric and a length longer than actual Data Object Length. Perform a test for each of the following DOLs: - PDOL - CDOL1
Pass Criteria	All Cases: The reader shall provide the DOL with portion of the DOL field representing the Data Object correctly padded with trailing hexadecimal zeroes (portion has the same length as the Data Object in DOL).

4M20-0420: DOL - Longer data object length, numeric format	
Objective	To ensure that, if the length specified is greater than the actual data, the reader pads the actual data with leading hexadecimal zeroes if the data has numeric format.
References	"DOL Handling"

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Conditions	Case 1: The DOL of the card contains a data object which has numeric format and a length longer than actual Data Object Length. Case 2: The DOL of the card contains a proprietary tag present in TLV database which has numeric format and a length longer than actual Data Object Length. Perform a test for each of the following DOLs: - PDOL - CDOL1
Pass Criteria	All Cases: The reader shall provide the DOL with portion of the DOL field representing the Data Object correctly padded with leading hexadecimal zeroes (portion has the same length as the Data Object in DOL)

4M20-0445: DOL - Constructed tag	
Objective	To verify that whenever the tag of any data object identified in the DOL represents a constructed data object, the reader provides a data element with the length specified and a value of all hexadecimal zeroes.
References	"DOL Handling"
Conditions	The DOL of the card contains a constructed Data Object. Perform a test for each of the following DOLs: - PDOL - CDOL1
Pass Criteria	The reader shall provide the DOL with portion of the DOL field representing the Data Object filled with hexadecimal zeroes (portion has the same length as the Data Object in DOL).

4M20-0515: DOL - longer data object length, compressed numeric format	
Objective	To ensure that, if the length specified is greater than the actual data, the reader pads the actual data with trailing hexadecimal FF's if the data has compressed numeric (cn) format.
References	"DOL Handling"
Conditions	Case 1: The DOL of the card contains a data object which has compressed numeric format and a length longer than actual Data Object Length (AIDtest1). Case 2: Same than Case 1 with AIDtest2. Case 3: The DOL of the card contains a proprietary tag which has compressed numeric format and a length longer than actual Data Object Length. Perform a test for each of the following DOLs: - PDOL - CDOL1

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Pass Criteria	All Cases: The card shall receive the DOL with portion of the DOL field representing the Data Object correctly padded with trailing hexadecimal FF's (portion has the same length as the Data Object in DOL).

4M25-0122: GetPO - Response parsing - length processing	
Objective	To ensure the reader correctly parses the GetPO response message with lengths on one or several bytes.
References	s20.14, s20.15-TRUE
Conditions	GetPO response includes AIP, AFL, CardKeyData and ATC. Case 1: template length is coded on 1 byte. Data object lengths are coded on 1 byte. Case 2: template length is coded on 1 byte. Data object lengths are coded on 2 bytes (lengths are less than 128 bytes). Case 3: template length is coded on 2 bytes while the data length is less than 128. Data object lengths are coded on 1 byte. Case 4: template length is coded on 2 bytes while the data length is more than 128. Data object lengths are coded on 2 bytes (lengths are less than 128 bytes).
Pass Criteria	The AIP value in the DataRecord must be in accordance with the value returned by the card. The transaction must successfully complete.

4M25-0123: GetPO - Response parsing - padding	
Objective	To ensure the reader ignores any '00' padding before, between or after TLV-coded data objects.
References	s20.14, s20.15-TRUE
Conditions	Case 1: the template contains AFL and AIP with a '0000' padding in between. AFL is placed before the AIP in the response. Case 2: the template contains AFL and AIP with a '00' padding of 50 bytes after the two objects Case 3: the template contains AFL and AIP with a '0000' padding before the two objects
Pass Criteria	The reader must issue ReadRecord commands in accordance to the AFL. The transaction must successfully complete.

4M25-0124: GetPO - Response parsing - storing data	
Objective	To ensure the reader stores the recognized data objects (Known and update conditions include RA Signal) and ignores the unrecognized ones.
References	s20.14, s20.15-TRUE, s20.18-TRUE

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Conditions	Case 0 (no optional tags): GetPO does not return any optional tag (tested in 4M80-4444). Case 1: The optional data objects '9F36' and '8C' are returned. '9F36' has same value as in GenAC. '9F36' is requested in the CDOL. Case 2: the CApk is returned in the GetPO response instead of the ReadRecord. Case 3: - the proprietary tag '9F05' (EMV tag "Application DiscretionaryData") is present in the TLV database with no value (i.e.: IsKnown=False, IsPresent=True, IsEmpty=True so update condition includes RA). This tag is returned in the GetPO response (with any value). '9F05' is listed in the CDOL. - the proprietary private class tag 'CF' is returned in GPO with same value. This tag is present in the TLV database with RA update condition. 'CF' is listed in the CDOL. Case 4: the proprietary private class tag 'DFED08' is returned with 3 bytes length '82 00 0x' and any value. It is placed first in the response. This tag is not present in the TLV database. Case 5: - the proprietary tag '9F05' (EMV tag "Application DiscretionaryData") is not present in the TLV database (i.e.: IsKnown=False, IsPresent=False, IsEmpty=False). This tag is returned in the GetPO response (with any value). '9F05' is listed in the CDOL. Case 6: '9F0A' is returned empty in the Select ADF response. It is then returned with any value in the GPO response. It is requested in the CDOL. Case 7: '84' is returned in the GPO response with same value as in Select. Case 8: CDOL is returned empty (in GetPO). Case 9: The reader data object "Amount Other" ('9F03' upd:K/RA) is not present in the TLV database but it is returned in the GetPO response Case 10: Tag '9F4E', upd=K (not RA) is present empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True upd=K (not RA)). This tag is returned in the GPO response Case 11: Tag '9F1E', upd=K (not RA) is present not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False upd=K (not RA)). This tag is returned in the GPO response with same V as in TLV database Case 12: Proprietary Tag 'C1', privateClass, upd=K/RA, not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False upd=K/RA). This tag is returned in the GPO response with same V as in TLV database Case 13: Proprietary Tag '9F08', upd=K/RA, not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False upd=K/RA). This tag is returned in the GPO response with same V as in TLV database Case 14: Tag '9F40', upd=K (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in the GPO response with different value and incorrect Length. Case 15: Proprietary tag '81' is present in the TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K (not RA)). This tag is returned in the GPO response with different value. Case 16: Proprietary tag '81' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in the GPO response with different value and incorrect Length.

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	Case 17: Tag '9F21', upd=K/ACT/DET (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the GPO response with different value and incorrect Length. Case 18: Tag '9F21', upd=K/ACT/DET (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the GPO response with different value and correct Length. Case 19: Proprietary tag '9F04' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the GPO response with same value. Case 20: Proprietary tag '9F04' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the GPO response with different value and incorrect Length. Case 21: Proprietary tag '9F04' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the GPO response with different value and correct Length. Case 22: Proprietary tag 'CE' ,privateClass, not RA (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in the GPO response with different value and correct Length. Case 23:Tag,'9F40', upd=K (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in the GPO response with different value. Case 24: Proprietary tag 'C3', privateClass, empty not RA(i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K (not RA)). This tag is returned in the GPO response response. Case 25: Proprietary tag 'C4', PrivateClass, upd=K/ACT/DET, Not empty (value updated in ACT) (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the GPO response with incorrect length. Case 26: Proprietary tag '9F29', upd=K/DET, empty in TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K/DET (not RA)). This tag is returned in the GPO response . Case 27: Proprietary tag '9F28', upd=K/ACT, not empty (updated in ACT) (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT (not RA)). This tag is returned in the GPO response with incorrect length . Case 28: Proprietary tag 'C7',Private Class, upd=K/DET/RA, empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True nd update conditions include RA). This tag is returned in the GPO response with correct length . Case 29: Proprietary tag 'C9',Private Class, with update condition K/ACT/DET/RA, empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True and update conditions include RA). This tag is returned in the GPO response with correct length. Case 30: Proprietary tags with with three-byte tag field, with update condition K/RA, empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True and update conditions include RA). These tags are returned in the GPO response with correct length.
Pass Criteria	The transaction must successfully complete. Cases 1, 3, 12, 13, 28 and 29: the CDOL populated data includes data value same as returned by the card in GPO response. Others Cases: reader must ignore the values of unrecognized tags returned by the card in GPO response, these values are not stored in TLV Database. The CDOL shall be correctly populated.

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4M25-0130: GetPO - No PDOL	
Objective	To ensure that, when the PDOL is not provided by the card, the reader constructs the data field of the GetPO command message as a template introduced by the tag '83' with a length field set to zero.
References	s2.18, s2.19
Conditions	EDA not performed Card returns a fake '9F8103' tag. Records are not encrypted. IADmac has a fixed value. Case 1: PDOL is not returned in the FCI of the selected ADF Case 2: empty PDOL tag
Pass Criteria	The reader issues a GetPO command with a data field consisting of a template introduced by the tag '83' with no value field and with length field having the value zero. The transaction then terminates because the EDAmac verification fails.

4M25-0224: GetPO - filling Data To Send/DataNeeded	
Objective	To ensure the reader correctly adds known data listed in TagsToRead Yet to DataToSend. To ensure the reader does not remove the tag from TagsToReadYet when it is empty or absent. To ensure the reader correctly adds TagsToRead and DataEnvelopesToWrite to DataNeeded when applicable.
Applicability	[DE/DS]
References	s1.13,s2.22,s2.23-TRUE,s2.24,s2.25
Conditions	Case 1: Tags To Read requests at least 5 tags. Just after sending the GPO command, - 1 tag listed in Tags To Read is absent (5A) - 1 tag listed in Tags To Read is empty (50) - 1 tag listed in Tags To Read is present in ACT (9A) - 2 tags listed in Tags To Read are present in SelectADF (5F2D and 84) Tags 5A and 50 are returned in the 1st record. Case 2: Tags To Read is provided empty in ACT Case 3: Data Envelopes To Write is provided empty in ACT Case 4: Tags To Read is provided not empty in ACT such that Data To Send is sent at GPO. Data Envelopes To Write is provided empty in ACT Case 5: Tags To Read requests a tag such that just after sending the GPO command, this tag is absent

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Pass Criteria	Transaction must complete successfully. Case 1: a 1st DEK is sent with 5F2D, 84 and 9A Case 2: a DEK is sent requesting Tags To Read in DataNeeded Case 3: a DEK is sent requesting Data Envelopes To Write in DataNeeded Case 4: a DEK is sent requesting Data Envelopes To Write in DataNeeded and sending tags identified in Tags To Read Case 5: no DEK is sent

4M25-2018: GetPO - card CDOL1	
Objective	To ensure the reader correctly uses the card CDOL1 when the default CDOL1 is not available.
References	s20.18-TRUE, s20.20-TRUE
Conditions	CDOL is returned in GetPO. It is slightly different from the usual default CDOL1. Default CDOL1 is not present in TLV database
Pass Criteria	The transaction must successfully complete.

4M25-2090: GetPO - DET received	
Objective	To ensure the reader correctly stores the DET received while waiting for the GetPO response.
Applicability	[DE/DS]
References	s20.9
Conditions	DEK is sent just after the GetPO command. GetPO response is delayed a DET is sent immediately requesting an envelope to be read RRP performed
Pass Criteria	ReadData shall be sent prior to the records
Note	**Some kernels might receive the DET after the GetPO response. If so, the DET should be stored prior to sending the RRP command so the reader should be able to meet the pass criteria.**

4M25-9001: GetPO - L1RSP	
Objective	To ensure the reader processing is stopped when a time out occurs (due to a mute card).
References	s20.1, s20.2, s20.3
Conditions	

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Procedure	A first transaction is performed. Upon reception of the GetPO command, the card stops the communication with the reader (mute card, i.e.: it does not send any R-APDU or S(WTX) blocks). A second (successful) transaction is performed.
Pass Criteria	First transaction: the reader must stop the transaction upon reception of the GetPO response. The reader must build an OUT signal. - in OUT\OPS, - 'Status' must indicate "TRY AGAIN" - 'Start' must indicate "B" - In OUT\DiscretionaryData\ErrorIndication, - 'L1' must indicate "TIME OUT ERROR" - 'Msg On Error' must indicate 'N/A' Second transaction: the transaction completes successfully.

4M25-9015: GetPO - parsing error	
Objective	To ensure the reader stops the transaction when the response message of the GetPO command is not correctly formatted (i.e.: data is absent from the response, parsing is not as per the encoding rules, tag already present in the TLV database and different value, tag not present in the TLV database but update conditions do not include RA, length is not in the specified range).
References	s20.15-FALSE, s20.16
Conditions	SW12 in GetPO response is '9000'. GetPO response is in Format 2 with the following formatting errors: Case 1: the length value indicated in the template '77' is greater than the data length Case 2: the following additional data is returned: 'DF 00' (tag indicates subsequent byte and subsequent byte is not greater than zero, plus length is absent.) Case 3: the following additional data is returned: 'DF01 02 03' (length is erroneous) Case 5a: the proprietary tag 'CF' is present in the TLV database. This tag is returned in the GPO response with length lower than expected length range(with any value). Case 5b: the proprietary tag 'C8' is present in the TLV database. This tag is returned in the GPO response with length higher than expected length range(with any value). Case 6: the card returns the template '77' correctly formatted followed by another constructed tag 'E5' including a tag 'C5'. 'E5' is not included in the template '77'. Case 7: AFL is 5 bytes long Case 8: the ATC tag is present but empty ('9F36 00') Case 9a: AIP is empty Case 9b: AIP ('82') length is 3 bytes Case 10: AFL is empty Case 11: ApplicationLabel returned empty in Select ADF. It is then returned with wrong length in GPO.

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	Case 12: '9F0A' returned not empty in Select ADF. It is then returned with different value in GPO. Case 13: Tag with Tag field length greater than 3 Bytes is returned (9FFF8F01) Case 20: primitive tag is returned without length
Pass Criteria	The reader must terminate the transaction after the GetPO response because the response does not parse correctly. The reader must build an OUT signal. In OUT\OPS, - 'Status' must indicate "END APPLICATION" - 'Start' must indicate "N/A" - 'UI Request on Outcome Present' is set In OUT\DiscretionaryData\ErrorIndication: - 'Msg On Error' must indicate "ERROR - OTHER CARD" - 'L2' must indicate "Parsing error". "CARD DATA ERROR" is also acceptable for case 6. In OUT\UIRD: - 'Message Identifier' must indicate "ERROR - OTHER CARD" - 'Status' must indicate "NOT READY".

4M25-9101: GetPO - SW <> 9000 - Next AID selection	
Objective	To ensure the reader selects the next AID in the candidate list when the status in the GetPO response is different from '90 00'. Test1 AID first.
References	s20.10-FALSE, s20.11, s20.12
Conditions	PPSE such that Test1 application has 1st priority, Test2 2nd. A) A payment transaction is started. The card returns a status different from '90 00' in the GetPO response as defined below. * Case 1: '6F 00' * Case 2: '62 83' * Case 3: '67 00' * Case 4: '63 C2' B) A cash transaction is started. The card returns a status different from '90 00' in the GetPO response as defined below. * Case 5: '69 83' * Case 6: '69 85' * Case 7: '90 01' * Case 8: '6A 81' C) A cashback transaction is started. The card returns a status different from '90 00' in the GetPO response as defined below.

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	* Case 9: '6A 82' * Case 10: '6A 83' * Case 11: '6A 86' * Case 12: '65 00' D) A refund transaction is started. The card returns a status different from '90 00' in the GetPO response as defined below. * Case 13: '64 00' * Case 14: 'EF EF'
Pass Criteria	Upon reception of the bad status word in response to the GetPO command, the reader must stop the transaction. The reader must build an OUT signal. In OUT\DiscretionaryData\ErrorIndication: - 'L2' indicates "Status bytes" - 'SW12' indicates [status bytes defined in the related case] In OUT\OPS, - 'Status' must indicate "SELECT NEXT" - 'Start' must indicate "C" - 'Field off request' must indicate "N/A". Then, The reader must switch back to Application Selection and selects the next AID in the list. The transaction must complete successfully.

4M25-9102: GetPO - SW <> 9000 - Test2 AID first	
Objective	To ensure the reader selects the next AID in the candidate list when the status in the GetPO response is different from '90 00'. Test2 AID first.
References	s20.10-FALSE, s20.11, s20.12
Conditions	PPSE such that Test2 application has 1st priority, Test1 2nd. A) A payment transaction is started. The card returns a status different from '90 00' in the GetPO response as defined below. * Case 1: '6F 00' * Case 2: '62 83' * Case 3: '63 00' * Case 4: '63 C2' B) A cash transaction is started. The card returns a status different from '90 00' in the GetPO response as defined below. * Case 5: '69 83' * Case 6: '69 84' * Case 7: '90 01' * Case 8: '6A 81'

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	C) A cashback transaction is started. The card returns a status different from '90 00' in the GetPO response as defined below. * Case 9: '6A 82' * Case 10: '6A 83' * Case 11: '6A 88' * Case 12: '65 00' D) A refund transaction is started. The card returns a status different from '90 00' in the GetPO response as defined below. * Case 13: '64 00' * Case 14: 'EF EF'
Pass Criteria	Upon reception of the bad status word in response to the GetPO command, the reader must stop the transaction. The reader must build an OUT signal. In OUT\DiscretionaryData\ErrorIndication: - 'L2' indicates "Status bytes" - 'SW12' indicates [status bytes defined in the related case] In OUT\OPS, - 'Status' must indicate "SELECT NEXT" - 'Start' must indicate "C" - 'Field off request' must indicate "N/A". Then, The reader must switch back to Application Selection and selects the next AID in the list. The transaction must complete successfully.

4M25-9130: GetPO - Template issue	
Objective	To ensure the reader aborts the transaction if GetPO template is not '77' or it is empty.
References	s20.13-FALSE, s20.16
Conditions	case 1: template is 70 case 2: template is 80 case 3: template is 80, it is coded as per format 1 case 4: only SW12 case 5: only 77 is returned
Pass Criteria	The reader must stop the transaction upon reception of the erroneous GetPO response.

4M25-9150: GetPO - AIP, AFL, CardKeyData or CDOL1 missing	
Objective	To ensure the reader stops the transaction when the mandatory objects are missing from the GPO response.

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References	s20.18-FALSE, s20.19, s202122232425.24-FALSE, s202122232425.25
Conditions	For all cases, the card returns several valid combinations in the PPSE response. Case 1: AIP tag is not returned Case 2: AFL tag is not returned Case 3: CardKeyData tag is not returned Case 4: CardKeyData tag is not returned. Optional ATC is returned. Case 5: Template is empty Case 6: CDOL1 absent and defaultCDOL1 absent Case 7: CDOL1 empty and defaultCDOL1 absent
Pass Criteria	Upon reception of the GetPO response, the reader must terminate the transaction. In OUT\OPS, - 'Status' must indicate "END APPLICATION" - 'Start' must indicate "N/A" - 'UI Request on Outcome Present' is set In OUT\DiscretionaryData\ErrorIndication: - 'Msg On Error' must indicate "ERROR - OTHER CARD" - 'L2' indicates "CARD DATA MISSING" In OUT\UIRD: - 'Message Identifier' must indicate "ERROR - OTHER CARD" - 'Status' must indicate "NOT READY"

4M25-9226: GPO - decryption failed	
Objective	To ensure that the reader stops the transaction when the Card Key Data length is different from 2 * NFIELD.
Applicability	Cases 02, 03: [DE/DS]
References	C.11-FALSE,C.21,s22.6, s22.51, s23.6, s23.34, s24.5, s24.28, s26.4, s26.28, s27.2, s27.10
Conditions	Card Key Data length != 2 * NFIELD Case 1: decryption failure is found in s22(or s27 if GPO privacy postponed after GenAC) first record encrypted Case 2: decryption failure is found in s23 Case 3: decryption failure is found in s24 Case 4: decryption failure is found in s22 first record not encrypted Case 5: decryption failure is found in s26
Pass Criteria	Reader must stop the transaction .

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4M25-9227: GPO - Blinded Public Key Recovered failed	
Objective	To ensure that the reader stops the transaction when blinded Public Key is not recovered.
Applicability	Cases 02, 03: [DE/DS]
References	C.11-TRUE, C.12a-FALSE, C12, C.14, C.21, s22.6, s22.51, s23.6, s23.34, s24.5, s24.28, s26.4, s26.28, s27.2, s27.10
Conditions	In all cases erroneous blinded Public Key is returned in in Card Key Data Case 1: decryption failure is found in s22 (or s27 if GPO privacy postponed after GenAC) first record encrypted Case 2: decryption failure is found in s23 Case 3: decryption failure is found in s24 Case 4: decryption failure is found in s22 first record not encrypted Case 5: decryption failure is found in s26
Pass Criteria	Reader must stop the transaction .

4M30-0022: RRP - not supported	
Objective	To ensure the reader correctly continues the transaction when RRP is not supported.
References	s20.24-FALSE, s20.25
Conditions	Case 1: AIP indicates 'Relay resistance protocol NOT supported'. KernelConfiguration indicates 'Relay resistance protocol supported' Case 2: AIP indicates 'Relay resistance protocol supported'. KernelConfiguration indicates 'Relay resistance protocol NOT supported'
Pass Criteria	* The reader must successfully process the transaction until completion. * TVR byte 5, b2-1 = '01' ("RRP not performed")

4M30-0180: RRP - RRPrMeasuredTime > Minimum Time	
Objective	To ensure the reader correctly calculates RRPrMeasuredTime. To ensure the reader correctly sets the RRPrTimeExcess. To ensure the reader continues the transaction when the RRPrMeasuredTime is greater than the expected minimum time.
References	s21.17,s21.18-FALSE
Conditions	AIP indicates 'Relay resistance protocol supported' KernelConfiguration indicates 'Relay resistance protocol supported' Case 1: RRPrMinGrace='0FA0' (400ms). RRPcMinTime= RRPrMinGrace+5. RRPcEstimated='0FA0'. RRPrExpectedRAPDU has default value

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	Case 2: RRPrMinGrace, RRPcMinTime and RRPcEstimated have default values. RRPrExpectedRAPDU='0FA0'. Case 3: RRPcMinTime=0. All other values are default. Case 4: RRPcMinTime=0. RRPrExpectedCAPDU='0FA0'. Case 5: RRPrExpectedCAPDU='0FA0', RRPcMinTime=RRPrMinGrace * Case 6: (RRPrTimeExcess>0) RRPcMaxTime=0, RRPrMaxGrace=400
Pass Criteria	The transaction completes successfully. Especially, the RRP command must be sent only once. RRPrTimeExcess provided in GenAC is zero, except in case 6.
Note	***: this case leads to RRPrMeasuredTime=0.**

4M30-2000: RRP - Thresholds	
Objective	To ensure the reader correctly uses the default values when the RRP data items are not present in the database.* To ensure the reader correctly checks the mismatch and accuracy thresholds.
References	KS.1, s21.24, s21.25
Conditions	AIP indicates 'Relay resistance protocol supported' KernelConfiguration indicates 'Relay resistance protocol supported' Case 1: RRPrAccuracyThreshold='0FA0' (400ms) Case 2: RRPrAccuracyThreshold='0FA0' (400ms). RRPcEstimated='000A' (1ms). Case 3: RRPrAccuracyThreshold='0FA0' (400ms). RRPcEstimated='0032' (5ms). Case 4: Default kernel values are used (so RRPrAccuracyThreshold='012C' (30ms)). TimeTaken must be 55ms minimum. Case 5: RRPcEstimated='0000' Case 6: RRPrExpectedRAPDU='0000'
Pass Criteria	The transaction completes successfully. Cases 1, 5, 6: In GenAC, TVR B5b4-1= 0010b ('RRP threshold NOT exceeded, RRP limits NOT exceeded, RRP performed') Cases 2 to 4: In GenAC, TVR B5b4-1= 1010b ('RRP threshold exceeded, RRP limits NOT exceeded, RRP performed')
Note	***: the values used by the reader cannot be verified accurately, however when the expected default values are used then the TVR should be set as per the above pass criteria.**

4M30-2110: RRP - Repeat RRP	
Objective	To ensure the reader repeats the RRP command when applicable. To ensure the reader does not repeat the RRP command when the Measured Time is smaller than the Maximum Time or the counter is more than 2.

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	To ensure the reader correctly indicates in the TVR if the RRP limits are exceeded or not.
References	s21.20-TRUE, s21.22, s21.23
Conditions	AIP indicates 'Relay resistance protocol supported' KernelConfiguration indicates 'Relay resistance protocol supported' RRPcEntropy is different in each RRP RAPDU. Case 1: RRPcMaxTime=RRPrMaxGrace=240ms in each RRP APDU. The card waits for at least (RRPcMaxTime + RRPrMaxGrace + 20ms) before returning the 1st RRP RAPDU (so a 2nd RRP CAPDU is sent). Case 2: RRPcMaxTime and RRPrMaxGrace have default values in each RRP APDU. The card waits for at least (RRPcMaxTime + RRPrMaxGrace + 20ms) before returning the RRP RAPDU to all RRP commands. Case 3: RRPcMaxTime=RRPrMaxGrace=240ms in each RRP APDU. The card waits for at least (RRPcMaxTime + RRPrMaxGrace + 20ms) before returning the 1st and 2nd RRP RAPDUs.
Pass Criteria	The reader sends 2 (case 1) or 3 (cases 2 and 3) RRP commands. In each RRP command the entropy has a different value. In GenAC, TVR B5b3 indicates: - 'RRP time limits NOT exceeded' (cases 1 and 2) - 'RRP time limits exceeded' (case 3) In GenAC, TVR B5b2-1='10' ('RRP performed'). The transaction completes successfully.

4M30-9010: RRP - L1RSP	
Objective	To ensure the reader processing is stopped when a time out occurs (due to a mute card).
References	s21.1, s21.2, s21.3, s21.4, s21.5
Conditions	AIP indicates 'Relay resistance protocol supported' KernelConfiguration indicates 'Relay resistance protocol supported'
Procedure	A) A first transaction is performed. Upon reception of the RRP command, the card stops the communication with the reader (mute card, i.e.: it does not send any R-APDU or S(WTX) blocks). A second (successful) transaction is performed to ensure the timer was stopped in the first transaction. B) A first transaction is performed. RRPcMaxTime is zero so the RRP command is sent again. Upon reception of the second RRP command, the card stops the communication with the reader (mute card, i.e.: it does not send any R-APDU or S(WTX) blocks). A second (successful) transaction is performed to ensure the timer was stopped in the first transaction.

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Pass Criteria	First transaction: the reader must stop the transaction upon reception of the first (A) or second (B) RRP response. The reader must build an OUT signal. - in OUT\OPS, - 'Status' must indicate "END APPLICATION" - 'Start' must indicate "B" - 'UI Request on Restart Present' is set - In OUT\DiscretionaryData\ErrorIndication, - 'L1' must indicate "TIME OUT ERROR" - 'Msg On Error' must indicate 'TRY AGAIN' - In OUT\UIRD, - 'MessageIdentifier' must indicate "TRY AGAIN" - 'Status' must indicate "Ready To Read" - 'Hold Time' must indicate '000000' Second transaction: the transaction completes successfully. Especially, the RRP command must be sent only once since the timer was reinitialized.

4M30-9120: RRP - SW <> 9000	
Objective	To ensure the reader terminates the transaction when the status in the RRP response is different from '9000'.
References	s21.12-FALSE, s21.13
Conditions	AIP indicates 'Relay resistance protocol supported' KernelConfiguration indicates 'Relay resistance protocol supported' * Case 1: the card returns status value '6300' in response to RRP * Case 2: the card returns status value '63C2' in response to RRP * Case 3: the card returns status value '6700' in response to RRP * Case 4: the card returns status value '6983' in response to RRP * Case 5: the card returns status value '6984' in response to RRP * Case 6: the card returns status value '6985' in response to RRP * Case 7: the card returns status value '6A81' in response to RRP * Case 8: the card returns status value '6A86' in response to RRP * Case 9: the card returns status value '6D00' in response to RRP
Procedure	A) A first transaction is performed. Upon reception of the RRP command, the card returns the status words defined in cases 1, 2, 3. A second (successful) transaction is performed to ensure the timer was stopped in the first transaction. B) A first transaction is performed. RRPcMaxTime is zero so the RRP command is sent again. Upon reception of the second RRP command, the card returns the status words defined in cases 4, 5, 6.

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	A second (successful) transaction is performed to ensure the timer was stopped in the first transaction. C) A first transaction is performed. RRPcMaxTime is zero so the RRP command is sent again twice.Upon reception of the third RRP command, the card returns the status words defined in cases 7, 8, 9. A second (successful) transaction is performed to ensure the timer was stopped in the first transaction.
Pass Criteria	First transaction: the reader must stop the transaction upon reception of the first (A) or second (B) or third (C) RRP response. The reader must build an OUT signal. In OUT\OPS, - 'Status' must indicate "END APPLICATION" - 'Start' must indicate "N/A" - 'UI Request on Outcome Present' is set In OUT\DiscretionaryData\ErrorIndication: - 'Msg On Error' must indicate 'Error other card' - 'L2' indicates "Status bytes" - 'SW12' indicates [status bytes defined in the related case] In OUT\UIRD, - 'MessageIdentifier' must indicate "ERROR OTHER CARD" - 'Status' must indicate "Not Ready" Second transaction: the transaction completes successfully. Especially, the RRP command must be sent only once since the timer was reinitialized.

4M30-9170: RRP - parsing error	
Objective	To ensure the reader terminates the transaction when the RRP command response is not correctly formatted.
References	s21.14-FALSE, s21.16, s202122232425.24, s202122232425.25
Conditions	AIP indicates 'Relay resistance protocol supported' KernelConfiguration indicates 'Relay resistance protocol supported' Case 1a: The length of response message data field is 11. Case 1b: The length of response message data field is 10. Case 2a: The response message data template is '77' Case 2b: The response message data template is '88' Case 3a: The length of value field of response message data field is 11. Case 3b: The length of value field of response message data field is '09'. Case 3c: The length of value field of response message data field is '810A'.
Pass Criteria	The reader must stop the transaction upon reception of the erroneous RRP response.

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4M30-9180: RRP - RRPrMeasuredTime < Minimum Time?	
Objective	To ensure the reader stops the transaction when the RRPrMeasuredTime is smaller than the expected minimum time.
References	s21.18-TRUE,s21.19
Conditions	AIP indicates 'Relay resistance protocol supported' KernelConfiguration indicates 'Relay resistance protocol supported' Case 1: RRPrExpectedCAPDU is 400ms ('0FA0'). RRPcMinTime = (RRPrMinGrace +1)* Case 2: RRPrExpectedRAPDU=RRPcEstimated='0FA0'. RRPcMinTime=(RRPrMinGrace +1)* Case 3: RRPcMinTime='0FA0'
Procedure	A) Upon reception of the RRP command, the card returns the response with the RRPcMinTime value defined in case 1. B) RRPcMaxTime is zero so the RRP command is sent again twice. Upon reception of the third RRP command, the card returns the values defined in cases 2 and 3.
Pass Criteria	The reader must terminate the transaction after the RRP response. The reader must build an OUT signal. In OUT\OPS, - 'Status' must indicate "END APPLICATION" - 'Start' must indicate "N/A" - 'UI Request on Outcome Present' is set In OUT\DiscretionaryData\ErrorIndication: - 'Msg On Error' must indicate "ERROR - OTHER CARD" - 'L2' must indicate "Card data error" In OUT\UIRD: - 'Message Identifier' must indicate "ERROR - OTHER CARD" - 'Status' must indicate "NOT READY"
Note	***: The result of (RRPcMinTime – RRPrMinGrace) will finally give 0.1 ms**

4M35-0241: ReadRecord - Response parsing - TLV length	
Objective	To ensure the reader correctly manages the TLV length when parsing the ReadRecord response message.
Applicability	Cases E|04: [DE/DS]
References	s22.29, s22.30-TRUE, s22.47, s22.48-TRUE, s26.22, s26.23-TRUE, s24.20, s24.21-FALSE
Conditions	The AFL identifies more than one record to be read. Case 00: record template length is coded on 1 byte. Data object lengths are coded on 1 byte. (Already done)

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	Case 02: record template length is coded on 1 byte. Data object lengths are coded on 2 bytes (lengths are less than 128 bytes). Case 03: record template length is coded on 2 bytes while the data length is less than 128. Data object lengths are coded on 1 byte. Case 04: record template length is coded on 2 bytes while the data length is more than 128. Data object lengths are coded on 2 bytes. Case 05: record template length is coded on 2 bytes while the data length is more than 128. All data object lengths are coded on 3 bytes. Case 06: a record is empty: 7000
Procedure	A) cases 2 and 3 are implemented in 2nd record, not encrypted B) cases 4 and 5 and 6 are implemented in last record, not encrypted C) cases 2 and 3 is implemented in first record, encrypted and signed D) cases 4 and 6 are implemented in the last record, encrypted. PTFWF is not present. No ReadData (s26.22). E) (DE/DS supported) case 4 is implemented in the last record, encrypted. PTFWF is empty (s24.20).
Pass Criteria	The transaction must successfully complete.

4M35-0242: ReadRecord - Response parsing - padding	
Objective	To ensure the reader ignores any '00' padding before, between or after TLV-coded data objects.
References	s22.29, s22.30-TRUE, [EMV Book 3]:Annex B
Conditions	Case 01: the 4th record (non encrypted) template contains '000000' padding between data objects Case 02: the 5th record template is right padded with 50 '00' bytes. ICC Certificate Exponent (9F47) and ICC Certificate Remainder (9F48) are included in this record Case 03: the 4th record template is left padded with '0000' Case 04: the 1st record template contains '0000' padding between data objects. It is encrypted and signed.
Pass Criteria	The transaction must successfully complete.

4M35-0243: ReadRecord - Response parsing - storing data	
Objective	To ensure the reader stores the recognized data objects (Known and update conditions include RA Signal) and ignores the non-recognized ones. To ensure the reader is able to manage non-recognized data objects in signed records. To ensure the reader correctly sends a DEK after the GenAC when data item is available in the last record.
References	s22.29, s22.30-TRUE, s22.47, s22.48-TRUE, s26.22, s26.23-TRUE, s24.20, s24.21-FALSE,s2627.3

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Conditions	All tags listed in the different cases are present in Discretionary Data Tag List ('9F821F') Case 1: Tag '9F8221' (i.e.: IsKnown=True, IsPresent=False, IsEmpty=False ie: upd=K (not RA)). This Tag is returned in ReadRecord response. Case 2: '9F12' is returned empty in the first Record response. It is then returned with any value in another Record response. Case 3: '5A' is returned with length on 1 Byte in the first Record response. It is then returned with same value in another Record response with length on 2 Bytes. Case 4: tag '9F40', upd=K (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned with different value. Case 5: Tag '9F4E', upd=K (not RA), empty in TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True upd=K (not RA)). This tag is returned in the ReadRecord response.
Procedure	A) cases 4 are implemented in 1st record, signed and encrypted. ECC. B) cases 1, 2, 3, 4, 5 are implemented in last record, not encrypted. C) cases 3, 5 are implemented in last record, signed and encrypted.
Pass Criteria	The transaction must successfully complete. Case 1: the DiscretionaryData includes '9F822100' Case 2: the DiscretionaryData is populated with the value of '9F12' returned in the response to ReadRecord command. Case 3: the DiscretionaryData is populated with the value of '5A' returned in the response to ReadRecord command. Case 4: the DiscretionaryData is populated with the value of '9F40' present in the configuration before Read Record Response. the Value returnes in Response to ReadRecord is ignored. Case 5: the DiscretionaryData includes '9F4E00'

4M35-0244: ReadRecord - Response parsing - storing data	
Objective	To ensure the reader stores the recognized data objects (Known and update conditions include RA Signal) and ignores the non-recognized ones. To ensure the reader is able to manage non-recognized data objects in signed records. To ensure the reader correctly sends a DEK after the GenAC when data item is available in the last record.
Applicability	Cases B|04b, B|19, C|04b: [DE/DS]
References	s22.29, s22.30-TRUE, s22.47, s22.48-TRUE, s26.22, s26.23-TRUE, s24.20, s24.21-FALSE,s2627.3
Conditions	All proprietary tags listed in the different cases are present in Discretionary Data Tag List ('9F821F') Case 1: the proprietary private class tag 'DFEE03' is returned with 3 bytes length '82 00 0x' and any value in a signed record. This tag is not present in the TLV database. Case 2: the proprietary tag '9F4A' not present in TLV database (i.e.: IsKnown=False, IsPresent=False)is returned . Case 3: the proprietary private class tag 'D1' (unknown) .

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	Case 4: the proprietary tags 'C8' and 9F05 (empty in TLV database) are returned with any value. These tags are present in the TLV database with RA update condition. Case 4b (DE/DS): same as case 4 plus: 'C8' is requested in Tags To Read (together with the PAN). Case 5: Proprietary tag '85' is not present in the TLV database (i.e.: IsKnown=False, IsPresent=False) . This tag is returned in the ReadRecord response Case 6: Proprietary tag '9F08', not privateClass, RA, not empty. Case 7: Proprietary tag 'C3', privateClass, empty not RA(i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K (not RA)). This tag is returned in the ReadRecord response. Case 8: Proprietary tag '81' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in the ReadRecord response with different value and incorrect Length. Case 9: Proprietary tag 'CF', privateClass, RA, not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/RA . This tag is returned in the ReadRecord response. Case 10: Known tag, proprietary tag 'CE' ,privateClass, not RA (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in the ReadRecord response with different value and correct Length. Case 11: Proprietary tag '9F04' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the ReadRecord response with same value. Case 12: Proprietary tag '9F04' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the ReadRecord response with different value and correct Length. Case 13: Proprietary tag 'C4', Private Class, upd=K/ACT/DET, Not empty (value updated in ACT) (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the ReadRecord response with maximum correct length. Case 14: Proprietary tag 'C4', Private Class, upd=K/ACT/DET, empty in TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K/ACT/DET (not RA)). This tag is returned in the ReadRecord response with maximum correct length. Case 15: Proprietary tag 'C5', Private Class, upd=K/ACT, empty in TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K/ACT (not RA)). This tag is returned in the ReadRecord response . Case 16: Proprietary tag 'C6', Private Class, upd=K/DET, empty in TLV database (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K/DET (not RA)). This tag is returned in the ReadRecord response . Case 17: Proprietary tag '5F19', Private Class, upd=K/ACT/RA, not empty in TLV database (updated in SELECT) (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/RA). This tag is returned in the ReadRecord response with same V . Case 18: Proprietary tag '9F04' is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the ReadRecord response with different value and incorrect Length. Case 19: Proprietary tag 'C6',private class upd=K/DET (not RA) is present in the TLV database not empty, value updated in DET signal (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/DET (not RA)). This tag is returned in the ReadRecord response with different value .

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Procedure	A) cases 1, 2, 4, 5, 10, 13, 16, 17, 18 are implemented in 1st record, signed and encrypted. ECC. B) cases 1, 3, 4, 4b, 5, 6, 7, 8, 9, 11, 12, 14, 15, 19 are implemented in last record, not encrypted. C) cases 4, 4b are implemented in last record, signed and encrypted. RSA.
Pass Criteria	The transaction must successfully complete. Cases 4, 4b: the DiscretionaryData includes the correct values for the tags '9F05' and 'C8'. Case 4, 4b, 9, 17: the DiscretionaryData populated data include the value of the tags returned in ReadRecord reponse. Case 1, 2, 3, 5:the DiscretionaryData populated data does not include the tags. Cases 6, 7, 8, 10, 11, 12,13, 14, 15, 16, 17, 18, 19 : the DiscretionaryData populated data include the value of the tags in TLV Database before the ReadRecord. values returned in ReadRecord response are ignored.. Case 4b: DEK must be sent after the GenAC with PAN and CF correct values. C): DiscretionaryData includes the correct values for the tags '9F05' and 'CF'; DataRecord includes the correct value for the tag '5A'.

4M35-1421: ReadRecord - Each entry in the AFL	
Objective	To ensure the reader is able to interpret the AFL and send ReadRecord commands for each record between the starting record number and the ending record number, inclusively.
References	s20.23
Conditions	AFL with a length of 128 bytes
Pass Criteria	The reader shall send a sequence of ReadRecord commands according to the AFL.

4M35-1422: ReadRecord - SFI range 1-10	
Objective	To ensure the reader is able to read data in file with SFI in range 1 to 10 (0x01 to 0x0A) using the ReadRecord command.
References	s20.23
Conditions	- PAN is located in file with SFI 01. - CA key is located in file with SFI 02. - ICC Certificate is located in file with SFI 03. - Issuer Country Code is located in file with SFI 04. - Application Currency Code is located in file with SFI 05. - Track2EquivalentData is located in file with SFI 06. - AVN Card is located in file with SFI 07.

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	- Application Expiry Date is located in file with SFI 08. - Pan Sequence Number is located in file with SFI 09. - IAC default is located in file with SFI 10.
Pass Criteria	The reader must successfully complete the transaction.

4M35-1423: ReadRecord - Card Message Counter	
Objective	To ensure the reader accepts Card Message Counter with high values.
References	s20.23,C.18
Conditions	40 Encrypted Records
Pass Criteria	The reader shall send a sequence of ReadRecord commands according to the AFL.

4M35-2021: ReadRecord - setting Active AFL	
Objective	To ensure the reader is able to interpret the AFL and send ReadRecord commands accordingly.
References	s20.23
Conditions	Case 1: AFL is '08010101 10010100 18010200 50010100 58010100' Case 2: AFL is '08010101 10010100 18010200 50010101 60010201' Case 3: AFL is '08010100 10010100 18010200 50010100 60010201'
Pass Criteria	The reader shall send a sequence of ReadRecord commands according to the AFL, excluding SFI>10. The DataRecord must correctly include the value of the tag (Card Capabilities Information) returned in the last record.

4M35-2218: ReadRecord - signed record	
Objective	To ensure the reader correctly manages the 'signed flag'.
References	s20.23
Conditions	Case 1: AFL indicates record to be signed, EDA NOT enabled in KernelQualifier, Local authentication supported in AIP Case 2: AFL indicates record to be signed, EDA enabled in KernelQualifier, Local authentication not supported in AIP Case 3: AFL indicates record NOT to be signed, EDA enabled in KernelQualifier, Local authentication supported in AIP Case 4: AFL indicates one record with SFI>10 to be signed, EDA enabled in KernelQualifier, Local authentication supported in AIP

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	Case 5: AFL indicates several records (including SFI>10) to be signed, EDA enabled in KernelQualifier, Local authentication supported in AIP Case 6: Last record is encrypted and signed. Case 7:AFL indicates SFI lower than 10 with multiple records to be signed (not all), EDA enabled in KernelQualifier, Local authentication not supported in AIP. Case 8: one record with SFI>10 is positionned between two records with SFI lower than 10
Procedure	A: ECC all cases B: RSA cases: 1,2,5,6,7 and 8
Pass Criteria	The transaction must complete successfully. Especially the SDAhash is correctly calculated.
Note	**Cases 5 and 6: CAUTION: the test tool shall not sign the records with SFI > 10 (the kernel will not read them).** **Case 6: tricky test: the CRYPTO process is supposed to manage a queue, so it should not recover the ICC certificate before decrypting the last record (that is signed!)**

4M35-9100: ReadRecord - SW <> 9000	
Objective	To ensure the reader ends the transaction when the status in the ReadRecord response is different from '90 00'.
References	s22.16-FALSE, s22.17, s202122232425.24, s202122232425.25
Conditions	A) A payment transaction is started. The AFL is '08010101 1001010018010200'. The card returns a SW<> '90 00' to the first ReadRecord command. Cases 1 to 4 are made with a non-encrypted record while cases 5 to 7 are made with an encrypted record, case 8: empty template: Case 1:the card returns status value '6D00' in response to ReadRecord Case 2:the card returns status value '6E00' in response to ReadRecord Case 3:the card returns status value '6F00' in response to ReadRecord Case 4:the card returns status value '6895' in response to ReadRecord Case 5:the card returns status value '6A82' in response to ReadRecord Case 6:the card returns status value '6A83' in response to ReadRecord Case 7:the card returns status value '6A86' in response to ReadRecord Case 8:the card returns status value '6985' in response to ReadRecord B) A payment transaction is started.The AFL is '08010100 1001010018010200'. The card returns SW<>'90 00' to the second ReadRecord command. Cases 9 to 11 are made with a non-encrypted record while cases 12 to 13 are made with an encrypted record: Case 9:the card returns status value '6283' in response to ReadRecord Case 10:the card returns status value '6300' in response to ReadRecord Case 11:the card returns status value '63C0' in response to ReadRecord Case 12:the card returns status value '6983' in response to ReadRecord

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	Case 13:the card returns status value '6984' in response to ReadRecord C) A payment transaction is started.The AFL is '08010100 1001010018010200'. The card returns SW<>'90 00' to the last ReadRecord command. Cases 14 to 15 are made with a non-encrypted record while cases 16 to 17 are made with an encrypted record: Case 14:the card returns status value '6A81' in response to ReadRecord Case 15:the card returns status value '6A82' in response to ReadRecord Case 16:the card returns status value '6A83' in response to ReadRecord Case 17:the card returns status value '6A88' in response to ReadRecord
Pass Criteria	Upon reception of the bad status word in response to the ReadRecord command, the reader must terminate the transaction after the first (case A), second (case B) and the last (case C) readrecord APDU. The reader must build an OUT signal. In OUT\OPS, - 'Status' must indicate "END APPLICATION" - 'Start' must indicate "N/A" - 'UI Request on Outcome Present' is set In OUT\DiscretionaryData\ErrorIndication: - 'Msg On Error' must indicate "ERROR - OTHER CARD" - 'L2' must indicate " Status bytes" - 'SW12' indicates [status bytes defined in the related case] In OUT\UIRD: - 'Message Identifier' must indicate "ERROR - OTHER CARD" - 'Status' must indicate "NOT READY".

4M35-9221: ReadRecord - L1RSP	
Objective	To ensure the reader processing is stopped when a time out occurs (due to a mute card).
References	s22.1, s22.10, s22.11, s22.12
Conditions	For 1st transaction: The card stops the communication with the reader upon reception of the first read record command (mute card, i.e. it does not send neither any R-APDU nor S(WTX) blocks.) For 2nd transaction: Amount,Authorized is greater than Reader CVM Required Limit 'EDA enabled' in Kernel Qualifier is set Reference Control Parameter in Generate AC command set to TC No Read Data/Write Data Card TVR not present

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Procedure	The TEI does not send an ACT prior to the 2nd transaction.
Pass Criteria	Since the card is mute, the reader must stop the 1st transaction. The reader must build an OUT signal. In UIRD, - 'Message Identifier' must indicate "Try again" - 'Status' must indicate "READY TO READ" - 'MessageHold Time' must indicate "000000" In OUT\OPS, - 'Status' must indicate "END APPLICATION" - 'Start' must indicate "B" In OUT\DiscretionaryData\ErrorIndication, - 'L1' must indicate "RETURN CODE" - 'Msg On Error' must indicate "Try again" The reader must then process the second transaction successfully

4M35-9228: ReadRecord - template check	
Objective	To ensure that the reader stops the transaction when the response message of the ReadRecord command is not correctly formatted.
References	s22.28-FALSE
Conditions	Amount,Authorized is greater than Reader CVM Required Limit Local authentication supported in AIP 'EDA enabled' in Kernel Qualifier is set Case 1: ReadRecord response does not include any data, only SW12='9000' Case 2: ReadRecord response is '00' Case 3: ReadRecord response includes only template tag 'DA' Case 4: ReadRecord response record includes only template tag '70' Case 5: ReadRecord response includes only template tag '77' Case 6: ReadRecord response is such that the record length >1 but first byte is '74'
Pass Criteria	Reader must stop the transaction upon reception of the erroneous record.

4M35-9260: ReadRecord - parsing error	
Objective	To ensure the reader stops the transaction when the ReadRecord response is not correctly formatted (i.e.: data is absent from the response, template tag is not 70, parsing is not as per the encoding rules, tag already present in the TLV database with

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	different value, tag not present in the TLV database but update conditions do not include RA, length is not in the specified range). Error is detected in s22.
Applicability	Cases E|04, E|05, E|12: [DE/DS]
References	s22.29, s22.30-FALSE, s22.47, s22.48-FALSE, s24.20, s24.21-FALSE, s26.22, s26.23-FALSE
Conditions	The card returns several valid combinations in the PPSE response. In all following cases, SW12 in ReadRecord response is '9000'. All SFIs referenced in the AFL are in the range 1 to 10. DataEnvelopesToWrite is provided in ACT. Amount is below the limit so OPS bit 'Receipt' is never set whenever the record parsing occurs after the GenAC. Case 2: the length value indicated in the template '70' is greater than the data length Case 3: the following data object is returned: 'DF 00' (tag indicates subsequent byte and subsequent byte is not greater than zero, plus length is absent) Case 4: the following additional data is returned: 'DF01 02 03' (length is erroneous) Case 5: PAN is returned twice in two different ReadRecord responses, with different values Case 6: the proprietary tag 'CF' is present in the TLV database. This tag is returned with length lower than expected length range(with any value). Case 8: the proprietary tag 'C8' is present in the TLV database. This tag is returned with length higher than expected length range(with any value). Case 9: the card returns the template '70' correctly formatted followed by another tag 'C5'. 'C5' is not included in the template '77' Case 10: tag '57' is returned on 20 bytes. The last byte is '0F'. (There must not be any other 'F' in the value). Case 11: '9F07' is returned with zero length Case 12: ApplicationLabel returned empty in Select ADF. It is then returned with wrong length in RR Case 14: Tag with Tag field length greater than 3 Bytes is returned (9FFF8F01) Case 16: PAN ('5A') length is 11 bytes Case 17: Proprietary tag 'C7', Private Class, upd=K/DET/RA, not empty in TLV database (updated in SELECT) (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/DET/RA (not RA)). This tag is returned in the ReadRecord response with different value . Case 19: tag is returned without length Case 21: the OPS '9F8210' is returned. Case 22: Proprietary tag 'C9', Private Class with update condition K/ACT/DET/RA, not empty in TLV database (updated in SELECT) (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False update conditions include RA). This tag is returned with different value.
Procedure	A) All odd cases above must be tested with the following condition: the parsing error occurs in the last record identified in the AFL. Record is not encrypted. The other records parse correctly. (s22.29) B) The even cases above must be tested with the following condition: the parsing error occurs in the second record identified in the AFL. (s22.29)

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	C) The even cases above must be tested with the following condition: the parsing error occurs in the first record, encrypted (s22.47) D) Cases 5, 6, 11 must be tested with the following condition: the parsing error occurs in the last record, encrypted. PTFWF is not present. No ReadData (s26.22). E) (DE/DS supported) Cases 4 and 5 and 12 above must be tested with the following condition: the parsing error occurs in the last record, encrypted. PTFWF is empty (s24.20).
Pass Criteria	The reader must terminate the transaction because the response does not parse correctly. The reader must not send the WriteData command. The reader must build an OUT signal. In OUT\OPS, - 'Status' must indicate "END APPLICATION" - 'Start' must indicate "N/A" - 'UI Request on Outcome Present' is set In OUT\DiscretionaryData\ErrorIndication: - 'Msg On Error' must indicate "ERROR - OTHER CARD" - 'L2' must indicate "PARSING ERROR" In OUT\UIRD: - 'Message Identifier' must indicate "ERROR - OTHER CARD" - 'Status' must indicate "NOT READY".
Note	**When the kernel parses the records after the GenAC, this test expects the kernel to go through s26.13. If such kernel would go through s22.32 instead, the test would raise an error with UIRD1.holdTime that is set to '000000' in s26.18.**

4M35-9329: ReadRecord - parsing error (detected in ReadData state)	
Objective	To ensure that the reader stops the transaction when the decrypted ReadRecord response does not parse correctly while in the state 'Waiting for Read Data Response'.
Applicability	[DE/DS]
References	s23.5,s23.30,s23.31-FALSE,s23.33
Conditions	Parsing error occurs in an encrypted record prior to the ReadData. So the parsing error is detected in s23 (not in s25). AFL identifies other records after the ReadData (the reader may send those records if it parses the records after the GenAC). Case 1: tag is returned with incorrect length
Pass Criteria	Reader must stop the transaction because of the erroneous record.

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4M36-2021: ReadData - ReadData before Records	
Objective	Ensure the reader correctly sends ReadData before the records, when applicable.
Applicability	[DE/DS]
References	s2021.1, s2021.2-FALSE, s2021.3
Conditions	Case 01: no RRP Case 02: RRP Case 03: RRP, returned envelope is empty Case 04: RRP, returned envelope is max length Case 05: RRP, returned envelope length is less than 128, but envelope length is on 2 bytes
Pass Criteria	Reader must send a ReadData prior to any Record.

4M36-2025: ReadData - ReadData without records	
Objective	Ensure the reader correctly processes the transaction with ReadData when no record is done.
Applicability	[DE/DS]
References	s2021.5
Conditions	- PAN is returned in GPO - AFL is such that no ReadRecord command is sent
Pass Criteria	Reader must send a ReadData and no record.

4M36-2238: ReadData - ReadData after a record	
Objective	Ensure the reader correctly processes the transaction with ReadData after reading a record. Ensure the reader correctly stores the DET received while waiting for the record response.
Applicability	[DE/DS]
References	s22.15, s22.36, s22.37-FALSE, s22.38
Conditions	Case 1: first record is encrypted. All envelopes requested. Case 2: first record is not encrypted
Pass Criteria	Reader must send a ReadData after record(s).

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4M36-2380: ReadData - DET	
Objective	Ensure the reader correctly stores a DET received while waiting for ReadData response. To ensure the reader correctly gets the next ReadData tag from the TagsToReadYet list. To ensure the reader does not add to TagsToReadYet a tag that is already present in TagsToReadYet. To ensure the reader correctly adds the retrieved envelope to DataToSend while waiting for the next ReadData response. To ensure the reader correctly updates a tag in a list when it is already present in that list. To ensure the reader does not send a DEK in s2021222324.4 when DataToSend is not empty but TagsToReadYet is not empty. To ensure the reader correctly sends envelope in a DEK after the GenAC when applicable.
Applicability	[DE/DS]
References	s23.14,s23.38,s2021222324.3-FALSE,s2627.3
Conditions	A dek is sent at GPO. Then several (6) envelopes to be read. A det is sent during s23, requesting another envelope. In TagsToRead, the envelopes tags are not consecutive. Envelope 3 is present twice.
Pass Criteria	Reader shall read 6 envelopes prior to any Record. Envelope3 is read once.

4M36-7080: ReadData - Message fails	
Objective	Ensure the reader correctly behaves when 'Message fails' upon reception of ReadData response.
Applicability	[DE/DS]
References	s22.8, s22.55, s23.8, s23.37
Conditions	Case 01 (s22.8, s22.55): error detected during ReadRecord Case 02 (s23.8, s23.37): Tags2Read includes at least 3 envelopes. Case 03 (s23.8, s23.37): MAC length is less than 8 bytes
Pass Criteria	Cases 1, 2, 3: Reader must clear the 'Completed' bit in ReadDataStatus.

4M36-7250: ReadData - SW!=9000	
Objective	Ensure the reader correctly behaves when SW!=9000 in ReadData response.
Applicability	[DE/DS]
References	s23.25-FALSE, s23.28

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Conditions	Tags2Read includes at least 3 envelopes.
Pass Criteria	Reader must clear the 'Completed' bit in ReadDataStatus.

4M36-7251: ReadData - incorrect tag	
Objective	Ensure the reader detects an error when the tag returned in the ReadData response is not correct. Ensure the reader removes the ReadData tag T from the list 'ReadDataTagsToValidateYet' upon reception of the response
Applicability	[DE/DS]
References	s22.51b-FALSE, s22.51c
Conditions	Case 01 (except B): Envelope tag does not correspond to the requested one - one envelope Case 02: Envelope tag does not correspond to the requested one - several envelopes A |02: last ReadData response returns the first requested envelope B |02: last but one ReadData response returns the first requested envelope Case 03: ApplicationLabel is returned with the same TLV as in the ADF (no envelope is returned)
Procedure	A) Error is detected in s22 (record state) B) Error is detected in s23 (readdata state) (except case 1)
Pass Criteria	ReadDataStatus must be '00'.

4M36-7253: ReadData - parsing error	
Objective	To ensure the reader stops the transaction when the ReadData response is not correctly formatted.
Applicability	[DE/DS]
References	s22.53-FALSE, s22.55
Conditions	The following errors are returned in the 1st ReadData. Case 1: the envelope length value is greater than the data length*. Case 2: the envelope does not have length*. Case 3: 2 envelopes are returned in the same response. Case 4: no envelope returned, only MAC. The following errors are returned in the 2nd ReadData. Case 5: Tag of data envelope is not returned in the answer (only data and MAC)

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	The following errors are returned in the 3rd ReadData. Case 6: same envelope returned twice (in 2 responses) with different TLV.
Pass Criteria	Reader must clear the 'Completed' bit in ReadDataStatus.
Note	***: those failures may be caught in s22.51a instead of s22.52, however the result is the same.**

4M36-9231: ReadData - L1RSP	
Objective	To ensure the reader processing is stopped when a time out occurs (due to a mute card).
Applicability	[DE/DS]
References	s23.1, s23.9, s23.10, s23.11
Conditions	For 1st transaction: The card stops the communication with the reader in the state 'Waiting for Read Data Response'(mute card, i.e. it does not send neither any R-APDU nor S(WTX) blocks.) Case 1: L1RSP occurs at 1st ReadData Case 2: L1RSP occurs at 2nd ReadData
Pass Criteria	First transaction must terminate since the card is mute. 2nd transaction must perform correctly.

4M38-1000: PTFWF - waiting for PTFWF	
Objective	Ensure the reader correctly behaves when the PTFWF is needed or not.
Applicability	[DE/DS]
References	s2021222324.6, s2021222324.7, s2021222324.8, s2021222324.9, s2021222324.10, s2021222324.11, s2021222324.13
Conditions	Case 0: PTFWF absent (already tested) Case 1: PTFWF is empty (DataToSend is empty and TagsToReadYet is not empty) => 2021222324.9-TRUE Case 1b: - PTFWF is empty, DataToSend not empty, TagsToReadYet not empty (=> 2021222324.9-TRUE). - Upon reception of the 1st DET (PTFWF='00' + 9F1F returned): PTFWF is not empty, DataNeeded empty, DataToSend not empty, TagsToReadYet not empty (=> 2021222324.9-FALSE). - PTFWF is returned in 2nd DET with value '02' Case 1c: - PTFWF is empty, DataToSend is not empty (2021222324.8) => 2021222324.9-TRUE - 1st DET returns PTFWF='00'

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	- 2nd DET returns PTFWF='02' Case 2: PTFWF is '00' (DataNeeded is empty, DataToSend is not empty and TagsToReadYet is empty) => 2021222324.9-TRUE Case 2b: PTFWF is '00' (DataNeeded is empty, DataToSend is not empty and TagsToReadYet is empty) => 2021222324.9-TRUE Upon reception of the 1st DET (PTFWF='00'): PTFWF is not empty, DataNeeded empty, DataToSend empty, TagsToReadYet empty (=> 2021222324.9-FALSE). PTFWF is returned in 2nd DET with value '02' Case 3: PTFWF is '01' Case 4: PTFWF is '80'
Pass Criteria	Transaction must complete successfully. Reader must request PTFWF in cases 1, 1b and 1c.

4M38-2424: PTFWF - Data To Send after encrypted record	
Objective	Ensure the reader correctly adds known data to Data To Send after decrypting a record.
Applicability	[DE/DS]
References	s24.24
Conditions	PTFWF is empty TagsToRead includes 9F1F Last record encrypted (so s24-D is reached) Case 1: 9F1F NOT returned Case 2: 9F1F returned in last record
Pass Criteria	Transaction must complete successfully. Case 1: reader must not send a DEK in s24.26 Case 2: reader must send a DEK in s24.26

4M38-2431: PTFWF - decrypted ReadData received in s24	
Objective	Ensure the reader correctly decrypts the ReadData while waiting for PTFWF. Ensure the reader stops the PTFWF timer upon reception of the DET.
Applicability	[DE/DS]
References	s24.12,s24.31,s24.32,s24.34,s24.35,s24.36
Conditions	PTFWF is empty Upon reception of a DEK requesting PTFWF, a DET is returned requesting an envelope Case 1: Upon reception of a DEK providing the requested envelope, a DET is returned with the PTFWF. The ReadData response is delayed to ensure the PTFWF timer was stopped in s24.12. Case 2: ReadData is wrongly encrypted. A 2nd DET is sent with PTFWF.

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Pass Criteria	Transaction must complete successfully.

4M38-4000: PTFWF - ReadData needed	
Objective	Ensure the reader correctly behaves when ReadData is requested while waiting for the PTFWF. Ensure the reader correctly stores a DET received while waiting for PTFWF. Ensure the reader decrypts the last ReadData before sending GenAC.
Applicability	[DE/DS]
References	s23.20-TRUE, s23.24, s24.11, s202122232425.2-FALSE
Conditions	PTFWF is empty Upon reception of a DEK requesting PTFWF, a DET is returned with the PTFWF and TagsToRead requesting an Envelope. Case 1: ReadData response is correct Case 2: ReadData response is not correct
Pass Criteria	Transaction must complete successfully.

4M38-9241: PTFWF - TimeOut	
Objective	To ensure the reader processing is stopped when a time out occurs while waiting for the PTFWF.
Applicability	[DE/DS]
References	s24.1, s24.8, s24.10, s24.12, s2021222324.12
Conditions	Case 1a: PTFWF is empty. No DET is ever returned. Case 1b: PTFWF is empty, a 1st DET is sent with empty PTFWF, a 2nd DET is sent with PTFWF='00' Case 2a: PTFWF is '00', no DEK sent Case 2b: PTFWF is '00', DataToSend is not empty so a DEK is sent
Pass Criteria	The reader shall stop the transaction and send an OUT Signal: In OUT\OPS, - Start shall indicate "N/A" - Status shall indicate "END APPLICATION" In OUT\DiscretionaryData\ErrorIndication, 'L3' shall indicate "TIME OUT".

4M40-1140: Pre-GenAC checks - Floor Limit	
Objective	To ensure that the terminal sets the bit 'Transaction exceeds floor limit' in TVR when Amount is greater than Reader Contactless Floor Limit.

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References	s202122232425.14-FALSE, s202122232425.14-TRUE, s202122232425.15
Conditions	Case 1: Amount is less than Floor Limit and greater than CVM required limit Case 2: Amount is equal to Floor Limit Case 3: Amount is above Floor Limit
Pass Criteria	The terminal must set the bit 'Transaction exceeds floor limit' in TVR when Amount is greater than Reader Contactless Floor Limit.

4M40-4171: Pre-GenAC checks - AID(Terminal) != DF Name	
Objective	To ensure terminal continues the transaction as expected when AID(Terminal) does not match DFName.
References	s202122232425.16-FALSE, s202122232425.17
Conditions	AID(Terminal) != DF Name Case 1: DFname='B00000000198' Case 2: DFname='B0000000019886' Case 3: DFname='B000000001988610' Case 4: DFname='B0000000012345' Case 5: DFname='B0000000019876' (AID(Terminal)='B0000000012345')
Pass Criteria	Reader must set the TVR bit 'AID mismatch between card and terminal'.

4M40-9140: Pre-GenAC checks - Amount/TransactionDate not present	
Objective	To ensure the reader terminates the transaction when the Amount or the TransactionDate are not present or empty.
Applicability	Cases 05: [DE/DS]
References	s2021222324.14-FALSE, s2021222324.15
Conditions	Local authentication supported in AIP, AIP of card indicates EMV profile supported 'EDA enabled' in Kernel Qualifier is set No Read Data/Write Data Case 1: Amount Authorized is not present. Case 2: Amount Authorized is empty. Case 3: Transaction Date is not present. Case 4: Transaction Date is empty. Case 5: Amount Authorized and Transaction Date are sent empty in a DET signal after receiving the PAN.
Pass Criteria	The reader must terminate the transaction before issuing the GenAC.

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4M50-0020: TAA - TVR AND TAC-Denial = '00 00 00 00 00'	
Objective	To ensure the reader correctly checks the TVR against the TAC-denial, and requests an ARQC when it is Online-Only.
References	TAA.1-TRUE, TAA.2-TRUE, TAA.7
Conditions	Data is such as at least one bit is set in the TVR. TAC-Online have all bits set to 0b. TAC-Denial has one bit set to 0b for the corresponding TVR bit is set to 1b. Other TAC-Denial bits are set to 1b. IAC-denial (unknown tag) is returned with value 'FFFFFFFFFF'. Following cases are done with OnlineOnly card: Case 1: TerminalType=11 Case 2: TerminalType=14 Case 3: TerminalType=21 Following cases are done with non-OnlineOnly card: Case 4: TerminalType=24 Case 5: TerminalType=34
Procedure	A) CID = ARQC B) CID = AAC (KernelDecision should still be set to Online in TAA#2, however this is not observable because reader shall decline anyway)
Pass Criteria	The card shall receive a GenAC command requesting ARQC. The Reader must successfully complete the transaction.

4M50-0030: TAA - OfflineOnly reader	
Objective	To ensure the reader requests a TC when it is Offline-Only and the card is offline capable.
References	TAA.1-TRUE, TAA.2-FALSE, TAA.3-TRUE, TAA.5-FALSE, TAA.8
Conditions	Data is such as at least one bit is set in the TVR. TAC-Denial has one bit set to 0b for the corresponding TVR bit is set to 1b. Other TAC-Denial bits are set to 1b. TAC-Online have all bits set to 1b. Card is not OnlineOnly IACs (unknown tags) are returned with value 'FFFFFFFFFF'. Case 1: TerminalType=13 Case 2: TerminalType=16 Case 3: TerminalType=23 Case 4: TerminalType=26 Case 5: TerminalType=36

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Procedure	A) CID = TC B) CID = ARQC (KernelDecision should be set to Approved in TAA#2, however this is not observable because reader shall go online anyway because of the ARQC)
Pass Criteria	The card shall receive a GenAC command requesting TC. The Reader must successfully complete the transaction.

4M50-0040: TAA - OfflineOnline reader - OnlineOnly card	
Objective	To ensure the reader requests an ARQC when it is OfflineOnline capable and the card is OnlineOnly.
References	TAA.1-TRUE, TAA.2-FALSE, TAA.3-FALSE, TAA.4-TRUE, TAA.7
Conditions	Data is such as at least one bit is set in the TVR. TAC-Denial has one bit set to 0b for the corresponding TVR bit is set to 1b. Other TAC-Denial bits are set to 1b. TAC-Online have all bits set to 0b. Card is OnlineOnly IACdenial and default (unknown tags) are returned with value 'FFFFFFFFFF'. Case 1: TerminalType=12 Case 2: TerminalType=15 Case 3: TerminalType=22 Case 4: TerminalType=25 Case 5: TerminalType=35
Procedure	A) CID = ARQC B) CID = AAC (KernelDecision should still be set to Online in TAA#2, however this is not observable because reader shall decline anyway)
Pass Criteria	The card shall receive a GenAC command requesting ARQC. The Reader must successfully complete the transaction.

4M50-0061: TAA - OfflineOnline reader - Offline card	
Objective	To ensure the reader requests a TC when it is OfflineOnline capable and the card is not OnlineOnly.
References	TAA.1-TRUE, TAA.2-FALSE, TAA.3-FALSE, TAA.4-FALSE, TAA.6-TRUE, TAA.8
Conditions	Data is such as at least one bit is set in the TVR. TAC-Denial has one bit set to 0b for the corresponding TVR bit is set to 1b. Other TAC-Denial bits are set to 1b. TAC-Online have all bits set to 0b. Card is not OnlineOnly IACs (unknown tags) are returned with value 'FFFFFFFFFF'.

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	Case 1: TerminalType=12 Case 2: TerminalType=15 Case 3: TerminalType=22 Case 4: TerminalType=25 Case 5: TerminalType=35
Procedure	A) CID = TC B) CID = ARQC (KernelDecision should be set to Approved in TAA#2, however this is not observable because reader shall go online anyway because of the ARQC)
Pass Criteria	The card shall receive a GenAC command requesting TC. The Reader must successfully complete the transaction.

4M50-0062: TAA - TVR AND TAC-online != '00 00 00 00 00'	
Objective	To ensure the reader issues a GenAC requesting an ARQC, if for a bit set to 1b in the TVR, the corresponding bit in the TAC Online is set to 1b.
References	TAA.6-FALSE, TAA.7
Conditions	TAC-denial has all bits set to 0b. IAC-denial (unknown tag) is returned with value 'FFFFFFFFFF'. Reader is OfflineOnline capable. Card is offline capable. TAC-Online has one bit set to 1b for the corresponding TVR bit is set to 1b. Other TAC-Denial bits are set to 0b. A different bit is tested in all cases below. Case 1 (only applies to A (TAA#1)): Use TVR bit 'Local authentication not performed' Case 2 (only applies to A (TAA#1)): Use TVR bit 'Kernel 8 processing and TVR format' Case 3: Use TVR bit 'floor limit exceeded' Case 4: Use TVR bit 'AID mismatch' Case 5: Use TVR bit 'RRP limit exceeded' Case 6: Use TVR bit 'RRP not performed' Case 7 (case A only): Use TVR bit 'RRP performed' Case 8 (case A only): Use TVR bit 'RRP threshold' Following cases only apply to B (TAA#2) (CID=TC): Case 10: Use TVR bit 'LocalAuthentication failed' Case 11: Use TVR bit 'different application versions' Case 12: Use TVR bit 'expired application' Case 13: Use TVR bit 'application not yet effective' Case 14: Use TVR bit 'service not allowed' Case 15: Use TVR bit 'CVM failed' (already tested in 4M65-2926) Case 16: Use TVR bit 'unrecognized CVM' Case 17: Use TVR bit 'online CVM captured' (already tested in 4M65-2942) Case 18: Use TVR bit 'RFU'

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Procedure	A) all tests (when applicable) are made with TAA#1: the related TVR bits must be set prior to TAA#1 B) all tests (except cases 1,2,7,8) are made with TAA#2: the related TVR bits must NOT be set prior to TAA#1. They are set in cardTVR when required.
Pass Criteria	The reader must successfully complete the transaction online. A) The card shall receive a GenAC command requesting an ARQC. B) The card shall receive a GenAC command requesting a TC.

4M50-8010: TAA - TVR AND TAC-Denial != '00 00 00 00 00'	
Objective	To ensure the reader issues a GenAC requesting an AAC, if for a bit set to 1b in the TVR, the corresponding bit in the TAC Denial is set to 1b.
References	TAA.1-FALSE, TAA.9
Conditions	TAC-Online have all bits set to 1b. IAC-online (unknown tag) is returned with value 'FFFFFFFFFF'. TAC-Denial has one bit set to 1b for the corresponding TVR bit is set to 1b. Other TAC-Denial bits are set to 0b. A different bit is tested in all cases below. Case 1: OnlineOnly terminal; Use TVR bit 'Local authentication not performed' Case 2: OnlineOnly terminal; OnlineOnly card; Use TVR bit 'Kernel 8 processing and TVR format' Case 3: OfflineOnly terminal; Use TVR bit 'floor limit exceeded' Case 4: OfflineOnly terminal, OnlineOnly card Use TVR bit 'AID mismatch' (case A only) Case 5: OfflineOnline terminal; Use TVR bit 'RRP limit exceeded' Case 6: OfflineOnline terminal, OnlineOnly card; Use TVR bit 'RRP not performed' Following cases only apply to B (TAA#2) (CID=TC or ARQC): Case 10: Use TVR bit 'LocalAuthentication failed' Case 11: Use TVR bit 'different application versions' Case 12: Use TVR bit 'expired application' Case 13: Use TVR bit 'application not yet effective' Case 14: Use TVR bit 'service not allowed' Case 15: Use TVR bit 'CVM failed' (already tested in 4M65-2926) Case 16: Use TVR bit 'unrecognized CVM' Case 17: Use TVR bit 'online CVM captured' (already tested in 4M65-2942) Case 18: Use TVR bit 'RFU'
Procedure	A) all tests (when applicable) are made with TAA#1: the related TVR bits must be set prior to TAA#1

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	B) all tests (except cases 1,2) are made with TAA#2: the related TVR bits must NOT be set prior to TAA#1. They are set in cardTVR when required.
Pass Criteria	The reader must decline the transaction. A) The card shall receive a GenAC command requesting an AAC in all cases.

4M50-8050: TAA - OfflineOnly reader - OnlineOnly card	
Objective	To ensure the reader requests an AAC when it is Offline-Only and the card is OnlineOnly.
References	TAA.1-TRUE, TAA.2-FALSE, TAA.3-TRUE, TAA.5-TRUE, TAA.9
Conditions	Data is such as at least one bit is set in the TVR. TAC-Denial has one bit set to 0b for the corresponding TVR bit is set to 1b. Other TAC-Denial bits are set to 1b. TAC-Online have all bits set to 1b. Card is OnlineOnly IAC-online (unknown tag) is returned with value 'FFFFFFFFFF'. Case 1: TerminalType=13 Case 2: TerminalType=16 Case 3: TerminalType=23 Case 4: TerminalType=26 Case 5: TerminalType=36
Pass Criteria	The card shall receive a GenAC command requesting AAC. The Reader must correctly complete the transaction.

4M55-0110: GenAC - CID RFU bits	
Objective	To ensure the reader only checks the CID bits 7 and 8.
References	s29.20-TRUE
Conditions	CID RFU bits are sets
Pass Criteria	Reader must complete the transaction as expected.

4M55-1202: GenAC - Response parsing - TLV length	
Objective	To ensure the reader correctly parses the GenAC format 2 response message.
References	s26.11, s26.12-TRUE
Conditions	Case 0: template length is coded on 1 byte. Data object lengths are coded on 1 byte. (already tested).

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	Case 2: template length is coded on 1 byte. Data object lengths are coded on 2 bytes. Case 3: template length is coded on 2 bytes while the data length is less than 128. Data object lengths are coded on 1 byte. Case 4: template length is coded on 2 bytes while the data length is more than 128. Data object lengths are coded on 2 bytes(lengths are less than 128).
Pass Criteria	In all cases the transaction must successfully complete.

4M55-1203: GenAC - Response parsing - padding	
Objective	To ensure the reader ignores any '00' padding before, between or after TLV-coded data objects.
References	s26.11, s26.12-TRUE
Conditions	Case 1: the template contains CID and ATC with a '0000' padding in between Case 2: the template contains a trailing '00' padding of 50 bytes Case 3: the template contains a heading '0000' padding
Pass Criteria	In all cases the transaction must successfully complete.

4M55-1210: GenAC - Response parsing - unexpected items	
Objective	To ensure the reader correctly behaves when receiving unexpected items in the GenAC response.
Applicability	Cases 09b: [DE/DS]
References	s26.11, s26.12-TRUE, s2627.3
Conditions	Case 1: the template contains unknown tag 'C1 02 99FE' in last position. AIP is returned in GPO and GenAC. Case 2: the template contains in 4th position unknown tag 'C1 02 99FE' followed by '00' padding of 50 bytes. EDA not performed. Case 3: the template contains unknown tag '9F817E 02 000E' followed by '00' padding of 50 bytes. EDA performed. Case 5: Application Preferred Name returned in GenAC Case 6: '50' is returned empty in the Select ADF response. It is then returned with any value in the GenAC response. Case 7: PAN is returned with same value as in records. Case 9: the proprietary private class tag 'CF'(no value) and proprietary tag '9F05' (any value not empty) are returned in GenAC . These tags are present in the TLV database with RA update condition. 'CF' and '9F05' are requested in DiscretionaryData. Case 9b (DE/DS): same as case 9 plus: '9F05' is requested in Tags2Read (together with Extended SDA tag list). Case 10: 9F4B Unknown tag,proprietary tag, not present in the TLV database(i.e.: IsKnown=False, IsPresent=False). This tag is returned in GenAC response.

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	Case 11: '9F4E' tag, upd=K (not RA) is present in the TLV database empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K (not RA)). This tag is returned in GenAC response and is requested in DiscretionaryData. Case 12: '9F40' tag, upd=K (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K (not RA)). This tag is returned in the GenAC response with different value and incorrect Length and is requested in DiscretionaryData. Case 13: '9F21' tag, upd=K/ACT/DET (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT/DET (not RA)). This tag is returned in the GenAC response with different value and correct Length. Case 14: Proprietary tag '9F28', upd=K/ACT (not RA) is present in the TLV database not empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=False ie: upd=K/ACT (not RA)). This tag is returned in the GenAC response with different value and incorrect length. Case 15: Proprietary tag 'C6',private class upd=K/DET (not RA) is present in the TLV database empty (i.e.: IsKnown=True, IsPresent=True, IsEmpty=True ie: upd=K/DET (not RA)). This tag is returned in the GenAC response .
Pass Criteria	In all cases the transaction must successfully complete. Case 5: AppPrefName value is correct in dataRecord Case 6: AppLabel value is correct in dataRecord Case 7: PAN value is correct in dataRecord Case 8: CF value is correct in DiscData Cases 9 and 9b: 9F05 and CF values are correct in DiscData Case 9b: 9F05 value is correct in DEK case 11:The reader must ignore the value of Tag 9F4E returned by the card in GenAC response, this value is not stored in TLV Database. 9F4E value is correct in DiscData (empty). case 12: The reader must ignore the value of Tag 9F40 returned by the card in GenAC response, this value is not stored in TLV Database. 9F40 value is correct in DiscData.

4M55-9010: GenAC - L1RSP	
Objective	To ensure that the reader stops the transaction when L1 error is received in the current transaction while waiting for GenAC response.
References	s26.1, s26.5, s26.6, s26.7
Conditions	The card stops the communication with the reader upon reception of the GenAC command (mute card, i.e. it does not send neither any R-APDU nor S(WTX) blocks.)
Pass Criteria	Since the card is mute, the reader must stop the transaction. The reader must build an OUT signal. In UIRD, - 'Message Identifier' must indicate "Try again" - 'Status' must indicate "READY TO READ" - 'MessageHold Time' must indicate "000000"

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	In OUT\OPS, - 'Status' must indicate "END APPLICATION" - 'Start' must indicate "B" In OUT\DiscretionaryData\ErrorIndication, - 'L1' must indicate "RETURN CODE" - 'Msg On Error' must indicate "Try again" The reader must then start a new transaction and completes it until completion.

4M55-9100: GenAC - SW != '9000'	
Objective	To ensure the reader rejects transaction when the status words in the GenAC response are different from '90 00'.
Applicability	Cases B|04, B|05, B|06, B|07: [DE/DS]
References	s26.8-FALSE, s26.9, s2627.13, s2627.14
Conditions	Case 1: the card returns status value '62 83' in response to GenAC Case 2: the card returns status value '63 00' in response to GenAC Case 3: the card returns status value '63 Cx' in response to GenAC Case 4: the card returns status value '69 83' in response to GenAC Case 5: the card returns status value '69 84' in response to GenAC Case 6: the card returns status value '69 85' in response to GenAC Case 7: the card returns status value '6A 81' in response to GenAC Case 8: the card returns status value '6A 82' in response to GenAC Case 9: the card returns status value '6A 83' in response to GenAC Case 10: the card returns status value '6A 88' in response to GenAC Case 11: the card returns status value '90 01' in response to GenAC Case 12: the card returns status value '64 00' in response to GenAC Case 13: the card returns status value '65 00' in response to GenAC Case 14: the card returns status value '67 00' in response to GenAC Case 15: the card returns status value '6A 86' in response to GenAC Case 16: the card returns status value '6D 00' in response to GenAC Case 17: the card returns status value '6E 00' in response to GenAC Case 18: the card returns status value '6F 00' in response to GenAC Case 19: the card returns status value 'AB 21' in response to GenAC
Procedure	A) Cases 1, 2, 3: EDA is not performed. B) Cases 4, 5, 6, 7: EDA is performed, Data Envelopes To Write is not null. C) Cases 8, 9: Refund transaction. D) Cases 10, 11, 12, 13: EDA is not performed, CVD Failed E) Cases 14, 15, 16, 17: EDA is performed, restart indicator not empty(i.e. NO RESTART). F) Cases 18, 19: Refund transaction, restart indicator not empty(i.e. RESTART B).

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Pass Criteria	The reader must terminate the transaction upon reception of the erroneous response. WRITE DATA must not be sent.

4M55-9120: GenAC - Response Message Data format	
Objective	To ensure the reader rejects transaction when the Response Message Data Field format is not correct.
References	s26.10-FALSE, s26.13, s2627.13, s2627.14
Conditions	Case 1: Response Message Template tag is 70 Case 2: Response is '779000' Case 3: Response Message Template is in Format 1 i.e. the template tag is 80 Case 4: Response Message has only SW12.
Pass Criteria	The reader must terminate the transaction upon reception of the erroneous response.

4M55-9140: GenAC - parsing error	
Objective	To ensure the reader stops the transaction when the response message of the GenAC command is not correctly formatted.
References	s26.12-FALSE,s26.13,s2627.13,s2627.14
Conditions	DataEnvelope2Write is provided in ACT signal. Case 1a: the length value indicated in the template '77' is greater than the data length Case 1b: the length value indicated in the template '77' is smaller than the data length Case 3: the following additional data is returned in the template '77': '9F' (tag indicates subsequent byte and subsequent byte is not greater than zero, plus length is absent.) Case 4: the following additional data is returned in the template '77': 'DF02 02 03' (length is erroneous) Case 5a: the proprietary tag 'CF' is present in the TLV database. This tag is returned in the GenAC response with length lower than expected length range(with any value). Case 5b: the proprietary tag 'C8' is present in the TLV database. This tag is returned in the GenAC response with length higher than expected length range(with any value). Case 6: the card returns the template '77' correctly formatted followed by another tag 'C5' that is not included in the template. Case 7a: CID(9F27) is on 2 bytes. Case 7b: ATC(9F36) is on 1 bytes. Case 7c: AC(9F26) is on 9 bytes. Case 7d: CVD(9F8102) is on 2 bytes. Case 7f: IAD(9F10) is on 33 bytes. IAD was empty in GPO. Case 7g: Enhanced Data Authentication MAC(9F8105) is on 9 bytes. Case 7h: CardTVR(9F8104) is on 06 bytes. Case 7i: Restart Indicator(9F8108) length is on 6 bytes.

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	Case 9a: CID(9F27) is empty Case 9b: ATC(9F36) is empty Case 9c: AC is empty Case 9d: CVD is empty Case 9e: EDAMAC is empty Case 10: 9F27 is present in GenAC response but not in template 77 Case 11: Tag with Tag field length greater than 3 Bytes is returned (9FFF8F01) Case 13: ATC is returned both in the GetPO and the GenAC responses with different values. Case 14: known tag, Proprietary, privateClass, RA, not empty (ie: already set in GPO)is returned in GenAC responses with different value and incorrect length. Case 15: Proprietary tag '9F05' not empty (ie: already set in GPO), (i.e.: IsKnown=True, IsPresent=True, IsEmpty=false upd=K/RA) is returned in GenAc response with different value. Case 16: Proprietary tag '5F19' not empty (ie: already set in ACT), (i.e.: IsKnown=True, IsPresent=True, IsEmpty=false upd=K/ACT/RA) is returned in GenAc response with different value. Case 17: tag is returned without length. Case 18: Proprietary tag 'C7', Private Class empty, (i.e.: IsKnown=True, IsPresent=True, IsEmpty=true upd=K/DET/RA) is returned in GenAc response with incorrect Length.
Pass Criteria	The reader must terminate the transaction upon reception of the erroneous response.

4M55-9160: GenAC - Mandatory data objects	
Objective	To ensure the reader stops the transaction when mandatory data object in the GenAC response is missing.
References	s26.14-FALSE,s26.15,s2627.13,s2627.14
Conditions	Case 1a: CID(9F27) is missing in GenAc response Case 1b: ATC(9F36) is missing in GenAc response Case 1c: AC(9F26) is missing in GenAc response Case 1d: CVD(9F8102) is missing in GenAC response Case 1e: IAD(9F10) is missing in GenAC response Case 1f: IAD(9F10) is empty in GenAC response Case 1g: EDAmac is missing in GenAC response Case 1h: GenAC response is 7700
Pass Criteria	The reader must terminate the transaction upon reception of the erroneous response.

4M55-9240: GenAC - Invalid CID	
Objective	To ensure the reader terminates the transaction when the CID is invalid.

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References	s29.20-FALSE,s29.22
Conditions	Data returned by card are such that the reader requests an AAC Case 1: Card returns CID = '40' Case 2: Card returns CID = '80' Data returned by card are such that the reader requests a TC Case 3: Card returns CID = 'C0' Case 4: Card returns CID = 'FF' Data returned by card are such that the reader requests an ARQC Case 5: Card returns CID = '40' Case 6: Card returns CID = '4F' Case 7: Card returns CID = 'C8' Case 8: Card returns CID = 'CA'. 'Local authentication supported' in Kernel Qualifier is not set. Case 9: Card returns CID = 'CB'. 'Local authentication supported' in Kernel Qualifier is not set.
Pass Criteria	The reader must terminate the transaction upon reception of the erroneous response.

4M55-9270: GenAC - Application PAN missing	
Objective	To ensure that the reader terminates the transaction if Application PAN is not available after receiving the GenAC response.
References	s2627.1-FALSE,s2627.2
Conditions	Last record is encrypted. Case 1: Application PAN not present, EDA enabled Case 2: Application PAN is empty, EDA disabled Case 3: AFL is such that active AFL is empty.
Pass Criteria	The reader must terminate the transaction.
Note	**Test tool must use a PAN value '00' when crypto requires it.**

4M59-1140: No local authentication - transaction without local authentication	
Objective	To ensure that when local authentication is disabled: - the reader correctly processes the transaction with or without RRP - the reader does not launch the ProcessRequestCertificates function - the reader properly sets the bit 'Local authentication was not performed' in TVR - the reader does not check the local authentication data objects.

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References	s202122232425.18-FALSE,s202122232425.20
Conditions	Case 1: 'local authentication' enabled in kernel qualifier is not set and 'local authentication' in AIP is set . Case 2: 'local authentication' in AIP is not set and 'local authentication' in kernel qualifier is set. CApk indicates an unknown key ('EE'). Case 3: - Only Amount, Transaction Type and TransactionTime are provided in the ACT signal - Local authentication not supported in AIP - RRP not supported in AIP - The card only returns the PAN in the records Case 4: similar to case 3: - PAN is returned in GPO - AFL is such that no ReadRecord command is sent
Pass Criteria	The reader must complete the transaction without local authentication.

4M60-0180: Certificates processing - RSA exponent	
Objective	To ensure the reader supports values '3' and '2^16+1' for the RSA Public Key exponents.
References	C.26 (ie: Book2, section 6.1)
Conditions	Case 0 (already tested): '3' is used for the Public Key exponents (CA, Issuer and ICC) Case 1: '2^16+1' is used for the Public Key exponents (CA, Issuer and ICC)
Pass Criteria	Reader must complete the transaction successfully.

4M60-0225: Certificates processing - Issuer Identifier	
Objective	To ensure the Reader correctly processes the Issuer Identifier, depending on the number of leftmost PAN digits used.
References	C.26.8 (ie: Book2, section 6.3, bullet 8)
Conditions	ECC certificates: Case 1: Issuer PK Certificate in the card is calculated using leftmost 3 PAN digits and right padded with 'F' up to a length of 10 digits. Case 2: Issuer PK Certificate in the card is calculated using leftmost 6 PAN digits and right padded with 'F' up to a length of 10 digits. Case 3: Issuer PK Certificate in the card is calculated using 10 PAN digits. RSA certificates:

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	Case 5: Issuer PK Certificate in the card is calculated using leftmost 3 PAN digits and right padded with 'F' up to a length of 8 digits. Case 6: Issuer PK Certificate in the card is calculated using leftmost 6 PAN digits and right padded with 'F' up to a length of 8 digits. Case 7: Issuer PK Certificate in the card is calculated using 8 PAN digits.
Procedure	Cases 2, 5, 7 are run with Test2 AID. Other cases are run with Test1 AID.
Pass Criteria	Reader must complete the transaction successfully.

4M60-0226: Certificates processing - ExpDate = CurrentDate	
Objective	To ensure the Reader correctly processes the certificates when the expiration date is same as the current date.
References	C.25.8, C.26.9 (ie: Book2, section 6.3, bullet 9)
Conditions	Issuer Certificate ExpDate is same as current date. ICC Certificate ExpDate and ExpTime (ECC only) are same as current date/time. Case 1: date is in 2022 Case 2: date is in 2050 Case 3: date is 29Feb2024 Case 4: date is 29 Feb 2076
Procedure	A: ECC B: RSA
Pass Criteria	Reader must complete the transaction successfully.

4M60-0227: Certificates processing - Length of all Moduli (RSA)	
Objective	To ensure the Reader supports Moduli with a length which is a multiple of 8 bytes.
References	C.26, C.34
Conditions	Length of moduli used is multiple of 8 bytes for the CAkey, Issuer key and ICC key.
Pass Criteria	Reader must complete the transaction successfully.

4M60-0248: Certificates processing - PAN (ICC RSA)	
Objective	To ensure the Reader correctly processes the PAN in the RSA ICC certificate.
References	C.34.8 (ie: Book2, section 6.4, bullet 8)
Conditions	Case 1: PAN is 9 digits long, padded with F

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	Case 2: PAN is 10 digits long Case 3: PAN is 19 digits long (max length) Case 4: PAN is 3 digits long
Pass Criteria	Reader must complete the transaction successfully.

4M60-0255: Certificates processing - Upper bound for size of moduli (RSA)	
Objective	To ensure the Reader supports Public key Moduli with maximum lengths as defined below: max NIC length is 247 bytes max NI length is 247 bytes max NCA length is 248 bytes.
References	C.26, C.34
Conditions	length NCA = 248 bytes, length NI = 247 bytes, Length NIC = 247 bytes
Pass Criteria	Reader must complete the transaction successfully.

4M60-0257: Certificates processing - storing keys	
Objective	To ensure the reader is able to store 6 RSA CA Public Keys and 10 ECC CA Public Keys and it is able, given RID and Certification Authority Public Key Index, to locate such key.
References	'Persistent Dataset for Kernel 8' table
Conditions	* Reader supports two RIDs. Case 1: * Reader is loaded with 6 RSA CA Public Keys for the RID 'B000000001' * At least one of the RID 'B000000001' RSA CA Public Keys indexes does not exist for RID 'B123456789' * 6 transactions are made in a raw using the 6 RSA CA Public Keys for the RID 'B000000001' * A 7th transaction is made with AID 'B123456789...' using a CA Public Key index that exists for the RID 'B000000001' but does not exist for the RID 'B123456789' Case 2: * Reader is loaded with 10 ECC CA Public Keys for the RID 'B000000001' * At least one of the RID 'B000000001' ECC CA Public Keys indexes does not exist for RID 'B123456789' * 10 transactions are made in a raw using the 10 ECC CA Public Keys for the RID 'B000000001' * A 11th transaction is made with AID 'B123456789...' using a CA Public Key index that exists for the RID 'B000000001' but does not exist for the RID 'B123456789'
Pass Criteria	First transactions must complete successfully. Last transaction must raise 'Local authentication failed' bit in the TVR.

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4M60-0259: Certificates processing - NIC=NI=NCA (RSA)	
Objective	To ensure the Reader supports Public key Moduli with mNIC=NI=NCA.
References	C.26, C.34
Conditions	length NCA = 240 bytes, length NI = 240 bytes, Length NIC = 240 bytes
Pass Criteria	Reader must complete the transaction successfully.

4M60-0262: Certificates processing - RSA - remainder not needed	
Objective	To ensure the reader correctly recovers the certificates when the PKRemainder is not returned because not needed.
References	C.26, C.34
Conditions	Case 1: Size of Issuer PK and CAkey is such as NI < NCA - 36. IssuerPKRemainder data object is not present. Case 2: Same as case 1 plus: IssuerPKRemainder data object is empty. Case 3: Size of ICC PK and Issuer PK is such as NICC < NI - 42. ICCPKRemainder data object is not present. Case 4: Same as case 3 plus: ICCPKRemainder data object is empty.
Pass Criteria	* The Reader shall process the transaction until completion.

4M60-1030: Certificates processing - No Extended SDA Tag List	
Objective	Ensure the reader correctly completes the transaction when the Extended SDA Tag List is not available.
References	RPC.13-FALSE, RPC.14
Conditions	A) ECC Case 1: Extended SDA Tag List not returned Case 2: Extended SDA Tag List is returned empty B) RSA Case 3a: Extended SDA Tag List not returned - ICC remainder present Case 3b: Extended SDA Tag List not returned - ICC remainder empty Case 4a: Extended SDA Tag List is returned empty - ICC remainder present Case 4b: Extended SDA Tag List is returned empty - ICC remainder not present
Pass Criteria	Transaction completes successfully.

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4M60-1031: Certificates processing - Extended SDA Tag List	
Objective	Ensure the reader correctly builds the Extended SDA Tag List.
References	s2627.8
Conditions	Extended SDA Tag List includes: - a tag that is present ('9F06') - a tag that is empty ('50') - a tag that is absent ('9F8106') - a 1-byte tag ('50') - a 2-bytes tag ('9F06') - a 3-bytes tag ('9F8106') - a tag with length on 2 bytes while length is less than 128 - CDOL (returned in GPO) Tags are returned in an order different from the one in the extended SDA tag list. A|01a: ECC and RRP and version 2 A|01b: ECC and RRP and version 0F A|02a: ECC no RRP and version 3 A|02b: ECC no RRP and version 2 B|01a: RSA and RRP and version 2 B|01b: RSA and RRP and version 3 B|02a: RSA no RRP and version 2 B|02b: RSA no RRP and version FF
Pass Criteria	Transaction completes successfully. Reader shall use an Extended SDA Tag List related data: 5A[Length][Value] || 9F810600 || 9F06[Length][Value] || 5000 || 8C[Length][Value].

4M60-2001: Certificates processing - last record encrypted (ECC)	
Objective	Positive test to verify that the transaction correctly completes with ECC based profile when the last record is encrypted. Flow will go through s26.3, or s27*. To ensure the kernel can resume RequestProcessCertificate procedure processing when the missing items become available.
References	RPC.2, RPC.12, RPC.15, RPC.16, RPC.19
Conditions	First and Last records are encrypted. Case 1: ICC Certificate is present in the last record (encrypted) (RPC.19-FALSE, 26.32 or not*) Case 2: Tag ‘8F’ is present in the last record (encrypted) (RPC.2-FALSE, 26.32 or not*)

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	Case 3: extended SDAtagList is present in the last record (encrypted) (RPC.12-FALSE, 26.32 or not*). Extended SDA Tag List is empty in first record. Case 4: extended SDAtagList and '8F' are present in the last record (encrypted) (RPC.2-FALSE, 26.32 or not*) Case 5: Data objects referenced in Extended SDA Tag List are returned empty (RPC.15-FALSE, RPC.16-FALSE) Case 6: Issuer Certificate is present in the last record (encrypted) (RPC.4-FALSE, 26.32 or not*) Case 7: PAN is present in the last record (encrypted) (RPC.4-FALSE, 26.32 or not*) Case 8: Last record encrypted; Extended Tag List data item empty in non encrypted record (RPC.16-FALSE, 26.32 or not*) Case 9: Last record encrypted; Extended Tag List data item absent (RPC.16-FALSE, 26.32 or not*)
Procedure	A) Subcases are run where GenAC response is delayed (trying to trigger s26.3) B) SubCases are run where GenAC response is not delayed (trying to trigger s27)
Pass Criteria	Transaction successfully completes online.
Note	***: depends on the product implementation**

4M60-2002: Certificates processing - last record encrypted (RSA)	
Objective	Positive test to verify that the transaction correctly completes with RSA based profile when the last record is encrypted. Flow will go through s26.3, or s27*. To ensure the kernel can resume RequestProcessCertificate procedure processing when the missing items become available.
References	RPC.2, RPC.12, RPC.15, RPC.16, RPC.19
Conditions	RSA is used (CApk='EF') Last record is encrypted Case 1: ICC Certificate is present in the last record (encrypted) (RPC.19-FALSE, 26.32 or not*) Case 2: Tag '8F' is present in the last record (encrypted) (RPC.2-FALSE, 26.32 or not*) Case 3: ExtSDAtagList is present in the last record (encrypted) (RPC.12-FALSE, 26.32 or not*). Extended SDA Tag List is empty in first record. Case 4: ExtSDAtagList and '8F' are present empty in 1st and 2nd records. Then, they are returned with correct values in the last encrypted record (encrypted) (RPC.2-FALSE, 26.32 or not*) Case 5: Data objects referenced in Extended SDA Tag List returned in last record, encrypted (RPC.15-FALSE, RPC.16-FALSE) Case 6: Issuer Certificate is present in the last record (encrypted) (RPC.6-FALSE, 26.32 or not*) Case 7: PAN is present in the last record (encrypted) (RPC.6-FALSE, 26.32 or not*) Case 9: Last record encrypted; Extended Tag List data item absent (RPC.16-FALSE, 26.32 or not*)

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	Case 10: Issuer Exponent is present in the last record (encrypted) (RPC.6-FALSE, 26.32 or not*) Case 11: Issuer Remainder is present in the last record (encrypted) (RPC.6-FALSE, 26.32 or not*) Case 12: ICC Exponent is present in the last record (encrypted) (RPC.6-FALSE, 26.32 or not*) Case 13: ICC Remainder is present in the last record (encrypted) (RPC.6-FALSE, 26.32 or not*)
Procedure	A) Subcases are run where GenAC response is delayed (trying to trigger 26.5) B) SubCases are run where GenAC response is not delayed
Pass Criteria	Transaction successfully completes online.
Note	***: depends on the product implementation**

4M60-3010: Certificates processing - Revocation List	
Objective	Test for ECC and RSA based profile to ensure that the reader correctly verifies if the certificate is revoked.
References	C.25, C.26, C.27-FALSE
Conditions	Case 1: Certificate Serial Number is present in Revoked List for ECC based profile i.e. RID B000000001. Case 2: Certificate Serial Number is present in Revoked List for ECC based profile i.e. RID B123456789. Case 3: Certificate Serial Number is present in Revoked List for RSA based profile i.e. RID B000000001. 'ReportLocalAuthenticationFailed' unset; CID=TC; TAA2 leads to decline. Case 4: Certificate Serial Number is present in Revoked List for RSA based profile i.e. RID B123456789. Case 5: Certificate Serial Number is not present in Revoked List for ECC based profile i.e. RID B000000001. Case 6: Certificate Serial Number is not present in Revoked List for RSA based profile i.e. RID B123456789.
Pass Criteria	Transaction must complete as expected.

4M60-3800: Certificates processing - locating ICC ECC Key	
Objective	Ensure the reader correctly locates the ICC ECC Key in the last record (signed).
References	C.38, C.39-TRUE
Conditions	RSA card profile. Case 1: ICC ECC Public Key '9F810B' returned in last record (signed)
Pass Criteria	Reader must successfully complete the transaction.

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4M60-7004: Certificates processing - various failures (ECC)	
Objective	Failure scenario's on call to Request Process Certificates for ECC based card profile.
References	RPC.2-FALSE,RPC.4-FALSE,RPC.10-FALSE,RPC.18,RPC.21-FALSE
Conditions	Case 1a: CA Public Key Index is not present (RPC.2-FALSE) Case 2a: Issuer Public Key Certificate not present (RPC.4-FALSE) Case 2b: Issuer Public Key Certificate is empty (RPC.4-FALSE) Case 7a: ICC Public Key Certificate is not present (RPC.19-FALSE) Case 7b: ICC Public Key Certificate is empty (RPC.19-FALSE). No RRP.
Pass Criteria	Reader must terminate.

4M60-7005: Certificates processing - various failures (RSA)	
Objective	Failure scenario's on call to Request Process Certificates for RSA based card profile.
References	RPC.2-FALSE,RPC.6-FALSE,RPC.21-FALSE
Conditions	Case 1a: CA Public Key Index card is not present (RPC.2) Case 2a: RSA Issuer Public Key Certificate not present (RPC.6-FALSE) Case 2b: RSA Issuer Public Key Exponent not present (RPC.6-FALSE) Case 2d: RSA Issuer Public Key Certificate is empty (RPC.6-FALSE) Case 6a: RSA ICC Public Key Certificate not present (RPC.21-FALSE) Case 6b: RSA ICC Public Key Exponent not present (RPC.21-FALSE) Case 6d: RSA ICC Public Key Certificate is empty (RPC.21-FALSE). No RRP.
Pass Criteria	Reader must terminate.

4M60-7008: Certificates processing - CA public key index not known	
Objective	Ensure the reader correctly reacts when the CA public key index is not found in the CA key database or the key type is unknown.
References	RPC.2a-FALSE, RPC.3-OTHER, s28.9-CAPKMISSING, s28.9a
Conditions	CA public key does not exist in the CA key database. Case 1: RSA supported, ECC profile Case 2: RSA supported, RSA profile Case 3: RSA not supported

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	Case 4: CA public key exists in the CA key database but the Algorithm Indicator is unknown.
Procedure	A: RRP B: no RRP
Pass Criteria	Reader must set the bit 'Local authentication failed' in TVR.

4M60-7011: Certificates processing - RSA certificates not enabled	
Objective	Negative test for RSA based profile when RSA certificates not enabled in Kernel Configuration.
References	C.24a-FALSE,C.29,C.30,s29.9
Conditions	'EDA enabled' in Kernel Qualifier is set, 'RSA certificates enabled' in Kernel Configuration is not set. Reference Control Parameter in Generate AC command set to TC
Pass Criteria	Transaction must complete.

4M60-7014: Certificates processing - Erroneous Issuer Certificate (ECC)	
Objective	Negative Test for ECC Issuer Certificate.
References	C.25, C.27-FALSE, C.29, C.30, s29.9
Conditions	Case 1a: Issuer Public Key Certificate is greater than '117' in length (i.e. '118') Case 1b: Issuer Public Key Certificate is less than '117' in length. (i.e. '116') Case 2a: Issuer Public Key Certificate Format is not '12' (i.e. '14') Case 2b: Issuer Public Key Certificate Format is not '12' (i.e. '02') Case 3a: 'Certificate Encoding' is not '00' (i.e. '01') Case 3b: 'Certificate Encoding' is not '00' (i.e. '10') Case 4: 'Issuer Identifier' does not match the IssuerIdentifier from Application PAN (i.e. 2213339000) Case 5: 'Certificate Expiry Date' is earlier than current date (i.e. 20220123) Case 5b: 'Certificate Expiry Date' is earlier than current date (i.e. 19990423) Case 5c: 'Certificate Expiry Date'(i.e 20491231) is earlier than current date (i.e. 500101) Case 5d: 'Certificate Expiry Date'(i.e 20760228) is earlier than current date (i.e. 760229) Case 6a: 'RID' does not match DF Name[1:5] (i.e. B000000002) Case 6b: 'RID' does not match DF Name[1:5] (i.e. B123456789) Case 7: 'CA Public Key Index' in certificate does not match with the 'CA Public Key Index' in tag '8F' Case 9a: 'Issuer Public Key Algorithm Suite Indicator' is not '10' (i.e. '01') Case 9b: 'Issuer Public Key Algorithm Suite Indicator' is not '10' (i.e. '00') Case 11a: 'Issuer Certificate Signature' in Issuer Public Key Certificate is not genuine (error in first byte)

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	Case 11b: 'Issuer Certificate Signature' in Issuer Public Key Certificate is not genuine (error in last byte)
Procedure	Case 5 is done with no RRP Case 5b is done with 'ReportLocalAuthenticationFailed' unset; CID=TC; TAA2 leads to online.
Pass Criteria	Reader must set the bit 'Local authentication failed' in TVR.

4M60-7015: Certificates processing - Erroneous Issuer Certificate (RSA)	
Objective	Negative Test for RSA Issuer Certificate.
References	C.26, C.27-FALSE, C.29, C.30, s29.9
Conditions	case 01a: IssuerPKCertificate length is greater than Certification Authority PK Modulus case 01b: IssuerPKCertificate length is less than Certification Authority PK Modulus case 02a: Data Trailer recovered from the IssuerPKCertificate does not equal 'BC'(i.e. 0C) case 02b: Data Trailer recovered from the IssuerPKCertificate does not equal 'BC'(i.e. B0) case 03a: Data Header recovered from the IssuerPKCertificate does not equal '6A'(i.e. 0A) case 03b: Data Header recovered from the IssuerPKCertificate does not equal '6A'(i.e. 60) case 04a: Certificate Format recovered from IssuerPKCertificate is not equal '02'(i.e. 12) case 04b: Certificate Format recovered from IssuerPKCertificate is not equal '02'(i.e. 20) case 05a: IssuerPKCertificate in card is calculated with Issuer Identifier different from leftmost 3-8 PAN digits: difference is on digit 5 case 05b: IssuerPKCertificate in card is calculated with Issuer Identifier different from leftmost 3-8 PAN digits: difference is on digit 3 case 05c: IssuerPKCertificate in card is calculated with Issuer Identifier different from leftmost 3-8 PAN digits: difference is on digit 8 case 06a: Issuer Certificate Expiration Date is earlier than the current date(i.e. 0122) case 06b: Issuer Certificate Expiration Date is earlier than the current date(i.e. 1200) case 06c: Issuer Certificate Expiration Date (i.e. 0149) is earlier than the current date(i.e. 500101) case 06d: Issuer Certificate Expiration Date (i.e. 0156) is earlier than the current date(i.e. 560229) case 07a: Issuer PK Algorithm is not supported (different from '01')(i.e. 10) case 07b: Issuer PK Algorithm is not supported (different from '01')(i.e. 02) case 08a1: Calculated Hash Result is different from the Hash Result recovered from the IssuerPKCertificate: first byte wrong case 08a2: Calculated Hash Result is different from the Hash Result recovered from the IssuerPKCertificate: last byte wrong case 08b: Calculated Hash Result is different from the Hash Result recovered from the IssuerPKCertificate: IssuerPKRemainder is absent case 08c: Calculated Hash Result is different from the Hash Result recovered from the IssuerPKCertificate: IssuerPKRemainder is empty

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	case 08d: Calculated Hash Result is different from the Hash Result recovered from the IssuerPKCertificate: IssuerPKRemainder returned in Read Record length is higher. case 08e: Calculated Hash Result is different from the Hash Result recovered from the IssuerPKCertificate: IssuerPKRemainder returned in Read Record length is lower. case 09: Issuer Public Key moduli length is higher than Certification Authority Public Key moduli length (Ni>NCA).
Pass Criteria	Reader must set the bit 'Local authentication failed' in TVR.

4M60-7040: Certificates processing - Erroneous ICC Certificate Format (ECC)	
Objective	Negative Test for ECC ICC Certificate Format.
References	RPC.20,C.33-ECC,C.36, C.37-FALSE,s29.10
Conditions	case 01a: ICC Public Key Certificate is not 145 bytes in length (i.e. 146) case 01b: ICC Public Key Certificate is not 145 bytes in length (i.e. 144) case 02a: 'ICC Certificate Format' in ICC Public Key Certificate is not '14' (i.e. 04) case 02b: 'ICC Certificate Format' in ICC Public Key Certificate is not '14' (i.e. 10) case 03a: 'ICC Certificate Encoding' in ICC Public Key Certificate is not '00' (i.e. 01) case 03b: 'ICC Certificate Encoding' in ICC Public Key Certificate is not '00' (i.e. 10) case 04a: ICC Public Key Certificate Expiration Date is earlier than the current date (i.e. 19990431) case 04b: ICC Public Key Certificate Expiration Date is earlier than the current date (i.e. 20200101) case 04c: ICC Public Key Certificate Expiration Date is earlier than the current date (i.e. 20201231) case 04d: ICC Public Key Certificate Expiration Time is earlier than the current time case 04e: ICC Public Key Certificate Expiration Date (i.e. 20491231) is earlier than the current date(i.e. 500101) case 04f: ICC Public Key Certificate Expiration Date (i.e. 20680228) is earlier than the current date(i.e. 680229) case 05a: 'ICC Public Key Algorithm Suite Indicator' in ICC Public Key Certificate is not '00' (i.e. 10) case 05b: 'ICC Public Key Algorithm Suite Indicator' in ICC Public Key Certificate is not '00' (i.e. 01) case 06a: 'ICCD Hash Encoding' in ICC Public Key Certificate is not '01' (i.e. 11) case 06b: 'ICCD Hash Encoding' in ICC Public Key Certificate is not '01' (i.e. 10) case 07a: 'ICCD Hash Algorithm Indicator' in ICC Public Key Certificate is not '02' (i.e. 20) case 07b: 'ICCD Hash Algorithm Indicator' in ICC Public Key Certificate is not '02' (i.e. 22) case 08a: Calculated Hash Result is different from the Hash Result recovered from the ICC Public Key Certificate: 1st byte wrong case 08b: Calculated Hash Result is different from the Hash Result recovered from the ICC Public Key Certificate: last byte wrong case 09a: 'ICC Public Key Certificate Signature' in ICC Public Key Certificate is not genuine: 1st byte wrong

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	case 09b: 'ICC Public Key Certificate Signature' in ICC Public Key Certificate is not genuine: last byte wrong
Pass Criteria	Reader must set the bit 'Local authentication failed' in TVR.

4M60-7041: Certificates processing - Erroneous ICC Certificate Format (RSA)	
Objective	Negative Test for RSA ICC Certificate Format.
References	RPC.22,C.33-RSA,C.34,C.35-FALSE,C.42,s29.10
Conditions	case 01a: ICC Public Key Certificate is greater than length of the Issuer Public Key Modulus. case 01b: ICC Public Key Certificate is less than length of the Issuer Public Key Modulus. case 02a: Data Trailer recovered from the ICC Public Key Certificate does not equal 'BC'(i.e. 0C). case 02b: Data Trailer recovered from the ICC Public Key Certificate does not equal 'BC'(i.e. B0). case 03a: Data Header recovered from the ICC Public Key Certificate does not equal '6A'(i.e. 0A). case 03b: Data Header recovered from the ICC Public Key Certificate does not equal '6A'(i.e. 60). case 04a: Certificate Format recovered from ICC Public Key Certificate is not equal '04'(i.e. 14). case 04b: Certificate Format recovered from ICC Public Key Certificate is not equal '04'(i.e. 40). case 06a: Calculated Hash Result is different from the Hash Result recovered from the ICC Public Key Certificate: first byte. Remainder not present. case 06b: Calculated Hash Result is different from the Hash Result recovered from the ICC Public Key Certificate: last byte. Remainder present. case 07: Recovered PAN from ICC Public Key Certificate is different than Application PAN read from the ICC. case 08a: ICC Public Key Certificate Expiration Date is earlier than the current date(i.e. 0120). case 08b: ICC Public Key Certificate Expiration Date is earlier than the current date(i.e. 1200). 'ReportLocalAuthenticationFailed' unset; TAA2 (and CID) lead to offline approval. case 08c: ICC Public Key Certificate Expiration Date (i.e. 1249) is earlier than the current date(i.e. 500101). case 08d: ICC Public Key Certificate Expiration Date (i.e. 0196) is earlier than the current date(i.e. 500101). case 09a: ICC Public Key Certificate in the card is calculated with ICC PK Algorithm value different than'01'(i.e. 10). case 09b: ICC Public Key Certificate in the card is calculated with ICC PK Algorithm value different than'01'(i.e. 11).
Pass Criteria	Reader must set the bit 'Local authentication failed' in TVR.

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4M60-7045: Certificates processing - ICC ECC Key not found (RSA)	
Objective	Ensure the reader correctly behaves when the ICC ECC Key is not found in signed records.
References	C.38, C.39-FALSE, C.42, C.50-FALSE, C.54, s29.3
Conditions	RSA card profile. Case 1: ICC ECC Public Key '9F810B' not returned Case 2: ICC ECC Public Key '9F810B' returned in 2nd record, non-signed Case 3: ICC ECC Public Key '9F810B' returned. No record is signed. 'ReportLocalAuthenticationFailed' unset; CID=ARQC; TAA2 leads to decline. case 4:ICC ECC Public Key '9F810B' returned in 2nd record, non-signed. ICC ECC Public Key requested in the Extended Tag List (so it is present in Data To Hash).
Pass Criteria	Reader must set the bit 'Local authentication failed' in TVR.
Note	**The test tool shall generate the blinding factor as usual, without considering the 9F810B tag value update above.**

4M60-7046: Certificates processing - ICC ECC Key not found (RSA)	
Objective	Ensure the reader correctly behaves when the ICC ECC Key is not found in signed record.
References	C.38, C.39-TRUE, C.40a-FALSE, C.42, C.50-FALSE, C.54, s29.3
Conditions	RSA card profile. Case 1: ICC ECC Public Key '9F810B' not recovered
Pass Criteria	Reader must set the bit 'Local authentication failed' in TVR.
Note	**The test tool shall generate the blinding factor as usual, without considering the 9F810B tag value update above.**

4M60-7100: Certificates processing - Erroneous BlindingFactor	
Objective	Negative test for PROCESS BLINDING FACTOR
References	RPC.24,C.6,C.50-TRUE,C.51,C.52-FALSE,C.54, s29.5, s29.11
Conditions	Issuer and ICC certificates are valid. First record is not encrypted. Case 3: ECC based test profile, recovered blinding factor is corrupted. 'ReportLocalAuthenticationFailed' unset; CID=TC; TAA2 leads to online. Case 4: RSA based test profile, recovered blinding factor is corrupted case 5: ECC: 'Card Public Key' in ICC Public Key Certificate is not correct to recover ICC ECC key

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	case 6: RSA: 'Card Public Key' in ICC Public Key Certificate is not correct to recover ICC ECC key
Pass Criteria	Reader must set the bit 'Local authentication failed' in TVR.

4M61-2830: IAD MAC - copy in IAD	
Objective	To ensure that the reader correctly copies the IAD MAC in the IAD, when applicable.
References	s28.3, s28.6
Conditions	01: 'Copy IAD MAC in IAD' in AIP = 00b 02a: 'Copy IAD MAC in IAD' in AIP = 01b (default IAD MAC offset= 0) 02b: 'Copy IAD MAC in IAD' in AIP = 01b (default IAD MAC offset= max value = IAD (max) length - 8 = 32-8 = 24) 02c: 'Copy IAD MAC in IAD' in AIP = 01b (default IAD MAC offset= IAD length - 8 = 26-8 = 18) 03a: 'Copy IAD MAC in IAD' in AIP = 10b (IAD MAC offset= 0); IAD length = 8 03b: 'Copy IAD MAC in IAD' in AIP = 10b (IAD MAC offset= max value = IAD (max) length - 8 = 32-8 = 24) 03c: 'Copy IAD MAC in IAD' in AIP = 10b (IAD MAC offset= IAD length - 8 = 30-8 = 22) 04: 'Copy IAD MAC in IAD' in AIP = 11b
Procedure	A) RRP B: no RRP
Pass Criteria	Reader must complete the transaction successfully. Reader must copy the IAD MAC in the IAD, except in cases 1 and 4.

4M61-9284: IAD MAC - invalid offset	
Objective	To ensure that the reader stops the transaction when IAD MAC offset is invalid.
References	s28.4-FALSE, s28.10
Conditions	01: 'Copy IAD MAC in IAD' in AIP = 10b; IAD MAC offset=0; IAD length = 7 02: 'Copy IAD MAC in IAD' in AIP = 10b; IAD MAC offset=25; IAD length = 32 03: 'Copy IAD MAC in IAD' in AIP = 01b; Default IAD MAC offset=18; IAD length = 25 04: 'Copy IAD MAC in IAD' in AIP = 01b; Default IAD MAC offset=18; IAD length = 17
Pass Criteria	Reader must stop the transaction after the GeneraceAC response.

4M62-7450: EDA MAC - MAC failure	
Objective	Negative test for Process EDA MAC

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References	C.45-TRUE,C.46,C.47-FALSE,C.49,s29.7,s29.12,s29.17,s29.18,s2930.46,s2930.47
Conditions	Issuer and ICC certificate are valid. EDA performed EDA MAC is wrong Case 1: ECC based profile Case 2: RSA based profile Case 3: ECC based profile + CID is invalid (to ensure s29 is completed before reaching CID verification) Case 5: 'Local authentication enabled' in Kernel Qualifier is NOT set Case 6: 'Local authentication enabled' in AIP is NOT set + CID is invalid (to ensure s29 is completed before reaching CID verification) Case 7: 'Local authentication failed' in TVR is set
Pass Criteria	Reader must terminate.

4M65-2926: CVM processing - 'CV Failed'	
Objective	To ensure the reader correctly performs Cardholder Verification Decision; case of 'CV Failed'. To ensure the reader correctly performs TAA after CVM processing. To ensure the reader correctly processes the CVM limit.
References	s29.25-TRUE,s29.26
Conditions	Cardholder Verification Decision is 'CV Failed' Case 1: CVM capability includes PINonline, SGN and CDCVM. Amount above the CVM limit. 'Cardholder verification was not successful' bit is set in TAConline. Case 2: CVM capability includes noCVM and CDCVM. Amount below the CVM limit. 'Cardholder verification was not successful' bit is set in TACdenial. Case 3: CVM capability includes noCVM and CDCVM. Amount equals the CVM limit. Purchase transaction.
Pass Criteria	Transaction must complete as expected.

4M65-2928: CVM processing - 'NO CVM'	
Objective	Cardholder Verification Decision is 'NO CVM'. To ensure the reader correctly processes the CVM limit.
References	s29.27-TRUE,s29.28
Conditions	Cardholder Verification Decision is set to '00'('NO CVM') 'No CVM required (Contactless)' in TRMD is set

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	case 1: Amount,Authorized is less than Reader CVM Required Limit case 2: Amount,Authorized is equal to Reader CVM Required Limit case 3: Amount,Authorized is greater than Reader CVM Required Limit
Pass Criteria	Transaction must complete as expected.

4M65-2930: CVM processing - 'SIGNATURE'	
Objective	Cardholder Verification Decision is 'SIGNATURE'.
References	s29.29-TRUE,s29.30
Conditions	Cardholder Verification Decision is set to '01'('SIGNATURE') Amount,Authorized is greater than Reader CVM Required Limit All CVMs are set in TRMD.
Pass Criteria	Transaction must complete as expected.

4M65-2932: CVM processing - 'CDCVM'	
Objective	Cardholder Verification Decision is 'CDCVM'.
References	s29.31-TRUE,s29.32
Conditions	Cardholder Verification Decision is set to '03' (CDCVM). case 1: Amount,Authorized is greater than Reader CVM Required Limit. All CVMs are set in TRMD. case 2: Amount,Authorized is less than Reader CVM Required Limit.
Pass Criteria	Transaction must complete as expected.

4M65-2934: CVM processing - 'N/A'	
Objective	Cardholder Verification Decision is 'N/A'.
References	s29.33-TRUE,s29.34
Conditions	Cardholder Verification Decision is set to '0F'('N/A') All CVMs are set in TRMD. Case 1: Amount,Authorized is greater than Reader CVM Required Limit Case 2: Amount,Authorized is less than Reader CVM Required Limit

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Pass Criteria	Transaction must complete as expected.

4M65-2941: CVM processing - 'Not Matching'	
Objective	CardholderVerificationDecision is not matching the TRMD. To ensure the reader correctly performs TAA after CVM processing.
References	s29.40-FALSE,s29.41
Conditions	case 1a: Amount,Authorized is less than Reader CVM Required Limit. CardholderVerificationDecision is '1F'. All CVMs are set in TRMD. case 1b: Amount,Authorized is greater than Reader CVM Required Limit. CardholderVerificationDecision is 'F0'. All CVMs are set in TRMD. case 2a: CardholderVerificationDecision is '10'. 'Cardholder verification was not successful' bit is set in TAConline. case 2b: CardholderVerificationDecision is '82'. 'Cardholder verification was not successful' bit is set in TACdenial. Case 3a: CardholderVerificationDecision is noCVM. TRMD does not include noCVM Case 3b: CardholderVerificationDecision is SGN. TRMD does not include SGN Case 3c: CardholderVerificationDecision is onlinePin. TRMD does not include onlinePin Case 3d: CardholderVerificationDecision is CDCVM. TRMD does not include CDCVM
Pass Criteria	Transaction must complete as expected.

4M65-2942: CVM processing - 'Online PIN'	
Objective	Cardholder Verification Decision is 'ONLINE PIN'. To ensure the reader correctly performs TAA after CVM processing.
References	s29.40-TRUE,s29.42
Conditions	Cardholder Verification Decision is set to '02'('ONLINE PIN'). 'Online PIN' in TRMD is set. case 1a: Amount,Authorized is less than Reader CVM Required Limit. All CVMs are set in TRMD. case 1b: Amount,Authorized is greater than Reader CVM Required Limit. All CVMs are set in TRMD. case 2a: 'Online CVM captured' bit is set in TAConline. case 2b: 'Online CVM captured' bit is set in TACdenial.
Pass Criteria	Transaction must complete as expected.

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4M70-2945: WriteData - positive test	
Objective	Ensure the reader correctly sends WriteData command when applicable. Ensure the reader correctly sets the P2 byte in WriteData command. Ensure the reader correctly sets WriteDataStatus when last WriteData is successful. Ensure the reader correctly updates List1 items that are already present in List2, when copying List1 to List2.
Applicability	[DE/DS]
References	s29.44-FALSE, s29.44a-FALSE, s29.45, s29.46, s30.2, s30.3, s306, s30.7, s30.8, s30.10, s30.11, s30.12, s30.13, s30.14, s30.15, s30.16
Conditions	Case 01: DataEnvelopesToWrite includes Envelope 1 (only) Case 02: DataEnvelopesToWrite includes Envelope 10 (only) Case 03: DataEnvelopesToWrite includes all envelopes, plus another one that is duplicate
Pass Criteria	Transaction must complete.

4M70-7090: WriteData - SW!=9000	
Objective	Ensure the reader correctly behaves when SW!=9000 in WriteData response.
Applicability	[DE/DS]
References	s30.9-FALSE
Conditions	Envelopes2Write includes at least 2 envelopes.
Pass Criteria	Transaction must complete.

4M70-7301: WriteData - L1RSP	
Objective	To ensure that the transaction succeeds when a time out occurs (due to a mute card) while in state Waiting for Write Data Response.
Applicability	[DE/DS]
References	s30.1,s30.15,s30.16
Conditions	Mute card at WriteData.
Pass Criteria	The reader must successfully complete the transaction.

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4M80-1005: Completion - No WriteData	
Objective	Reader does not send WriteData when Local Authentication Failed in TVR is set or Data Envelopes To Write Yet is empty. Ensure the reader correctly sets GenAC.P2 to '80' when Data Envelopes To Write Yet is not empty.
Applicability	[DE/DS]
References	s202122232425.22, s29.44-TRUE, s29.44a-TRUE
Conditions	Case 1: Data Envelopes to Write Yet is Empty and 'Local Authentication failed' in TVR is not set Reference Control Parameter in Generate AC command set to TC Case 2: Data Envelopes to Write Yet is not Empty and 'Local Authentication failed' in TVR is set Reference Control Parameter in Generate AC command set to ARQC Case 3: DataEnvelopesToWrite includes all envelopes. Local Authentication is not performed.
Pass Criteria	Reader does not send the WriteData command. Transaction must complete successfully.

4M80-1021: Completion - no restart	
Objective	To ensure the reader correctly behaves when no restart is requested.
References	s2930.2-FALSE
Conditions	Case 1: RestartIndicator is absent Case 3: RestartIndicator B1.b8-5 is '0000' (RFU) - decline Case 4: RestartIndicator B1.b8-5 is '0001' (NO RESTART) - online Case 5: RestartIndicator B1.b8-5 is '0011' (RFU) - accept
Pass Criteria	The reader must complete the transaction.

4M80-3010: Completion - Kernel decision	
Objective	To ensure the reader correctly completes the transaction depending on the CID and KernelDecision.
References	s2930.12, s2930.18
Conditions	A) Online (TC requested) A-1: CID=ARQC, KernelDecision=Accept

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	A-2: CID=ARQC, KernelDecision=Online A-3: CID=TC, KernelDecision=Online B) Online (ARQC requested) B-1: CID=ARQC, KernelDecision=Accept B-2: CID=ARQC, KernelDecision=Online C) Decline (TC requested) C-1: CID=AAC, KernelDecision=Decline C-2: CID=AAC, KernelDecision=Accept C-3: CID=AAC, KernelDecision=Online C-4: CID=TC, KernelDecision=Decline C-5: CID=ARQC, KernelDecision=Decline D) Decline (ARQC requested) D-1: CID=AAC, KernelDecision=Decline D-2: CID=AAC, KernelDecision=Accept D-3: CID=AAC, KernelDecision=Online D-5: CID=ARQC, KernelDecision=Decline E) Decline (AAC requested) E-1: CID=AAC, KernelDecision=Decline E-2: CID=AAC, KernelDecision=Accept E-3: CID=AAC, KernelDecision=Online F) Offline (TC requested-CID=TC) F-1: Signature F-2: no Signature F-3: noCVM F-4: CVM=N/A F-5: CVM=FAILED F-6: OnlinePIN
Pass Criteria	Reader must decline (C, D, E) or complete online (A, B) or complete offline (F).

4M80-3019: Completion - Decline vs TransactionType	
Objective	To ensure the reader correctly declines the transaction based on the TransactionType value. To ensure the reader correctly checks the RestartIndicator in case of decline.
References	s2930.21, s2930.23, s2930.24, s2930.31, s2930.38, s2930.40, s2930.41, s2930.25, s2930.26
Conditions	Transaction declines (see procedure). Tests are made for TransactionType='00' (Purchase) and following RestartIndicator details: Case 1a: RestartIndicator not returned Case 1b: restartRequested='1' (NO RESTART) | alternateInterface='1' (ContactMchip) | messageID = '00' (default)

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Case 1c: restartRequested='4' (RFU) | alternateInterface='2' (Mag stripe) | messageID = '21' (TRY AGAIN)
Case 1d: restartRequested='1' (NO RESTART) | alternateInterface='F' (NA) | messageID = '18' (TRY ANOTHER INTERFACE)
Case 1e: restartRequested='8' (RFU) | alternateInterface='4' (RFU) | messageID = '20' (SEE PHONE)
Case 1f: restartRequested='1' (NO RESTART) | alternateInterface='1' (ContactMchip) | messageID = '23' (RFU)

Case 1j: restartRequested='3' (OtherIntfce) | alternateInterface='1' (ContactMchip) | messageID = '00' (default)
Case 1k: restartRequested='3' (OtherIntfce) | alternateInterface='2' (Mag stripe) | messageID = '21' (TRY AGAIN)
Case 1l: restartRequested='3' (OtherIntfce) | alternateInterface='F' (NA) | messageID = '18' (TRY ANOTHER INTERFACE)
Case 1m: restartRequested='3' (OtherIntfce) | alternateInterface='4' (RFU) | messageID = '20' (SEE PHONE)
Case 1n: restartRequested='3' (OtherIntfce) | alternateInterface='8' (RFU) | messageID = '00' (default)
Case 1o: restartRequested='3' (OtherIntfce) | alternateInterface='1' (ContactMchip) | messageID = '80' (RFU)

Tests are made for TransactionType='01' (Cash) and following RestartIndicator details:
Case 2a: RestartIndicator not returned
Case 2b: restartRequested='C' (RFU) | alternateInterface='1' (ContactMchip) | messageID = '00' (default)
Case 2c: restartRequested='1' (NO RESTART) | alternateInterface='2' (Mag stripe) | messageID = '21' (TRY AGAIN)
Case 2d: restartRequested='9' (RFU) | alternateInterface='F' (NA) | messageID = '18' (TRY ANOTHER INTERFACE)
Case 2e: restartRequested='1' (NO RESTART) | alternateInterface='C' (RFU) | messageID = '20' (SEE PHONE)
Case 2f: restartRequested='7' (RFU) | alternateInterface='1' (ContactMchip) | messageID = 'FF' (RFU)

Case 2j: restartRequested='3' (OtherIntfce) | alternateInterface='1' (ContactMchip) | messageID = '00' (default)
Case 2k: restartRequested='3' (OtherIntfce) | alternateInterface='2' (Mag stripe) | messageID = '21' (TRY AGAIN)
Case 2l: restartRequested='3' (OtherIntfce) | alternateInterface='F' (NA) | messageID = '18' (TRY ANOTHER INTERFACE)
Case 2m: restartRequested='3' (OtherIntfce) | alternateInterface='7' (RFU) | messageID = '20' (SEE PHONE)
Case 2n: restartRequested='3' (OtherIntfce) | alternateInterface='3' (RFU) | messageID = '00' (default)
Case 2o: restartRequested='3' (OtherIntfce) | alternateInterface='2' (Mag stripe) | messageID = '41' (RFU)

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	Test is made for TransactionType='09' (Purchase with Cashback) Case 3: RestartIndicator not returned Test is made for TransactionType='17' (Manual cash) and following RestartIndicator details: Case 4: restartRequested='1' (NO RESTART) | alternateInterface='1' (ContactMchip) | messageID = '00' (default) Tests are made for TransactionType='20' (refund) and following RestartIndicator details: Case 5a: restartRequested='1' (NO RESTART) | alternateInterface='F' (NA) | messageID = '18' (TRY ANOTHER INTERFACE) Case 5b: restartRequested='0' (RFU) | alternateInterface='2' (Mag stripe) | messageID = '20' (SEE PHONE) Case 5c: restartRequested='3' (OtherIntfce) | alternateInterface='1' (ContactMchip) | messageID = '20' (SEE PHONE) Case 5d: restartIndicator not returned
Procedure	A) case n: card returns AAC B) case n+1: CID=ARQC, KernelDecision is turned into DECLINE in TAA#2 C) case n+2: CID=TC, KernelDecision is turned into DECLINE in TAA#2
Pass Criteria	Reader correctly declines.

4M80-4444: Completion - DataRecord and DiscretionaryData	
Objective	To ensure the reader correctly initializes OPS, UIRD1 and ErrorIndication as per KS.2. To ensure the reader correctly sets/pads the Language Preference, when needed (case of UIRD1). To ensure the reader correctly builds the DataRecord and DiscretionaryData.
References	KS.2,s1.10,s2930.1
Conditions	In the following sub-cases, transaction data are such that the DataRecord and DiscretionaryData include all possible items. Case 1: data have default length. All optional items returned. Test1 AID Following cases require changes to the variable length DOs: Reader side: - None Card side, DataRecord: - AppLabel - AppPAN - AppPrefName - DFname

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	- IAD - tr2eq - ThirdPartyData Case 1b: similar to case 1, data are returned in a different order in the RAPDUs. Case 2a: data have minimum length (or length='00' when applicable). LanguagePreference length is 2. RRP is performed. Case 2b: data have minimum length (or length='00' when applicable). LanguagePreference length is 2. RRP is NOT performed. Case 3: data have maximum length. Test2 AID. In the two following sub-cases, optional data objects are not returned. Case 4a: RRP is performed. Case 4b: RRP is NOT performed. Test2 AID. Case 5: some data items are returned empty. Case 6: Empty Tag is included in Discretionary Data Tag List
Pass Criteria	Reader must correctly fill OPS, UIRD1 and ErrorIndication. Especially: - In OPS, 'Data Record Present' is set - in ErrorIndication, 'Msg On Error' = "N/A" In UIRD1, LanguagePreference has the correct value. The DataRecord and DiscretionaryData include all possible items listed in the specification tables, with the correct values.

4M80-5000: Completion - Tag Mapping	
Objective	To ensure the reader correctly implements tag mapping.
References	A.1.112
Conditions	Tests are done with several tag lengths (1, 2, 3 bytes): - '9F8106' is mapped with a 1-byte long tag - '57' is mapped with a 2-bytes long tag - '9A' is mapped with a 3-bytes long tag - '9F24' is mapped with a 3-bytes long tag - '9F42' is mapped with a 1-byte long tag - '9F821C' is mapped with a 1-byte long tag
Pass Criteria	Reader must correctly fill the DataRecord and DiscretionaryData.

4MX8-0023: Stress tests - transaction types	
Objective	To ensure the reader performs the CVM limit checking, the CVM selection, the floor limit checking and the terminal action analysis irrespective of the TransactionType.

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References	Various
Conditions	Case 1: Amount is below the CVM limit. 'CVD = No CVM' Case 2: Amount is above the CVM limit. 'CVD = SIGNATURE' Case 3: Amount is below the CVM limit. 'CVD = N/A' Case 4: Amount is above the CVM limit. 'CVD = CDCVM'
Procedure	The cases must be run with: A) TransactionType = '00' (payment) B) TransactionType = '01' (cash) C) TransactionType = '09' (cashback) D) TransactionType = '20' (refund)
Pass Criteria	The transaction must successfully complete. Case 1: In the transaction record, CVMResults is set to '1F 00 02'. In OUT\OPS, 'CVM' indicates "NO CVM". Cases 2: In the transaction record, CVMResults is set to '1E 00 00'. In OUT\OPS, 'Receipt' indicates "YES". In OUT\OPS, 'CVM' indicates "OBTAIN SIGNATURE". Case 3: In the transaction record, CVMResults is set to '3F 00 00'. In OUT\OPS, 'CVM' indicates "N/A". Cases 4: In the transaction record, CVMResults is set to '01 00 02'. In OUT\OPS, 'Receipt' indicates "YES". In OUT\OPS, 'CVM' indicates "CONFIRMATION CODE VERIFIED".

4MX8-0030: Stress tests - RFU bits	
Objective	To ensure the reader performs the transaction correctly when RFU bits are set. To ensure the reader performs the transaction correctly when amount is big.
References	Various
Conditions	RFU/unused bits are set: Card side: - AIP - Card Capabilities Information - Card Qualifier Reader side: - Additional TermCap - Card Data Input Capability - CVM Capability – CVM Required - CVM Capability – No CVM Required - Kernel Configuration - Security Capability - Terminal Risk Management Data
Pass Criteria	The transaction must successfully complete.

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4MX8-0035: Stress tests - encrypted records	
Objective	To ensure the reader correctly decrypts the records.
References	Various
Conditions	Case 1: 1st record comes after GPO (ie: no RRP) and is encrypted with a single tag, on two bytes (ensuring the reader does not decrypt this record before having the sessions keys) Case 2: 16 records are encrypted.
Pass Criteria	The transaction must successfully complete.

4MX8-0040: Stress test - no DE/DS	
Objective	To ensure the reader not implementing DE/DS correctly discard the DE/DS items.
Applicability	NOT([DE/DS])
References	Numerous
Conditions	All possible ACT tags sent
Pass Criteria	All must go well.

4MX8-0200: Stress test - complex transactions #1	
Objective	Stress test.
Applicability	[DE/DS]
References	Numerous including: s21.14-TRUE
Conditions	Transaction is done with a big amount. Case 01: upon PTFWF, the terminal immediately returns PTFWF plus it requests READDATA and WRITEDATA Case 02: upon PTFWF, the terminal requests READDATA and WRITEDATA before returning PTFWF
Pass Criteria	All must go well.

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4MX8-0666: Stress test - complex transactions #2	
Objective	Stress test.
Applicability	[DE/DS]
References	Numerous
Conditions	trx1: - All possible ACT tags sent - 9F5D (unknown tag) returned with wrong value and wrong length - max length ReadData and WriteData - Parsing issue in (one) ReadData - All possible DET tags returned (even envelopes). - Several non-DET data returned in DETs - Send DET AFTER GenAC trx2: - ACT.Tags2Read is empty - ACT.DataEnvlp2Write is empty - DET requests Tag2Read with Envlop2, Tag2Write includes Envlp2 - ReadData returns Envlp2 with a value. PutData is sent with the Envlp2 vale as requested in Tag2Write. - 1st PutData fails
Pass Criteria	All must go well.

4MX8-0900: Maximum Data Length	
Objective	To ensure the reader correctly manage TAGS with maximum length.
References	s20.14, s20.15-TRUE, s20.18-TRUE
Conditions	Case 1: CDOL1 : 176 Bytes Length. Case 2: Default CDOL1 : 250 Bytes Length. Case 3: Cards returns '6F' max length (250 bytes), 'A5' max length (240 bytes) and 'BF0C' max length (220 bytes). Case 4: PDOL: 216 Bytes Length Case 5: AFL: 204 Bytes Length
Pass Criteria	In all cases the EMV transaction must successfully complete.

4MX8-1000: Test with Real AIDs	
Objective	To ensure the reader performs the transaction correctly when real AIDs are used.

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References	KS.1
Conditions	Subcases: Case1: Mastercard AID Case2: Maestro AID Case3: AMEX1 AID Case4: AMEX2 AID Case5: VISA1 AID Case6: VISA2 AID Case7: JCB AID Case8: DFS AID Case9: Cirrus AID
Pass Criteria	The transaction must successfully complete.

4MX8-1001: Discretionary Data	
Objective	To ensure the reader performs the transaction correctly when real AIDs are used.
References	KS.1
Conditions	Subcases: Case1: - The following proprietary Tags are present in the Data base and are part of the Discretionary Data Tag List ('9F821F'): - Tags with update condition K/RA: 9F5A, 9F7C, 9F6E, 9F7A, 9F7B, '9FFF01' through '9FFF0F' (15 tags) and 'DF01' through 'DF7F (127 tags). These Tags are returned in Read Record responses. - Tags with update condition K/ACT/DET: Tag '9F61' is present in the data base with length of 2 Bytes. -'9F8221' (Tag Mapping List) is '9F8106 9F60' - Test Performed with VISA1 AID Case2: Same as Case 1 but Tag '9F61' is 5 Bytes. Case3: Same as Case 1 but Tag '9F61' is 7 Bytes. Case4: Same as Case 1 but Tag '9F61' is 9 Bytes. Case5: Same as Case 1 but Tag '9F61' is 11 Bytes. Case6: Same as Case 1 but Tag '9F61' is 16 Bytes.
Pass Criteria	The transaction must successfully complete.

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4 System Test Cases

4S00-0248: Certificates processing - PAN (ICC RSA)	
Objective	To ensure the Reader correctly processes the PAN in the RSA ICC certificate.
References	C.34.8 (ie: Book2, section 6.4, bullet 8)
Conditions	Default Transaction Data Case 4: PAN is 3 digits long
Pass Criteria	The transaction must successfully complete .
Note	**This test is derivated from 4M60-0248**

4S00-0900: Maximum Data Length	
Objective	To ensure the reader correctly manage TAGS with maximum length.
References	s20.14, s20.15-TRUE, s20.18-TRUE
Conditions	Default Transaction Data Case 1: CDOL1 : 176 Bytes Length. Case 2: Default CDOL1 : 250 Bytes Length. Case 3: Cards returns '6F' max length (250 bytes), 'A5' max length (240 bytes) and 'BF0C' max length (220 bytes). Case 4: PDOL: 216 Bytes Length Case 5: AFL: 204 Bytes Length
Pass Criteria	In all cases the EMV transaction must successfully complete.
Note	**This test is derivated from 4MX8-0900**

4S00-1001: Discretionary Data	
Objective	To ensure the reader performs the transaction correctly when real AIDs are used.
References	KS.1

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Conditions	Default Transaction Data Case6: - The following proprietary Tags are present in the Data base and are part of the Discretionary Data Tag List ('9F821F'): - Tags with update condition K/RA: 9F5A, 9F7C, 9F6E, 9F7A, 9F7B, '9FFF01' through '9FFF0F' (15 tags) and 'DF01' through 'DF7F (127 tags). These Tags are returned in Read Record responses. - Tags with update condition K/ACT/DET: Tag '9F61' is present in the data base with length of 16 Bytes. -'9F8221' (Tag Mapping List) is '9F8106 9F60' - Test Performed with VISA1 AID
Pass Criteria	The transaction must successfully complete.
Note	**This test is derivated from 4MX8-1001**

4S00-1423: ReadRecord - Card Message Counter	
Objective	To ensure the reader accepts Card Message Counter with high values.
References	s20.23,C.18
Conditions	Default Transaction Data. 40 Encrypted Records
Pass Criteria	Transaction must successfully complete. The reader shall send a sequence of ReadRecord commands according to the AFL.
Note	**This test is derivated from 4M25-1423**

4S00-2021: ReadData - ReadData before Records	
Objective	Ensure the reader correctly sends ReadData before the records, when applicable.
Applicability	[DE/DS]
References	s2021.1, s2021.2-FALSE, s2021.3
Conditions	Default Transaction Data plus Tags To Read = '9F8112' Case 04: RRP, returned envelope is max length
Pass Criteria	Reader must send a ReadData prior to any Record.

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Note	**This test is derivated from 4M36-2021**

4S00-4444: Completion - DataRecord and DiscretionaryData	
Objective	To ensure the reader correctly initializes OPS, UIRD1 and ErrorIndication as per KS.2. To ensure the reader correctly sets/pads the Language Preference, when needed (case of UIRD1). To ensure the reader correctly builds the DataRecord and DiscretionaryData.
References	KS.2,s1.10,s2930.1
Conditions	In the following sub-cases, transaction data are such that the DataRecord and DiscretionaryData include all possible items. Deafault Transaction Data with following updates: - Amount, Authorised='000000550000' - Account Type='30' - Amount, Other (Numeric)='000000000199' Following cases require changes to the variable length DOs: Reader side: - None Card side, DataRecord: - AppLabel - AppPAN - AppPrefName - DFname - IAD - tr2eq - ThirdPartyData Case 3: data have maximum length. Test2 AID.
Pass Criteria	Reader must correctly fill OPS, UIRD1 and ErrorIndication. Especially: - In OPS, 'Data Record Present' is set - in ErrorIndication, 'Msg On Error' = "N/A" In UIRD1, LanguagePreference has the correct value. The DataRecord and DiscretionaryData include all possible items listed in the specification tables, with the correct values.
Note	**This test is derivated from 4M80-4444**

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A. List of sub-cases

Sub-Case ID	Configuration	Applicability Conditions
4M10-0010 | | 01	C8_ExtraTags	
4M10-0011 | | 01	C8_prof00	
4M10-0012 | | 01	C8_prop	
4M10-0012 | | 02	C8_prop	
4M10-0070 | | 01	C8_prof00	
4M10-0071 | | 01	C8_prof00	
4M10-0071 | | 02	C8_prof00	
4M10-0071 | | 03	C8_prof00	
4M10-0071 | | 04	C8_prof00	
4M10-0071 | | 05	C8_prof00	
4M10-0071 | | 06	C8_prof00	
4M10-0071 | | 07	C8_prof00	
4M10-0072 | | 01	C8_prof00	
4M10-0072 | | 02	C8_prof00	
4M10-0072 | | 03	C8_prof00	
4M10-0073 | | 01	C8_prof00	
4M10-0073 | | 02a	C8_prof00	
4M10-0073 | | 02b	C8_prof00	
4M10-0073 | | 03	C8_prof00	
4M10-0073 | | 04	C8_prop	
4M10-0073 | | 05	C8_prop	
4M10-0073 | | 06	C8_prof00	
4M10-0073 | | 07	C8_prop	
4M10-0073 | | 08	C8_prop	
4M10-0073 | | 09	C8_prof00	
4M10-0073 | | 10	C8_prof00	
4M10-0073 | | 11	C8_prof00	
4M10-0073 | | 12	C8_prop	
4M10-0073 | | 13	C8_prop	
4M10-0073 | | 14	C8_prop	
4M10-0073 | | 15	C8_prop	
4M10-0073 | | 16	C8_prop	
4M10-0073 | | 17	C8_prof00	
4M10-0073 | | 18	C8_prof00	
4M10-0073 | | 19	C8_prof00	
4M10-0073 | | 20	C8_prof00	
4M10-0073 | | 21	C8_prop	
4M10-0073 | | 22	C8_prop	
4M10-0073 | | 23	C8_prop	

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Sub-Case ID	Configuration	Applicability Conditions
4M10-0073 | | 24	C8_prop	
4M10-0073 | | 25	C8_prop	
4M10-0073 | | 26	C8_prop	
4M10-0108 | | 01	C8_prof00	
4M10-0108 | | 02	C8_prof00	
4M10-0108 | | 03	C8_prof00	
4M10-0108 | | 04	C8_prof00	
4M10-0112 | | 01a	C8_prof00	not [DE/DS]
4M10-0112 | | 01b	C8_T2R_ASRPD_PTFWFempty_Mask	[DE/DS]
4M10-0112 | | 02	C8_prop	
4M10-0112 | | 03	C8_prof00_unattendedTerminal	
4M10-0112 | | 05	C8_prof00	
4M10-0112 | | 06	C8_prop	
4M10-0112 | | 07	C8_prof00	
4M10-0112 | | 08	C8_prof00	
4M10-0112 | | 09	C8_prop	
4M10-0113 | | 01	C8_prof00	not [DE/DS]
4M10-0113 | | 02	C8_prof00	[DE/DS]
4M10-0115 | | 01	TAA2_decline_SecurityCapability	
4M10-0130 | | 01	C8_FieldOff4sec	
4M10-0130 | | 02	C8_FieldOff4sec	
4M10-0130 | | 03	C8_FieldOff4sec	
4M10-0130 | | 04	C8_FieldOff4sec	
4M10-0130 | | 05	C8_FieldOff4sec	
4M10-0130 | | 06	C8_FieldOff4sec	
4M10-0130 | | 07	C8_FieldOff4sec	
4M10-0130 | | 08	C8_FieldOff4sec	
4M10-0130 | | 09	C8_FieldOff4sec	
4M10-2019 | | 01	C8_prof00	
4M10-2019 | | 02	C8_prof00_RSACertDisabled	
4M10-2019 | | 03	C8_prof00	
4M10-9017 | | 01	C8_MandatoryConfData_Missing	
4M10-9017 | | 02	C8_MandatoryConfData_Empty	
4M10-9017 | | 03	C8_MandatoryConfData_wrongLength	
4M10-9019 | | 01	C8_prof00	
4M10-9019 | | 02	C8_prof00_RSACertDisabled	
4M10-9019 | | 03	C8_prof00	
4M10-9019 | | 04	C8_prof00	
4M10-9081 | | 01	C8_prof00	
4M10-9081 | | 02	C8_prof00	
4M10-9081 | | 03	C8_prof00	
4M10-9081 | | 04	C8_prof00	
4M10-9081 | | 05a	C8_prop	

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Sub-Case ID	Configuration	Applicability Conditions
4M10-9081 | | 05b	C8_prop	
4M10-9081 | | 06	C8_prof00	
4M10-9081 | | 08	C8_prof00	
4M10-9081 | | 09	C8_prof00	
4M10-9081 | | 10	C8_prof00	
4M10-9081 | | 11	C8_prof00	
4M10-9081 | | 16	C8_prof00	
4M10-9081 | | 17	C8_prof00	
4M10-9081 | | 18	C8_prof00	
4M10-9081 | | 19	C8_prof00	
4M10-9081 | | 20	C8_prof00	
4M10-9081 | | 21	C8_prof00	
4M10-9081 | | 22	C8_prof00	
4M10-9081 | | 23	C8_prof00	
4M10-9081 | | 24	C8_prof00	
4M10-9110 | | 01	C8_prof00	
4M10-9110 | | 02a	C8_prof00	
4M10-9110 | | 02b	C8_prof00	
4M10-9110 | | 02c	C8_prof00	
4M10-9110 | | 03	C8_prof00	
4M20-0310 | | 01	C8_prof00	
4M20-0310 | | 02	C8_prof00	
4M20-0310 | | 03	C8_prof00	
4M20-0310 | | 04	C8_ExtraTags	
4M20-0310 | | 05	C8_ExtraTags	
4M20-0310 | | 06	C8_ExtraTags	
4M20-0340 | | 01	C8_prof00	
4M20-0355 | | 01	C8_prof00	
4M20-0405 | | 01	C8_prof00	
4M20-0410 | | 01	C8_prof00	
4M20-0410 | | 02	C8_prop	
4M20-0412 | | 01	C8_prof00	
4M20-0412 | | 02	C8_prop	
4M20-0415 | | 01	C8_prof00	
4M20-0415 | | 02	C8_prop	
4M20-0420 | | 01	C8_prof00	
4M20-0420 | | 02	C8_prop	
4M20-0445 | | 01	C8_prof00	
4M20-0515 | | 01	C8_prof00	
4M20-0515 | | 02	C8_prof00	
4M20-0515 | | 03	C8_prop	
4M25-0122 | | 01	C8_prof00	
4M25-0122 | | 02	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M25-0122 | | 03	C8_prof00	
4M25-0122 | | 04	C8_prof00	
4M25-0123 | | 01	C8_prof00	
4M25-0123 | | 02	C8_prof00	
4M25-0123 | | 03	C8_prof00	
4M25-0124 | | 01	C8_prof00	
4M25-0124 | | 02	C8_prof00	
4M25-0124 | | 03	C8_prop	
4M25-0124 | | 04	C8_prof00	
4M25-0124 | | 05	C8_prof00	
4M25-0124 | | 06	C8_prof00	
4M25-0124 | | 07	C8_prof00	
4M25-0124 | | 08	C8_prof00	
4M25-0124 | | 09	C8_prof00	
4M25-0124 | | 10	C8_prof00	
4M25-0124 | | 11	C8_prof00	
4M25-0124 | | 12	C8_prop	
4M25-0124 | | 13	C8_prop	
4M25-0124 | | 14	C8_prop	
4M25-0124 | | 15	C8_prop	
4M25-0124 | | 16	C8_prop	
4M25-0124 | | 17	C8_prof00	
4M25-0124 | | 18	C8_prof00	
4M25-0124 | | 19	C8_prop	
4M25-0124 | | 20	C8_prop	
4M25-0124 | | 21	C8_prop	
4M25-0124 | | 22	C8_prop	
4M25-0124 | | 23	C8_prop	
4M25-0124 | | 24	C8_prop	
4M25-0124 | | 25	C8_prop	
4M25-0124 | | 26	C8_prop	
4M25-0124 | | 27	C8_prop	
4M25-0124 | | 28	C8_prop	
4M25-0124 | | 29	C8_prop	
4M25-0124 | | 30	C8_prop	
4M25-0130 | | 01	C8_prof00	
4M25-0130 | | 02	C8_prof00	
4M25-0224 | | 01	C8_prof00	[DE/DS]
4M25-0224 | | 02	C8_prof00	[DE/DS]
4M25-0224 | | 03	C8_prof00	[DE/DS]
4M25-0224 | | 04	C8_prof00	[DE/DS]
4M25-0224 | | 05	C8_prof00	[DE/DS]
4M25-2018 | | 01	C8_defaultCDOL1absent	

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Sub-Case ID	Configuration	Applicability Conditions
4M25-2090 | | 01	C8_prof00	[DE/DS]
4M25-9001 | | 01	C8_prof00	
4M25-9015 | | 01	C8_prof00	
4M25-9015 | | 02	C8_prof00	
4M25-9015 | | 03	C8_prof00	
4M25-9015 | | 05a	C8_prop	
4M25-9015 | | 05b	C8_prop	
4M25-9015 | | 06	C8_prof00	
4M25-9015 | | 07	C8_prof00	
4M25-9015 | | 08	C8_prof00	
4M25-9015 | | 09a	C8_prof00	
4M25-9015 | | 09b	C8_prof00	
4M25-9015 | | 10	C8_prof00	
4M25-9015 | | 11	C8_prof00	
4M25-9015 | | 12	C8_prof00	
4M25-9015 | | 13	C8_prof00	
4M25-9015 | | 20	C8_prof00	
4M25-9101 | | 01	C8_prof00	
4M25-9101 | | 02	C8_prof00	
4M25-9101 | | 03	C8_prof00	
4M25-9101 | | 04	C8_prof00	
4M25-9101 | | 05	C8_prof00	
4M25-9101 | | 06	C8_prof00	
4M25-9101 | | 07	C8_prof00	
4M25-9101 | | 08	C8_prof00	
4M25-9101 | | 09	C8_prof00	
4M25-9101 | | 10	C8_prof00	
4M25-9101 | | 11	C8_prof00	
4M25-9101 | | 12	C8_prof00	
4M25-9101 | | 13	C8_prof00	
4M25-9101 | | 14	C8_prof00	
4M25-9102 | | 01	C8_prof00	
4M25-9102 | | 02	C8_prof00	
4M25-9102 | | 03	C8_prof00	
4M25-9102 | | 04	C8_prof00	
4M25-9102 | | 05	C8_prof00	
4M25-9102 | | 06	C8_prof00	
4M25-9102 | | 07	C8_prof00	
4M25-9102 | | 08	C8_prof00	
4M25-9102 | | 09	C8_prof00	
4M25-9102 | | 10	C8_prof00	
4M25-9102 | | 11	C8_prof00	
4M25-9102 | | 12	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M25-9102 | | 13	C8_prof00	
4M25-9102 | | 14	C8_prof00	
4M25-9130 | | 01	C8_prof00	
4M25-9130 | | 02	C8_prof00	
4M25-9130 | | 03	C8_prof00	
4M25-9130 | | 04	C8_prof00	
4M25-9130 | | 05	C8_prof00	
4M25-9150 | | 01	C8_prof00	
4M25-9150 | | 02	C8_prof00	
4M25-9150 | | 03	C8_prof00	
4M25-9150 | | 04	C8_prof00	
4M25-9150 | | 05	C8_prof00	
4M25-9150 | | 06	C8_defaultCDOL1absent	
4M25-9150 | | 07	C8_defaultCDOL1absent	
4M25-9226 | | 01	C8_prof00	
4M25-9226 | | 02	C8_prof00	[DE/DS]
4M25-9226 | | 03	C8_prof00	[DE/DS]
4M25-9226 | | 04	C8_prof00	
4M25-9226 | | 05	C8_prof00	
4M25-9227 | | 01	C8_prof00	
4M25-9227 | | 02	C8_prof00	[DE/DS]
4M25-9227 | | 03	C8_prof00	[DE/DS]
4M25-9227 | | 04	C8_prof00	
4M25-9227 | | 05	C8_prof00	
4M30-0022 | | 01	C8_prof00	
4M30-0022 | | 02	C8_prof00_noRRP	
4M30-0180 | | 01	C8_RRP	
4M30-0180 | | 02	C8_RRP	
4M30-0180 | | 03	C8_prof00	
4M30-0180 | | 04	C8_RRP	
4M30-0180 | | 05	C8_RRP	
4M30-0180 | | 06	C8_RRP2	
4M30-2000 | | 01	C8_prof00	
4M30-2000 | | 02	C8_prof00	
4M30-2000 | | 03	C8_prof00	
4M30-2000 | | 04	C8_RRP2	
4M30-2000 | | 05	C8_RRP2	
4M30-2000 | | 06	C8_RRP2	
4M30-2110 | | 01	C8_RRP	
4M30-2110 | | 02	C8_prof00	
4M30-2110 | | 03	C8_RRP	
4M30-9010 | A |	C8_prof00	
4M30-9010 | B |	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M30-9120 | A | 01	C8_prof00	
4M30-9120 | A | 02	C8_prof00	
4M30-9120 | A | 03	C8_prof00	
4M30-9120 | B | 04	C8_prof00	
4M30-9120 | B | 05	C8_prof00	
4M30-9120 | B | 06	C8_prof00	
4M30-9120 | C | 07	C8_prof00	
4M30-9120 | C | 08	C8_prof00	
4M30-9120 | C | 09	C8_prof00	
4M30-9170 | | 01a	C8_prof00	
4M30-9170 | | 01b	C8_prof00	
4M30-9170 | | 02a	C8_prof00	
4M30-9170 | | 02b	C8_prof00	
4M30-9170 | | 03a	C8_prof00	
4M30-9170 | | 03b	C8_prof00	
4M30-9170 | | 03c	C8_prof00	
4M30-9180 | A | 01	C8_RRP	
4M30-9180 | B | 02	C8_RRP	
4M30-9180 | B | 03	C8_prof00	
4M35-0241 | A | 02	C8_prof00	
4M35-0241 | A | 03	C8_prof00	
4M35-0241 | B | 04	C8_prof00	
4M35-0241 | B | 05	C8_prof00	
4M35-0241 | B | 06	C8_prof00	
4M35-0241 | C | 02	C8_DiscretionaryDataTagList_9F24	
4M35-0241 | C | 03	C8_prof00	
4M35-0241 | D | 04	C8_prof00	
4M35-0241 | D | 06	C8_prof00	
4M35-0241 | E | 04	C8_prof00	[DE/DS]
4M35-0242 | | 01	C8_prof00	
4M35-0242 | | 02	C8_prof00	
4M35-0242 | | 03	C8_prof00	
4M35-0242 | | 04	C8_prof00	
4M35-0243 | A | 04	C8_prof00_Discretionary	
4M35-0243 | B | 01	C8_prof00_Discretionary	
4M35-0243 | B | 02	C8_prof00_Discretionary	
4M35-0243 | B | 03	C8_prof00_Discretionary	
4M35-0243 | B | 04	C8_prof00_Discretionary	
4M35-0243 | B | 05	C8_prof00_Discretionary	
4M35-0243 | C | 03	C8_prof00_Discretionary	
4M35-0243 | C | 05	C8_prof00_Discretionary	
4M35-0244 | A | 01	C8_prop_Discretionary	
4M35-0244 | A | 02	C8_prop_Discretionary	

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Sub-Case ID	Configuration	Applicability Conditions
4M35-0244 | A | 04	C8_prop_Discretionary	
4M35-0244 | A | 05	C8_prop_Discretionary	
4M35-0244 | A | 10	C8_prop_Discretionary	
4M35-0244 | A | 18	C8_prop_Discretionary	
4M35-0244 | A | 13	C8_prop_Discretionary	
4M35-0244 | A | 16	C8_prop_Discretionary	
4M35-0244 | A | 17	C8_prop_Discretionary	
4M35-0244 | B | 03	C8_prop_Discretionary	
4M35-0244 | B | 04	C8_prop_Discretionary	
4M35-0244 | B | 04b	C8_prop_Discretionary	[DE/DS]
4M35-0244 | B | 05	C8_prop_Discretionary	
4M35-0244 | B | 06	C8_prop_Discretionary	
4M35-0244 | B | 07	C8_prop_Discretionary	
4M35-0244 | B | 08	C8_prop_Discretionary	
4M35-0244 | B | 09	C8_prop_Discretionary	
4M35-0244 | B | 11	C8_prop_Discretionary	
4M35-0244 | B | 12	C8_prop_Discretionary	
4M35-0244 | B | 14	C8_prop_Discretionary	
4M35-0244 | B | 15	C8_prop_Discretionary	
4M35-0244 | B | 19	C8_prop_Discretionary	[DE/DS]
4M35-0244 | C | 04	C8_prop_Discretionary	
4M35-0244 | C | 04b	C8_prop_Discretionary	[DE/DS]
4M35-1421 | | 01	C8_prof00	
4M35-1422 | | 01	C8_prof00	
4M35-1423 | | 01	C8_prof00	
4M35-2021 | | 01	C8_prof00	
4M35-2021 | | 02	C8_prof00	
4M35-2021 | | 03	C8_prof00	
4M35-2218 | A | 01	C8_NoEda	
4M35-2218 | A | 02	C8_prof00	
4M35-2218 | A | 03	C8_prof00	
4M35-2218 | A | 04	C8_prof00	
4M35-2218 | A | 05	C8_prof00	
4M35-2218 | A | 06	C8_prof00	
4M35-2218 | A | 07	C8_prof00	
4M35-2218 | A | 08	C8_prof00	
4M35-2218 | B | 01	C8_NoEda	
4M35-2218 | B | 02	C8_prof00	
4M35-2218 | B | 05	C8_prof00	
4M35-2218 | B | 06	C8_prof00	
4M35-2218 | B | 07	C8_prof00	
4M35-2218 | B | 08	C8_prof00	
4M35-9100 | | 01	C8_prof00	

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4M35-9100 | | 02	C8_prof00	
4M35-9100 | | 03	C8_prof00	
4M35-9100 | | 04	C8_prof00	
4M35-9100 | | 05	C8_prof00	
4M35-9100 | | 06	C8_prof00	
4M35-9100 | | 07	C8_prof00	
4M35-9100 | | 08	C8_prof00	
4M35-9100 | | 09	C8_prof00	
4M35-9100 | | 10	C8_prof00	
4M35-9100 | | 11	C8_prof00	
4M35-9100 | | 12	C8_prof00	
4M35-9100 | | 13	C8_prof00	
4M35-9100 | | 14	C8_prof00	
4M35-9100 | | 15	C8_prof00	
4M35-9100 | | 16	C8_prof00	
4M35-9100 | | 17	C8_prof00	
4M35-9221 | | 01	C8_prof00	
4M35-9228 | | 01	C8_prof00	
4M35-9228 | | 02	C8_prof00	
4M35-9228 | | 03	C8_prof00	
4M35-9228 | | 04	C8_prof00	
4M35-9228 | | 05	C8_prof00	
4M35-9228 | | 06	C8_prof00	
4M35-9260 | A | 03	C8_prof00	
4M35-9260 | A | 05	C8_prof00	
4M35-9260 | A | 09	C8_prof00	
4M35-9260 | A | 11	C8_prof00	
4M35-9260 | A | 17	C8_prop	
4M35-9260 | A | 19	C8_prof00	
4M35-9260 | A | 21	C8_prof00	
4M35-9260 | B | 02	C8_prof00	
4M35-9260 | B | 04	C8_prof00	
4M35-9260 | B | 06	C8_prop	
4M35-9260 | B | 08	C8_prop	
4M35-9260 | B | 10	C8_prof00	
4M35-9260 | B | 12	C8_prof00	
4M35-9260 | B | 14	C8_prof00	
4M35-9260 | B | 16	C8_prof00	
4M35-9260 | C | 02	C8_prof00	
4M35-9260 | C | 04	C8_prof00	
4M35-9260 | C | 06	C8_prop	
4M35-9260 | C | 08	C8_prop	
4M35-9260 | C | 10	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M35-9260 | C | 12	C8_prof00	
4M35-9260 | C | 14	C8_prof00	
4M35-9260 | C | 16	C8_prof00	
4M35-9260 | C | 22	C8_prop	
4M35-9260 | D | 05	C8_prof00	
4M35-9260 | D | 06	C8_prop	
4M35-9260 | D | 11	C8_prof00	
4M35-9260 | E | 04	C8_prof00	[DE/DS]
4M35-9260 | E | 05	C8_prof00	[DE/DS]
4M35-9260 | E | 12	C8_prof00	[DE/DS]
4M35-9329 | | 01	C8_prof00	[DE/DS]
4M36-2021 | | 01	C8_prof00	[DE/DS]
4M36-2021 | | 02	C8_prof00	[DE/DS]
4M36-2021 | | 03	C8_prof00	[DE/DS]
4M36-2021 | | 04	C8_prof00	[DE/DS]
4M36-2021 | | 05	C8_prof00	[DE/DS]
4M36-2025 | | 01	C8_prof00	[DE/DS]
4M36-2238 | | 01	C8_prof00	[DE/DS]
4M36-2238 | | 02	C8_prof00	[DE/DS]
4M36-2380 | | 01	C8_prof00	[DE/DS]
4M36-7080 | | 01	C8_prof00	[DE/DS]
4M36-7080 | | 02	C8_prof00	[DE/DS]
4M36-7080 | | 03	C8_prof00	[DE/DS]
4M36-7250 | | 01	C8_prof00	[DE/DS]
4M36-7250 | | 02	C8_prof00	[DE/DS]
4M36-7250 | | 03	C8_prof00	[DE/DS]
4M36-7251 | A | 01	C8_prof00	[DE/DS]
4M36-7251 | A | 02	C8_prof00	[DE/DS]
4M36-7251 | A | 03	C8_prof00	[DE/DS]
4M36-7251 | B | 02	C8_prof00	[DE/DS]
4M36-7251 | B | 03	C8_prof00	[DE/DS]
4M36-7253 | | 01	C8_prof00	[DE/DS]
4M36-7253 | | 02	C8_prof00	[DE/DS]
4M36-7253 | | 03	C8_prof00	[DE/DS]
4M36-7253 | | 04	C8_prof00	[DE/DS]
4M36-7253 | | 05	C8_prof00	[DE/DS]
4M36-7253 | | 06	C8_prof00	[DE/DS]
4M36-9231 | | 01	C8_prof00	[DE/DS]
4M36-9231 | | 02	C8_prof00	[DE/DS]
4M38-1000 | | 01	C8_prof00	[DE/DS]
4M38-1000 | | 01b	C8_prof00	[DE/DS]
4M38-1000 | | 01c	C8_prof00	[DE/DS]
4M38-1000 | | 02	C8_prof00	[DE/DS]

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Sub-Case ID	Configuration	Applicability Conditions
4M38-1000 | | 02b	C8_prof00	[DE/DS]
4M38-1000 | | 03	C8_prof00	[DE/DS]
4M38-1000 | | 04	C8_prof00	[DE/DS]
4M38-2424 | | 01	C8_prof00	[DE/DS]
4M38-2424 | | 02	C8_prof00	[DE/DS]
4M38-2431 | | 01	C8_prof00	[DE/DS]
4M38-2431 | | 02	C8_prof00	[DE/DS]
4M38-4000 | | 01	C8_prof00	[DE/DS]
4M38-4000 | | 02	C8_prof00	[DE/DS]
4M38-9241 | | 01a	C8_prof00	[DE/DS]
4M38-9241 | | 01b	C8_prof00	[DE/DS]
4M38-9241 | | 02a	C8_prof00	[DE/DS]
4M38-9241 | | 02b	C8_prof00	[DE/DS]
4M40-1140 | | 01	C8_prof00	
4M40-1140 | | 02	C8_prof00	
4M40-1140 | | 03	C8_prof00	
4M40-4171 | | 01	C8_prof00	
4M40-4171 | | 02	C8_prof00	
4M40-4171 | | 03	C8_prof00	
4M40-4171 | | 04	C8_prof00	
4M40-4171 | | 05	C8_prof00	
4M40-9140 | | 01	C8_prof00	
4M40-9140 | | 02	C8_prof00	
4M40-9140 | | 03	C8_prof00	
4M40-9140 | | 04	C8_prof00	
4M40-9140 | | 05	C8_prof00	[DE/DS]
4M50-0020 | A | 01	TAA_TACdenial_OnlineOnly	
4M50-0020 | A | 02	TAA_TACdenial_OnlineOnly	
4M50-0020 | A | 03	TAA_TACdenial_OnlineOnly	
4M50-0020 | A | 04	TAA_TACdenial_OnlineOnly	
4M50-0020 | A | 05	TAA_TACdenial_OnlineOnly	
4M50-0020 | B | 01	TAA_TACdenial_OnlineOnly	
4M50-0020 | B | 02	TAA_TACdenial_OnlineOnly	
4M50-0020 | B | 03	TAA_TACdenial_OnlineOnly	
4M50-0020 | B | 04	TAA_TACdenial_OnlineOnly	
4M50-0020 | B | 05	TAA_TACdenial_OnlineOnly	
4M50-0030 | A | 01	TAA_TACdenial_OfflineOnly	
4M50-0030 | A | 02	TAA_TACdenial_OfflineOnly	
4M50-0030 | A | 03	TAA_TACdenial_OfflineOnly	
4M50-0030 | A | 04	TAA_TACdenial_OfflineOnly	
4M50-0030 | A | 05	TAA_TACdenial_OfflineOnly	
4M50-0030 | B | 01	TAA_TACdenial_OfflineOnly	
4M50-0030 | B | 02	TAA_TACdenial_OfflineOnly	

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Sub-Case ID	Configuration	Applicability Conditions
4M50-0030 | B | 03	TAA_TACdenial_OfflineOnly	
4M50-0030 | B | 04	TAA_TACdenial_OfflineOnly	
4M50-0030 | B | 05	TAA_TACdenial_OfflineOnly	
4M50-0040 | A | 01	TAA_TACdenial_OfflineOnline	
4M50-0040 | A | 02	TAA_TACdenial_OfflineOnline	
4M50-0040 | A | 03	TAA_TACdenial_OfflineOnline	
4M50-0040 | A | 04	TAA_TACdenial_OfflineOnline	
4M50-0040 | A | 05	TAA_TACdenial_OfflineOnline	
4M50-0040 | B | 01	TAA_TACdenial_OfflineOnline	
4M50-0040 | B | 02	TAA_TACdenial_OfflineOnline	
4M50-0040 | B | 03	TAA_TACdenial_OfflineOnline	
4M50-0040 | B | 04	TAA_TACdenial_OfflineOnline	
4M50-0040 | B | 05	TAA_TACdenial_OfflineOnline	
4M50-0061 | A | 01	TAA_TACdenial_OfflineOnline	
4M50-0061 | A | 02	TAA_TACdenial_OfflineOnline	
4M50-0061 | A | 03	TAA_TACdenial_OfflineOnline	
4M50-0061 | A | 04	TAA_TACdenial_OfflineOnline	
4M50-0061 | A | 05	TAA_TACdenial_OfflineOnline	
4M50-0061 | B | 01	TAA_TACdenial_OfflineOnline	
4M50-0061 | B | 02	TAA_TACdenial_OfflineOnline	
4M50-0061 | B | 03	TAA_TACdenial_OfflineOnline	
4M50-0061 | B | 04	TAA_TACdenial_OfflineOnline	
4M50-0061 | B | 05	TAA_TACdenial_OfflineOnline	
4M50-0062 | A | 01	TAA_TAConline_OfflineOnline	
4M50-0062 | A | 02	TAA_TAConline_OfflineOnline	
4M50-0062 | A | 03	TAA_TAConline_OfflineOnline	
4M50-0062 | A | 04	TAA_TAConline_OfflineOnline	
4M50-0062 | A | 05	TAA_TAConline_OfflineOnline	
4M50-0062 | A | 06	TAA_TAConline_OfflineOnline	
4M50-0062 | A | 07	TAA_TAConline_OfflineOnline	
4M50-0062 | A | 08	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 03	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 04	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 05	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 08	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 10	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 11	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 12	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 13	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 14	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 16	TAA_TAConline_OfflineOnline	
4M50-0062 | B | 18	TAA_TAConline_OfflineOnline	
4M50-8010 | A | 01	TAA_TACdenial_OnlineOnly_decline	

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Sub-Case ID	Configuration	Applicability Conditions
4M50-8010 | A | 02	TAA_TACdenial_OnlineOnly_decline	
4M50-8010 | A | 03	TAA_TACdenial_OfflineOnly	
4M50-8010 | A | 04	TAA_TACdenial_OfflineOnly	
4M50-8010 | A | 05	TAA_TACdenial_OfflineOnline	
4M50-8010 | A | 06	TAA_TACdenial_OfflineOnline	
4M50-8010 | B | 03	TAA_TACdenial_OfflineOnly	
4M50-8010 | B | 04	TAA_TACdenial_OfflineOnly	
4M50-8010 | B | 05	TAA_TACdenial_OfflineOnline	
4M50-8010 | B | 06	TAA_TACdenial_OfflineOnline	
4M50-8010 | B | 10	TAA_TACdenial_OfflineOnline	
4M50-8010 | B | 11	TAA_TACdenial_OfflineOnline	
4M50-8010 | B | 12	TAA_TACdenial_OfflineOnline	
4M50-8010 | B | 13	TAA_TACdenial_OfflineOnline	
4M50-8010 | B | 14	TAA_TACdenial_OfflineOnline	
4M50-8010 | B | 16	TAA_TACdenial_OfflineOnline	
4M50-8010 | B | 18	TAA_TACdenial_OfflineOnline	
4M50-8050 | | 01	TAA_TACdenial_OfflineOnly	
4M50-8050 | | 02	TAA_TACdenial_OfflineOnly	
4M50-8050 | | 03	TAA_TACdenial_OfflineOnly	
4M50-8050 | | 04	TAA_TACdenial_OfflineOnly	
4M50-8050 | | 05	TAA_TACdenial_OfflineOnly	
4M55-0110 | | 01	C8_prof00	
4M55-0110 | | 02	C8_prof00	
4M55-0110 | | 03	C8_prof00	
4M55-1202 | | 02	C8_prof00	
4M55-1202 | | 03	C8_prof00	
4M55-1202 | | 04	C8_prof00	
4M55-1203 | | 01	C8_prof00	
4M55-1203 | | 02	C8_prof00	
4M55-1203 | | 03	C8_prof00	
4M55-1210 | | 01	C8_prof00	
4M55-1210 | | 02	C8_prof00	
4M55-1210 | | 03	C8_prof00	
4M55-1210 | | 05	C8_prof00	
4M55-1210 | | 06	C8_prof00	
4M55-1210 | | 07	C8_prof00	
4M55-1210 | | 09	C8_prop	
4M55-1210 | | 09b	C8_prop	[DE/DS]
4M55-1210 | | 10	C8_prof00	
4M55-1210 | | 11	C8_DiscretionaryDataTagList_emptyTag	
4M55-1210 | | 12	C8_DiscretionaryDataTagList_various	
4M55-1210 | | 13	C8_prof00	
4M55-1210 | | 14	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M55-1210 | | 15	C8_prop	
4M55-9010 | | 01	C8_prof00	
4M55-9100 | A | 01	C8_prof00	
4M55-9100 | A | 02	C8_prof00	
4M55-9100 | A | 03	C8_prof00	
4M55-9100 | B | 04	C8_prof00	[DE/DS]
4M55-9100 | B | 05	C8_prof00	[DE/DS]
4M55-9100 | B | 06	C8_prof00	[DE/DS]
4M55-9100 | B | 07	C8_prof00	[DE/DS]
4M55-9100 | C | 08	C8_prof00	
4M55-9100 | C | 09	C8_prof00	
4M55-9100 | D | 10	C8_prof00	
4M55-9100 | D | 11	C8_prof00	
4M55-9100 | D | 12	C8_prof00	
4M55-9100 | D | 13	C8_prof00	
4M55-9100 | E | 14	C8_prof00	
4M55-9100 | E | 15	C8_prof00	
4M55-9100 | E | 16	C8_prof00	
4M55-9100 | E | 17	C8_prof00	
4M55-9100 | F | 18	C8_prof00	
4M55-9100 | F | 19	C8_prof00	
4M55-9120 | | 01	C8_prof00	
4M55-9120 | | 02	C8_prof00	
4M55-9120 | | 03	C8_prof00	
4M55-9120 | | 04	C8_prof00	
4M55-9140 | | 01a	C8_prof00	
4M55-9140 | | 01b	C8_prof00	
4M55-9140 | | 03	C8_prof00	
4M55-9140 | | 04	C8_prof00	
4M55-9140 | | 05a	C8_prop	
4M55-9140 | | 05b	C8_prop	
4M55-9140 | | 06	C8_prof00	
4M55-9140 | | 07a	C8_prof00	
4M55-9140 | | 07b	C8_prof00	
4M55-9140 | | 07c	C8_prof00	
4M55-9140 | | 07d	C8_prof00	
4M55-9140 | | 07f	C8_prof00	
4M55-9140 | | 07g	C8_prof00	
4M55-9140 | | 07h	C8_prof00	
4M55-9140 | | 07i	C8_prof00	
4M55-9140 | | 09a	C8_prof00	
4M55-9140 | | 09b	C8_prof00	
4M55-9140 | | 09c	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M55-9140 | | 09d	C8_prof00	
4M55-9140 | | 09e	C8_prof00	
4M55-9140 | | 10	C8_prof00	
4M55-9140 | | 11	C8_prof00	
4M55-9140 | | 13	C8_prof00	
4M55-9140 | | 14	C8_prop	
4M55-9140 | | 15	C8_prop	
4M55-9140 | | 16	C8_prop	
4M55-9140 | | 17	C8_prof00	
4M55-9140 | | 18	C8_prop	
4M55-9160 | | 01a	C8_prof00	
4M55-9160 | | 01b	C8_prof00	
4M55-9160 | | 01c	C8_prof00	
4M55-9160 | | 01d	C8_prof00	
4M55-9160 | | 01e	C8_prof00	
4M55-9160 | | 01f	C8_prof00	
4M55-9160 | | 01g	C8_prof00	
4M55-9160 | | 01h	C8_prof00	
4M55-9240 | | 01	TAA_TACdenial_OfflineOnly	
4M55-9240 | | 02	TAA_TACdenial_OfflineOnly	
4M55-9240 | | 03	C8_prof00	
4M55-9240 | | 04	C8_prof00	
4M55-9240 | | 05	C8_prof00	
4M55-9240 | | 06	C8_prof00	
4M55-9240 | | 07	C8_prof00	
4M55-9240 | | 08	C8_NoEda	
4M55-9240 | | 09	C8_NoEda	
4M55-9270 | | 01	C8_prof00	
4M55-9270 | | 02	C8_NoEda	
4M55-9270 | | 03	C8_prof00	
4M59-1140 | | 01	C8_NoEda	
4M59-1140 | | 02	C8_prof00	
4M59-1140 | | 03	C8_prof00	
4M59-1140 | | 04	C8_prof00	
4M60-0180 | | 01	C8_prof00	
4M60-0225 | | 01	C8_prof00	
4M60-0225 | | 02	C8_prof00	
4M60-0225 | | 03	C8_prof00	
4M60-0225 | | 05	C8_prof00	
4M60-0225 | | 06	C8_prof00	
4M60-0225 | | 07	C8_prof00	
4M60-0226 | A | 01	C8_prof00	
4M60-0226 | A | 02	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M60-0226 | A | 03	C8_prof00	
4M60-0226 | A | 04	C8_prof00	
4M60-0226 | B | 01	C8_prof00	
4M60-0226 | B | 02	C8_prof00	
4M60-0226 | B | 03	C8_prof00	
4M60-0226 | B | 04	C8_prof00	
4M60-0227 | | 01	C8_prof00	
4M60-0248 | | 01	C8_prof00	
4M60-0248 | | 02	C8_prof00	
4M60-0248 | | 03	C8_prof00	
4M60-0248 | | 04	C8_prof00	
4M60-0255 | | 01	C8_prof00	
4M60-0257 | | 01	C8_prof00	
4M60-0257 | | 02	C8_prof00	
4M60-0259 | | 01	C8_prof00	
4M60-0262 | | 01	C8_prof00	
4M60-0262 | | 02	C8_prof00	
4M60-0262 | | 03	C8_prof00	
4M60-0262 | | 04	C8_prof00	
4M60-1030 | A | 01	C8_prof00	
4M60-1030 | A | 02	C8_prof00	
4M60-1030 | B | 03a	C8_prof00	
4M60-1030 | B | 03b	C8_prof00	
4M60-1030 | B | 04a	C8_prof00	
4M60-1030 | B | 04b	C8_prof00	
4M60-1031 | A | 01a	C8_prof00	
4M60-1031 | A | 01b	C8_prof00	
4M60-1031 | A | 02a	C8_prof00	
4M60-1031 | A | 02b	C8_prof00	
4M60-1031 | B | 01a	C8_prof00	
4M60-1031 | B | 01b	C8_prof00	
4M60-1031 | B | 02a	C8_prof00	
4M60-1031 | B | 02b	C8_prof00	
4M60-2001 | A | 01	C8_prof00	
4M60-2001 | A | 02	C8_prof00	
4M60-2001 | A | 03	C8_prof00	
4M60-2001 | A | 04	C8_prof00	
4M60-2001 | A | 05	C8_prof00	
4M60-2001 | A | 06	C8_prof00	
4M60-2001 | A | 07	C8_prof00	
4M60-2001 | A | 08	C8_prof00	
4M60-2001 | A | 09	C8_prof00	
4M60-2001 | B | 01	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M60-2001 | B | 02	C8_prof00	
4M60-2001 | B | 03	C8_prof00	
4M60-2001 | B | 04	C8_prof00	
4M60-2001 | B | 05	C8_prof00	
4M60-2001 | B | 06	C8_prof00	
4M60-2001 | B | 07	C8_prof00	
4M60-2001 | B | 08	C8_prof00	
4M60-2001 | B | 09	C8_prof00	
4M60-2002 | A | 01	C8_prof00	
4M60-2002 | A | 02	C8_prof00	
4M60-2002 | A | 03	C8_prof00	
4M60-2002 | A | 04	C8_prof00	
4M60-2002 | A | 05	C8_prof00	
4M60-2002 | A | 06	C8_prof00	
4M60-2002 | A | 07	C8_prof00	
4M60-2002 | A | 09	C8_prof00	
4M60-2002 | A | 10	C8_prof00	
4M60-2002 | A | 11	C8_prof00	
4M60-2002 | A | 12	C8_prof00	
4M60-2002 | A | 13	C8_prof00	
4M60-2002 | B | 01	C8_prof00	
4M60-2002 | B | 02	C8_prof00	
4M60-2002 | B | 03	C8_prof00	
4M60-2002 | B | 04	C8_prof00	
4M60-2002 | B | 05	C8_prof00	
4M60-2002 | B | 06	C8_prof00	
4M60-2002 | B | 07	C8_prof00	
4M60-2002 | B | 09	C8_prof00	
4M60-2002 | B | 10	C8_prof00	
4M60-2002 | B | 11	C8_prof00	
4M60-3010 | | 01	C8_prof00	
4M60-3010 | | 02	C8_prof00	
4M60-3010 | | 03	TAA_TACdenial_OfflineOnline	
4M60-3010 | | 04	C8_prof00	
4M60-3010 | | 05	C8_prof00	
4M60-3010 | | 06	C8_prof00	
4M60-3800 | | 01	C8_prof00	
4M60-7004 | | 01a	C8_prof00	
4M60-7004 | | 02a	C8_prof00	
4M60-7004 | | 02b	C8_prof00	
4M60-7004 | | 07a	C8_prof00	
4M60-7004 | | 07b	C8_prof00	
4M60-7005 | | 01a	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M60-7005 | | 02a	C8_prof00	
4M60-7005 | | 02b	C8_prof00	
4M60-7005 | | 02d	C8_prof00	
4M60-7005 | | 06a	C8_prof00	
4M60-7005 | | 06b	C8_prof00	
4M60-7005 | | 06d	C8_prof00	
4M60-7008 | A | 01	C8_prof00	
4M60-7008 | A | 02	C8_prof00	
4M60-7008 | A | 03	C8_prof00_RSACertDisabled	
4M60-7008 | A | 04	C8_prof00	
4M60-7008 | B | 01	C8_prof00	
4M60-7008 | B | 02	C8_prof00	
4M60-7008 | B | 03	C8_prof00_RSACertDisabled	
4M60-7008 | B | 04	C8_prof00	
4M60-7011 | | 01	C8_prof00_RSACertDisabled	
4M60-7014 | | 01a	C8_prof00	
4M60-7014 | | 01b	C8_prof00	
4M60-7014 | | 02a	C8_prof00	
4M60-7014 | | 02b	C8_prof00	
4M60-7014 | | 03a	C8_prof00	
4M60-7014 | | 03b	C8_prof00	
4M60-7014 | | 04	C8_prof00	
4M60-7014 | | 05	C8_prof00	
4M60-7014 | | 05b	TAA_TAConline_OfflineOnline	
4M60-7014 | | 05c	C8_prof00	
4M60-7014 | | 05d	C8_prof00	
4M60-7014 | | 06a	C8_prof00	
4M60-7014 | | 06b	C8_prof00	
4M60-7014 | | 07	C8_prof00	
4M60-7014 | | 09a	C8_prof00	
4M60-7014 | | 09b	C8_prof00	
4M60-7014 | | 11a	C8_prof00	
4M60-7014 | | 11b	C8_prof00	
4M60-7015 | | 01a	C8_prof00	
4M60-7015 | | 01b	C8_prof00	
4M60-7015 | | 02a	C8_prof00	
4M60-7015 | | 02b	C8_prof00	
4M60-7015 | | 03a	C8_prof00	
4M60-7015 | | 03b	C8_prof00	
4M60-7015 | | 04a	C8_prof00	
4M60-7015 | | 04b	C8_prof00	
4M60-7015 | | 05a	C8_prof00	
4M60-7015 | | 05b	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M60-7015 | | 05c	C8_prof00	
4M60-7015 | | 06a	C8_prof00	
4M60-7015 | | 06b	C8_prof00	
4M60-7015 | | 06c	C8_prof00	
4M60-7015 | | 06d	C8_prof00	
4M60-7015 | | 07a	C8_prof00	
4M60-7015 | | 07b	C8_prof00	
4M60-7015 | | 08a1	C8_prof00	
4M60-7015 | | 08a2	C8_prof00	
4M60-7015 | | 08b	C8_prof00	
4M60-7015 | | 08c	C8_prof00	
4M60-7015 | | 08d	C8_prof00	
4M60-7015 | | 08e	C8_prof00	
4M60-7015 | | 09	C8_prof00	
4M60-7040 | | 01a	C8_prof00	
4M60-7040 | | 01b	C8_prof00	
4M60-7040 | | 02a	C8_prof00	
4M60-7040 | | 02b	C8_prof00	
4M60-7040 | | 03a	C8_prof00	
4M60-7040 | | 03b	C8_prof00	
4M60-7040 | | 04a	C8_prof00	
4M60-7040 | | 04b	C8_prof00	
4M60-7040 | | 04c	C8_prof00	
4M60-7040 | | 04d	C8_prof00	
4M60-7040 | | 04e	C8_prof00	
4M60-7040 | | 04f	C8_prof00	
4M60-7040 | | 05a	C8_prof00	
4M60-7040 | | 05b	C8_prof00	
4M60-7040 | | 06a	C8_prof00	
4M60-7040 | | 06b	C8_prof00	
4M60-7040 | | 07a	C8_prof00	
4M60-7040 | | 07b	C8_prof00	
4M60-7040 | | 08a	C8_prof00	
4M60-7040 | | 08b	C8_prof00	
4M60-7040 | | 09a	C8_prof00	
4M60-7040 | | 09b	C8_prof00	
4M60-7041 | | 01a	C8_prof00	
4M60-7041 | | 01b	C8_prof00	
4M60-7041 | | 02a	C8_prof00	
4M60-7041 | | 02b	C8_prof00	
4M60-7041 | | 03a	C8_prof00	
4M60-7041 | | 03b	C8_prof00	
4M60-7041 | | 04a	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4M60-7041 | | 04b	C8_prof00	
4M60-7041 | | 06a	C8_prof00	
4M60-7041 | | 06b	C8_prof00	
4M60-7041 | | 07	C8_prof00	
4M60-7041 | | 08a	C8_prof00	
4M60-7041 | | 08b	noLocalAuthFailureReport	
4M60-7041 | | 08c	C8_prof00	
4M60-7041 | | 08d	C8_prof00	
4M60-7041 | | 09a	C8_prof00	
4M60-7041 | | 09b	C8_prof00	
4M60-7045 | | 01	C8_prof00	
4M60-7045 | | 02	C8_prof00	
4M60-7045 | | 03	TAA_TACdenial_OfflineOnline	
4M60-7045 | | 04	C8_prof00	
4M60-7046 | | 01	C8_prof00	
4M60-7100 | | 03	TAA_TAConline_OfflineOnline	
4M60-7100 | | 04	C8_prof00	
4M60-7100 | | 05	C8_prof00	
4M60-7100 | | 06	C8_prof00	
4M61-2830 | A | 01	C8_prof00	
4M61-2830 | A | 02a	confIadMacOffset	
4M61-2830 | A | 02b	confIadMacOffset	
4M61-2830 | A | 02c	C8_prof00	
4M61-2830 | A | 03a	C8_prof00	
4M61-2830 | A | 03b	C8_prof00	
4M61-2830 | A | 03c	C8_prof00	
4M61-2830 | A | 04	C8_prof00	
4M61-2830 | B | 01	C8_prof00	
4M61-2830 | B | 02a	confIadMacOffset	
4M61-2830 | B | 02b	confIadMacOffset	
4M61-2830 | B | 02c	C8_prof00	
4M61-2830 | B | 03a	C8_prof00	
4M61-2830 | B | 03b	C8_prof00	
4M61-2830 | B | 03c	C8_prof00	
4M61-2830 | B | 04	C8_prof00	
4M61-9284 | | 01	C8_prof00	
4M61-9284 | | 02	C8_prof00	
4M61-9284 | | 03	C8_prof00	
4M61-9284 | | 04	C8_prof00	
4M62-7450 | | 01	C8_prof00	
4M62-7450 | | 02	C8_prof00	
4M62-7450 | | 03	C8_prof00	
4M62-7450 | | 05	C8_NoEda	

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Sub-Case ID	Configuration	Applicability Conditions
4M62-7450 | | 06	C8_prof00	
4M62-7450 | | 07	C8_prof00	
4M65-2926 | | 01	TAA_TAConline_OfflineOnline	
4M65-2926 | | 02	TAA_TACdenial_OnlineOnly_decline	
4M65-2926 | | 03	C8_prof00	
4M65-2928 | | 01	C8_prof00	
4M65-2928 | | 02	C8_prof00	
4M65-2928 | | 03	C8_prof00	
4M65-2930 | | 01	confCVM_allCVMs	
4M65-2932 | | 01	confCVM_allCVMs	
4M65-2932 | | 02	C8_prof00	
4M65-2934 | | 01	confCVM_allCVMs	
4M65-2934 | | 02	confCVM_allCVMs	
4M65-2941 | | 01a	C8_prof00	
4M65-2941 | | 01b	C8_prof00	
4M65-2941 | | 02a	TAA_TAConline_OfflineOnline	
4M65-2941 | | 02b	TAA_TACdenial_OnlineOnly_decline	
4M65-2941 | | 03a	C8_prof00	
4M65-2941 | | 03b	C8_prof00	
4M65-2941 | | 03c	C8_prof00	
4M65-2941 | | 03d	C8_prof00	
4M65-2942 | | 01a	confCVM_allCVMs	
4M65-2942 | | 01b	confCVM_allCVMs	
4M65-2942 | | 02a	TAA_TAConline_OfflineOnline	
4M65-2942 | | 02b	TAA_TACdenial_OfflineOnline	
4M70-2945 | | 01	C8_prof00	[DE/DS]
4M70-2945 | | 02	C8_prof00	[DE/DS]
4M70-2945 | | 03	C8_prof00	[DE/DS]
4M70-7090 | | 01	C8_prof00	[DE/DS]
4M70-7090 | | 02	C8_prof00	[DE/DS]
4M70-7090 | | 03	C8_prof00	[DE/DS]
4M70-7090 | | 04	C8_prof00	[DE/DS]
4M70-7301 | | 01	C8_prof00	[DE/DS]
4M80-1005 | | 01	C8_prof00	[DE/DS]
4M80-1005 | | 02	C8_prof00	[DE/DS]
4M80-1005 | | 03	C8_NoEda	[DE/DS]
4M80-1021 | | 01	C8_prof00	
4M80-1021 | | 03	C8_prof00	
4M80-1021 | | 04	C8_prof00	
4M80-1021 | | 05	C8_prof00	
4M80-3010 | A | 01	C8_prof00	
4M80-3010 | A | 02	TAA2_online	
4M80-3010 | A | 03	TAA2_online	

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Sub-Case ID	Configuration	Applicability Conditions
4M80-3010 | B | 01	TAA1_online_cardTVR	
4M80-3010 | B | 02	TAA_TAConline_OfflineOnline	
4M80-3010 | C | 01	TAA2_decline	
4M80-3010 | C | 02	C8_prof00	
4M80-3010 | C | 03	TAA2_online	
4M80-3010 | C | 04	TAA2_decline	
4M80-3010 | C | 05	TAA2_decline	
4M80-3010 | D | 01	TAA1_online_cardTVR	
4M80-3010 | D | 02	TAA1_online_cardTVR	
4M80-3010 | D | 03	TAA1_online_cardTVR	
4M80-3010 | D | 05	TAA1_online_cardTVR	
4M80-3010 | E | 01	TAA1_decline_cardTVR	
4M80-3010 | E | 02	TAA1_decline_cardTVR	
4M80-3010 | E | 03	TAA1_decline_cardTVR	
4M80-3010 | F | 01	C8_prof00	
4M80-3010 | F | 02	C8_prof00	
4M80-3010 | F | 03	C8_prof00	
4M80-3010 | F | 04	C8_prof00	
4M80-3010 | F | 05	C8_prof00	
4M80-3010 | F | 06	C8_prof00	
4M80-3019 | A | 01a	C8_prof00	
4M80-3019 | A | 01d	C8_prof00	
4M80-3019 | A | 01j	C8_prof00	
4M80-3019 | A | 01m	C8_prof00	
4M80-3019 | A | 02a	C8_prof00	
4M80-3019 | A | 02d	C8_prof00	
4M80-3019 | A | 02j	C8_prof00	
4M80-3019 | A | 02m	C8_prof00	
4M80-3019 | A | 03	C8_prof00	
4M80-3019 | A | 05b	C8_prof00	
4M80-3019 | B | 01b	TAA2_decline	
4M80-3019 | B | 01e	TAA2_decline	
4M80-3019 | B | 01k	TAA2_decline	
4M80-3019 | B | 01n	TAA2_decline	
4M80-3019 | B | 02b	TAA2_decline	
4M80-3019 | B | 02e	TAA2_decline	
4M80-3019 | B | 02k	TAA2_decline	
4M80-3019 | B | 02n	TAA2_decline	
4M80-3019 | B | 04	TAA2_decline	
4M80-3019 | B | 05c	TAA2_decline	
4M80-3019 | C | 01c	TAA2_decline	
4M80-3019 | C | 01f	TAA2_decline	
4M80-3019 | C | 01l	TAA2_decline	

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Sub-Case ID	Configuration	Applicability Conditions
4M80-3019 | C | 01o	TAA2_decline	
4M80-3019 | C | 02c	TAA2_decline	
4M80-3019 | C | 02f	TAA2_decline	
4M80-3019 | C | 02l	TAA2_decline	
4M80-3019 | C | 02o	TAA2_decline	
4M80-3019 | C | 05a	TAA2_decline	
4M80-3019 | C | 05d	TAA2_decline	
4M80-4444 | | 01	C8_prof00	
4M80-4444 | | 01b	C8_prof00	
4M80-4444 | | 02a	C8_prof00	
4M80-4444 | | 02b	C8_prof00	
4M80-4444 | | 03	C8_prof00	
4M80-4444 | | 04a	C8_prof00	
4M80-4444 | | 04b	C8_prof00	
4M80-4444 | | 05	C8_prof00	
4M80-4444 | | 06	C8_DiscretionaryDataTagList_emptyTag	
4M80-5000 | | 01	C8_tagMapping	
4MX8-0023 | A | 01	C8_prof00	
4MX8-0023 | A | 02	C8_prof00	
4MX8-0023 | A | 03	C8_prof00	
4MX8-0023 | A | 04	C8_prof00	
4MX8-0023 | B | 01	C8_prof00	
4MX8-0023 | B | 02	C8_prof00	
4MX8-0023 | B | 03	C8_prof00	
4MX8-0023 | B | 04	C8_prof00	
4MX8-0023 | C | 01	C8_prof00	
4MX8-0023 | C | 02	C8_prof00	
4MX8-0023 | C | 03	C8_prof00	
4MX8-0023 | C | 04	C8_prof00	
4MX8-0023 | D | 01	C8_prof00	
4MX8-0023 | D | 02	C8_prof00	
4MX8-0023 | D | 03	C8_prof00	
4MX8-0023 | D | 04	C8_prof00	
4MX8-0030 | | 01	C8_RFUbits	
4MX8-0035 | | 01	C8_prof00	
4MX8-0035 | | 02	C8_prof00	
4MX8-0040 | | 01	C8_prof00	not [DE/DS]
4MX8-0200 | | 01	C8_T2R_ASRPD_PTFWFempty_Mask	[DE/DS]
4MX8-0200 | | 02	C8_T2R_ASRPD_PTFWFempty_Mask	[DE/DS]
4MX8-0666 | | 01	C8_T2R_ASRPD_PTFWFempty_Mask	[DE/DS]
4MX8-0900 | | 01	C8_prop	
4MX8-0900 | | 02	C8_defaultCDOL1MaxLength	
4MX8-0900 | | 03	C8_prof00	

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Sub-Case ID	Configuration	Applicability Conditions
4MX8-0900 | | 04	C8_prof00	
4MX8-0900 | | 05	C8_prof00	
4MX8-1000 | | 01	C8_RealAIDs	
4MX8-1000 | | 02	C8_RealAIDs	
4MX8-1000 | | 03	C8_RealAIDs	
4MX8-1000 | | 04	C8_RealAIDs	
4MX8-1000 | | 05	C8_RealAIDs	
4MX8-1000 | | 06	C8_RealAIDs	
4MX8-1000 | | 07	C8_RealAIDs	
4MX8-1000 | | 08	C8_RealAIDs	
4MX8-1000 | | 09	C8_RealAIDs	
4MX8-1001 | | 01	C8_VISA1	
4MX8-1001 | | 02	C8_VISA2	
4MX8-1001 | | 03	C8_VISA3	
4MX8-1001 | | 04	C8_VISA4	
4MX8-1001 | | 05	C8_VISA5	
4MX8-1001 | | 06	C8_VISA6	
4S00-0248 | | 04	C8_prof00	
4S00-0900 | | 01	C8_prop	
4S00-0900 | | 02	C8_defaultCDOL1MaxLength	
4S00-0900 | | 03	C8_prof00	
4S00-0900 | | 04	C8_prof00	
4S00-0900 | | 05	C8_prof00	
4S00-1001 | | 06	C8_VISA6	
4S00-1423 | | 01	C8_prof00	
4S00-2021 | | 04	C8_prof00	[DE/DS]
4S00-4444 | | 03	C8_prof00	

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